ISOLATED MICRO-GRIDS AND ELECTROMAGNETIC COMPATIBILITY: ANALYSIS, MITIGATION TECHNIQUES, AND MEASUREMENTS OF RAPID LOAD CHANGES

Alexander Matthee

ISOLATED MICRO-GRIDS AND ELECTROMAGNETIC COMPATIBILITY: ANALYSIS, MITIGATION TECHNIQUES, AND MEASUREMENTS OF RAPID LOAD CHANGES

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Alexander Matthee

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This dissertation has been approved by:

Promotors
prof.dr.ir. F.B.J. Leferink
prof. dr. ir. R. Smolenski

Co-promotor
dr.ir. D.J.G. Moonen

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Promotors:	prof.dr.ir. F.B.J. Leferink Universiteit Twente, EEMCS, Radio Systems prof. dr. ir. R. Smolenski University of Zielona Gora, Poland
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	prof. dr. ing. Garbe Leibniz University Hannover, Institut für Grundlagen der Elektrotechnik und Messtechnik
	prof. dr. J.F.G. Cobben TU/e

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Contents

1 *Summary* 19

2 *Samevatting* 21

3 *Introduction* 23

3.1 *Description of the ETOPIA Project* 23

3.2 *Marie Sklodowska-Curie projects* 25

3.3 *Research Objectives* 25

3.4 *Informal Mission Statement* 26

3.5 *Development of The Electrical Grid and the Impacts of Emissions* 27

3.5.1 *Conducted* 27

3.5.2 *Radiated - Relevant in Terms of Isolated Micro-grids* 30

3.6 *Setting the Stage: Structure and Thesis Overview* 32

4 *Compliance Standards, Measurement Techniques and Grid Characteristics* 35

4.1 *Rational and Significance* 35

4.2 *EMC Compliance Standards* 36

4.2.1 *Relevant EMC Compliance Standards* 37

4.3 *Factors of Concern* 38

4.3.1 *Inrush Current of SPECs Devices* 38

4.3.2 *Standardized Grid Impedance Assumptions - Z_{ref}* 41

4.3.3 *Grid Inertia* 45

4.4 *Measurement Techniques* 47

5 *Measurement System Development* 49

5.1 *State Of The Art* 49

5.2 *Design Parameter Wish List* 50

5.3 *Components and Interfaces* 51

5.4 *Software* 54

5.4.1 *Software Event Timeline* 56

5.5 *Applications* 57

5.5.1 *Case Study - Carré Building (UT) - Long Duration Monitoring* 57

5.5.2 *Short Duration High Band-Width (1 MSps) Measurements* 60

6 *Analysis of Real-World Systems - Effects of Rapid Load Changes* 69

6.1 *Reasoning and History* 69

6.2 *Case studies* 73

6.2.1 *Indonesia - Rapid Load Changes on Static Power Supplies (Inductive Dominant Load)* 73

6.2.2 *Netherlands - Rapid Load Changes on Diesel Generator (Inductive Dominant Load)* 76

6.2.3 *Static Power Supplies and Capacitive Dominant Loads - Lab Micro-grid Emulation* 84

6.2.4 *Factors of Impulsive load Changes and Micro-grids* 86

6.2.5 *Active Switching Component SOA* 87

6.3 *Inrush Current* 89

6.3.1 *Emulated Grid Setup and Measurements* 89

6.3.2 *Simulation of Setup* 91

6.3.3 *Recap: Understanding the Implications of Rapid Load Changes on Micro-Grids and Generators* 97

7 *Active EMI Filter Design* 99

7.1 *Overview* 99

7.2 *Architecture* 100

7.3 FPGA Design and Implementation 101
7.3.1 Design Process for Programming Xilinx Artix-7 FPGA 101
7.3.2 Simulation - Proof of Concept 103
7.3.3 System components and software 105
7.3.4 Layout and Block Creation 108
7.3.5 Control block design 108
7.3.6 HDL Simulation/Emulation 109
7.3.7 Custom IP Block Creation and Porting 111
7.3.8 Vivado Block, Synthesis, Implementation and Programming 113
7.3.9 Control signals 117
7.4 Full System output measurement 118
7.5 Results Discussion 119

8 Research Conclusion & Future Work 123
8.1 Overview 123
8.1.1 Development of Models, Measurement Tools and Software 124
8.1.2 Analysis and Core Problems Identified 126
8.1.3 Proposed Solutions 127
8.1.4 Wrap Up 128
8.2 Recommendations for Future Work 129
8.2.1 1. Experimental Validation and Field Testing 129
8.2.2 2. System-Level Studies 129
8.2.3 3. Optimal Control Strategies 130
8.2.4 4. Integration of Energy Storage Systems 130
8.2.5 5. Standards and Guidelines 130

9 List Of Publications 133
9.1 Peer Reviewed Journals 133
9.2 Peer Reviewed Conference Papers 133

A Appendix A 135
A.1 HDL Coder Sub-blocks 135

Bibliography 137

Biography 149

List of Figures

3.1 Image of grid infrastructure in New York City - late 1800's. [1] 27
3.2 Lightning strike near electrical infrastructure [2] 30

4.1 LED inrush current plots. Opaque plot for all data and solid line for average values per cluster [3] 39
4.2 Measurement setup for LED inrush current. 39
4.3 Measured and simulated inrush current root mean square (RMS) value over a 10 ms period [3] 41
4.4 Inrush current waveforms for Cinergia Grid Emulator (GE), Laboratory power, LISN 1, LISN 2 as well as Z_{ref} impedance in accordance with [4] [5]. 43
4.5 Steady state current wave-forms for Cinergia GE R_{int}[0 - 1 Ω], Laboratory power, LISN 1, LISN 2 . 44
4.6 Chubu Electric Nishi-Nagoya power plant Block-1 in Japan — powered by GE's 7HA gas turbine [6] 45
4.7 Equations for the rotational inertia of some simple shapes under rotation [7] 46
4.8 Siemens phasor measurement units (PMU) [8] 47

5.1 PowerQube power analyzer [9] 49
5.2 Node deployment in multi-zone micro-grid [10] 50
5.3 Measurement node components and interfaces [10] 52
5.4 Measurement node components and interfaces [10] 53
5.5 Measurement node components and interfaces [11] 53
5.6 Software command paths 54
5.7 Basic software stack 55
5.8 Basic software stack 55
5.9 Time line of software events during measurement 56
5.10 Carré Building - The University of Twente, Enschede, The Netherlands 57
5.11 Phase voltage input waveforms a) No load, b) Energized 57
5.12 Single line diagram of Measurement points 58

5.13 Phase voltage input waveforms a) No load, b) Energized 59
5.14 Single line diagram of test setup and measurement points 1, 2 and 3 60
5.15 Current profile vs measurement window time for branch (1), (2) and (3) [12] 61
5.16 Current plot in branch (1), (2) and (3) [12] 62
5.17 Inrush period voltage - branch (1), (2), (3) [12] 63
5.18 Sub-millisecond window inrush voltage deviation [12] 64
5.19 Voltages in branch (1), (2) and (3) with additional capacitor on (3) [12] 64
5.20 Current in branch (1), (2) and (3) [12] 65
5.21 voltages in branch (1), (2) and (3) with inrush on peak of voltage cycle [12] 66
5.22 Sub-cycle voltage vs time in branch (1), (2) and (3) [12] 66

6.1 ESKOM load shedding histogram from 2007 - 2020 [13] 70
6.2 Solar PV panels at Baron TechnoPark, Indonesia (BRIN) 73
6.3 SMA micro-grid Baron Technopark (BRIN) Indonesia 73
6.4 Micro-grid output voltage and current 74
6.5 Apparent power (top) and RMS voltage with frequency (below) 75
6.6 Measurement configuration on generator micro-grid 77
6.7 Measurement setup on radar platform. 77
6.8 Startup sequence measurement of voltage and current, with multiple highlighted events 79
6.9 3 Phase voltage and current during transient event on diesel generator 80
6.10 Apparent power (top) and RMS voltage and frequency (below) 81
6.11 Apparent power (top) and frequency (below) 82
6.12 Operational mode measurement of voltage and current, with instantaneous and apparent power as labeled. 83
6.13 RMS voltage and frequency for operational mode 84
6.14 Cinergia GE, 15 kVA (5 kVA per phase) [14] 84
6.15 All measured loop impedances compared with IEC 61000-4-7 and percentile curves [15] 85
6.16 Impulse response of converter type power supply 85
6.17 Measurement points for internal (inductor current) and output current. Internal components of Cinergia GE obtained from datasheet [14] 86
6.18 Inrush current resulting in inverter failure [16] (t_{delay} = time from switch event to inverter deactivation) 87
6.19 Safe operating area for a MOSFET [17] 88

6.20 Potential failure points diagram [5] 89
6.21 Cinergia GE with LED light bank 90
6.22 Software controlled relay diagram (1 channel out of 8 total) 91
6.23 Inrush currents for clusters of 2W LED lamps [3] 92
6.24 2W LED simulation model - Simulink 92
6.25 Simulink model of an H-bridge inverter 93
6.26 Simulation vs measured data of light emitting diode (LED) devices. Measured data with solid lines, simulation data with dashed lines. 94
6.27 Simulation of inrush current trend with increased number of LED devices 95
6.28 Voltage deviations caused by switching currents for different amount of LED lights (Simulated) 95
6.29 Measured and simulated inrush current RMS value over a 10 ms period 96

7.1 Diagram of active EMI filter (AEF) as auxiliary high frequency power source [18] 103
7.2 Effects on voltage stability and current delivery with and without compensation [18] 104
7.3 Vivado field Programable Gate Array (FPGA) IP block design 106
7.4 AEF system component diagram 107
7.5 Control system flow diagram 108
7.6 Control system for grid deviation - Simulink 109
7.7 Simulation output voltage waveform with input measurement of current (Voltages scaled down for input voltage range of ADC and output voltage of control signal FPGA). 110
7.8 Timing information and worst negative slack (WNS) 114
7.9 Design timing summary 114
7.10 Vivado visualization of implemented FPGA logic elements 115
7.11 Estimates of power usage and system temperature 116
7.12 Physical arbitrary waveform generator (AWG) input (orange) and FPGA control output signal (red) measured, Simulink simulation output signal (blue) 117
7.13 Response time of AEF to a rapid load change, small time window measurement (Error signal - Blue, AEF response - RED, time scale on x-axis in ms) 118
7.14 AEF response to pulse load current and grid voltage deviation 119
7.15 AEF response to pulse load current - small time scale 120

8.1 Project path 123

A.1 Simulink sub-block which represents all internal function blocks 136

List of Tables

4.1 Number of LED vs inrush current characteristics 40
4.2 Current characteristics per source impedance [5] 44
4.3 Current peak and voltage sag during inrush. 44

5.1 Picoscope specifications 51
5.2 Measurement node components 52
5.3 THD_V measurement results 59
5.4 Branch loads 61
5.5 Current and power values per branch [12] 62
5.6 Current and power values per branch with added capacitor on branch (3) 67

6.1 Number of LED vs inrush current characteristics 91
6.2 Current deviation and time derivatives 93

7.1 Load profile of input current data 111

Glossary

ADC analogue to digital converter. 101, 105, 110, 112, 117
AEF active EMI filter. 13, 19, 99, 100, 103–105, 109–111, 117–121, 124, 127, 128
API application programming interface. 51, 52, 55
AWG arbitrary waveform generator. 13, 117, 121

CF crest factor. 38, 40, 91

EMC Electromagnetic Compatibility. 19–23, 25, 27, 28, 30, 35–37, 40, 41, 46, 50, 69, 70, 89, 94, 99, 101, 119, 120, 123–126, 130
EMI electromagnetic interference. 30, 34, 36, 38, 47–50, 58, 59, 84, 89, 99
EU European Union. 25, 36

FFT fast Fourier transform. 47, 48
FPGA field Programable Gate Array. 13, 99, 100, 105, 106, 108, 109, 111–115, 117, 121, 124

GE Grid Emulator. 11–13, 39, 42–44, 84–86, 90, 96, 103, 118
GFM grid-forming. 29
GPIO general purpose input output. 54
GPS global positioning system. 50

HEMPHigh-Altitude Electromagnetic Pulse. 30
HIL hardware in-the-loop. 105
HVACheating, ventilation, and air conditioning. 77

I/O input output. 105, 108
IEMI intentional electromagnetic interference. 30
IGBT insulated-gate bipolar transistor. 35, 87

LAN local area network. 54, 55, 125
LED light emitting diode. 13, 39, 40, 42, 44, 57, 60, 91–95, 103, 110, 118
LISN line impedance stabilization network. 42–44
LTI Linear Time-Invariant. 85
LV low voltage. 58, 85

MCR main control room. 58, 73
MN measurement node. 60, 63
MOSFET metal-oxide semiconductor field-effect transistor. 35
MSCA Marie Skłodowska-Curie Actions. 25

NL non-linear. 27
NTC negative temperature coefficient. 39, 90

PC personal computer. 52–55, 105
PMU phasor measurement units. 11, 47, 50
POC point of connection. 57–59
PQ power quality. 57, 58
PTP precision Time Protocol. 50
PWM pulse width modulation. 117

RAM random access memory. 112
RMS root mean square. 11, 13, 38, 40, 41, 62, 66, 74, 80, 90, 94, 96
RO reverse osmosis. 33, 71, 73
RP4 Raspbery Pi 4. 51, 52, 54, 55, 63
RVC rapid voltage changes. 47

SiC Silicium-Carbide. 100
SOA safe operating area. 8, 87
SPECs switching power electronics converters. 7, 27, 38, 46, 69–71, 86, 89, 91, 94, 96

THD total harmonic distortion. 57–59, 84

USB universal serial bus. 51, 55, 117, 125

VFDs variable frequency drives. 77

WNS worst negative slack. 13, 113, 114

1 Summary



This thesis focuses on investigating the Electromagnetic Compatibility (EMC) challenges in isolated micro-grids, specifically analyzing the impact of rapid load changes on the reliability of power supplies. The research approach consisted of three key steps: developing a measurement system, conducting measurements on real-world diesel generators and inverters, and proposing a solution to mitigate the identified issues.

The first step involved the development of a sophisticated measurement system capable of capturing high-bandwidth data to comprehensively understand the extent of the EMC problem. This system facilitated the gathering of a comprehensive dataset, enabling detailed analysis of the challenges associated with rapid load changes in complex micro-grids.

Using the measurement system, measurements were performed on real-world diesel generators and inverters as well as lab-based grid emulations and software models, providing insights into the nature and causes of the EMC issues in isolated systems. The research discovered that isolated micro-grids lacked the necessary inertia to effectively handle rapid load changes, leading to reliability concerns.

Building upon the findings from the measurements, a solution was proposed to address the identified problems. The solution involved the design and construction of a high-speed FPGA-controlled h-bridge inverter or AEF. These advanced control technologies aimed to compensate for the rapid rise in power caused by load fluctuations, thereby mitigating the adverse effects on the power supply's reliability in isolated micro-grids. This thesis

presents a comprehensive approach to understanding and addressing the EMC challenges arising from rapid load changes in isolated micro-grids.

By developing a measurement system, conducting real-world measurements, and proposing an innovative solution, this research provides valuable insights and recommendations for improving the reliability and electromagnetic compatibility of power supplies in isolated micro-grids. The findings contribute to the advancement of knowledge in the field and offer practical guidance for enhancing the performance of isolated micro-grids in various applications.

2 *Samevatting*



Deze scriptie richt zich op het onderzoeken van de EMC-uitdagingen in geïsoleerde micro-grids, waarbij specifiek de impact van snelle belastingsveranderingen op de betrouwbaarheid van de stroomvoorziening wordt geanalyseerd. De onderzoeksbenadering bestond uit drie hoofdstappen: het ontwikkelen van een meetsysteem, het uitvoeren van metingen aan dieselgeneratoren en omvormers in de praktijk, en het voorstellen van een oplossing om de geïdentificeerde problemen te verminderen.

De eerste stap omvatte de ontwikkeling van een geavanceerd meetsysteem dat in staat is om gegevens met een hoge bandbreedte vast te leggen om een grondig inzicht te krijgen in de omvang van het EMC-probleem. Dit systeem vergemakkelijkte het verzamelen van een uitgebreide dataset, waardoor gedetailleerde analyses van de uitdagingen die gepaard gaan met snelle belastingsveranderingen in complexe micro-grids mogelijk waren.

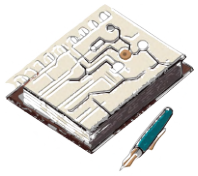
Met behulp van het meetsysteem werden metingen uitgevoerd aan dieselgeneratoren en omvormers in de praktijk, evenals laboratoriumopstellingen en softwaremodellen van netemulaties. Dit verschafte inzichten in de aard en oorzaken van de EMC-problemen in geïsoleerde systemen. Het onderzoek ontdekte dat geïsoleerde micro-grids niet over voldoende inertie beschikten om snelle belastingsveranderingen effectief op te vangen, wat leidde tot betrouwbaarheidsproblemen.

Voortbouwend op de bevindingen uit de metingen werd een oplossing voorgesteld om de geïdentificeerde problemen aan te pakken. De oplossing omvatte het ontwerp en de constructie van een snelheidsregelende h-brugomvormer

met FPGA-besturing of een actieve elektromagnetische interferentiefilter. Deze geavanceerde regeltechnologieën hadden als doel om de snelle stijging in vermogen als gevolg van belastingsfluctuaties te compenseren, waardoor de nadelige effecten op de betrouwbaarheid van de stroomvoorziening in geïsoleerde micro-grids werden verminderd.

Deze scriptie presenteert een uitgebreide aanpak voor het begrijpen en aanpakken van de EMC-uitdagingen die voortkomen uit snelle belastingsveranderingen in geïsoleerde micro-grids. Door de ontwikkeling van een meetsysteem, het uitvoeren van metingen in de praktijk en het voorstellen van een innovatieve oplossing, biedt dit onderzoek waardevolle inzichten en aanbevelingen om de betrouwbaarheid en elektromagnetische compatibiliteit van stroomvoorzieningen in geïsoleerde micro-grids te verbeteren. De bevindingen dragen bij aan de vooruitgang van kennis op dit gebied en bieden praktische richtlijnen om de prestaties van geïsoleerde micro-grids in verschillende toepassingen te verbeteren.

3 Introduction



3.1 Description of the ETOPIA Project

ETOPIA - **E**uropean **T**raining network **O**f **P**hD researchers on **I**nnovative **E**MI analysis and power Applications. ETOPIA project research is funded by the European Union's Horizon 2020 research and innovation programme under the Marie Sklodowska-Curie grant agreement No 812753. The project is based at six universities around Europe and is focused on studying the EMC of new technologies. It is intended that the project aims to increase knowledge of the impact of these technologies on the electrical grid and to identify ways to improve the EMC of micro-grids and other electrical systems that incorporate these technologies. ETOPIA is organized along three research lines which are organised in work packages (WP):

- WP1-4 Management, training, dissemination and outreach.
- WP5 Non-conventional modeling and measurements.
- WP6 Multi-domain optimization and,
- WP7 EM-Coexistence.

The project includes co-operation and scientific training with industrial partners across Europe.

- THALES
- Nederlandse EMC-ESD Vereniging
- IEEE EMC
- SOFTRONIC
- URSUS Bus
- Lambda Engineering
- Ekoenergetyka
- RSE Ricerca sul Sistema Energetico
- CESi Automation S.R.L
- RH MARINE
- JAGUAR LAND ROVER
- ENEA OPERATOR
- NETWORK RAIL
- ABB
- ATKINS CONSULTANTS LIMITED
- SOLARIS BUS & COACH
- TAURON
- IAV Automotive Engineering



The project partnership includes collaboration between six universities across Europe, each hosting two researchers.

- University of Nottingham - United Kingdom
- University of Zielona Góra - Poland
- Leibniz University - Germany
- Politecnico di Milano - Italy
- University of Craiova - Romania
- University of Twente - The Netherlands

UNIVERSITY OF CRAIOVA
POLITECNICO MILANO 1863
Leibniz Universität Hannover
UNIVERSITY OF ZIELONA GÓRA
UNIVERSITY OF TWENTE.
University of Nottingham UK | CHINA | MALAYSIA

3.2 *Marie Sklodowska-Curie projects*



The Marie Skłodowska-Curie Actions (MSCA) are a set of European Union (EU) funding programs that support research and innovation in a variety of fields. The MSCA are named after Marie Skłodowska Curie, a pioneering scientist and the first woman to win a Nobel Prize.

The MSCA provide funding for research projects, training programs, and fellowships to researchers at all stages of their careers, from doctoral students to experienced researchers. The MSCA are designed to support the mobility of researchers within the EU and beyond, and to encourage the exchange of knowledge and ideas between different research groups and institutions. The main motivation for the MSCA is to support excellence in research and innovation, and to help Europe maintain its position as a global leader in these areas. The MSCA aim to address societal challenges, such as climate change, health, and energy, and to foster economic growth and competitiveness in the EU. The MSCA also aim to promote the inclusion and diversity of the research workforce, and to support the career development of researchers.

3.3 *Research Objectives*

The objectives of this specific work package within the ETOPIA project are to provide cutting-edge measurements and analysis of conducted emissions' propagation and aggregation within the power system. The focus is on very fast, short-duration transient events and developing techniques to mitigate their negative impact. As distributed and embedded power generation continues to grow and new technologies emerge in loads and supplies, the EMC of micro-grids is compromised. The project aims to develop application-specific measurement systems, analyze micro-grid conditions, and create new mitigation strategies to address these challenges.

3·4 *Informal Mission Statement*

In this age of rapid technological advancement, information is more freely accessible than ever before, with the ability to reach beyond human boundaries. Although this statement may seem obvious, it is imperative that we acknowledge the pace at which we are moving forward and the potential consequences of our actions. We may not be prepared for the changes that are happening, and our current methods of protection may not be applicable in the near future. This thesis will not solve all of the world's problems, it aims to explore specific issues in depth.

The electrical grid has become essential to our way of life, providing power and connectivity to a plethora of devices that have become ingrained in our daily routines. The sheer number of devices we interact with on a daily basis is staggering, and the rate at which devices become obsolete is equally astonishing. Our expectations and perception of technology are heavily influenced by marketing trends, leaving many unaware of the potential risks associated with these devices. It is our duty, as those with the knowledge and expertise, to move forward responsibly and protect those who may not fully understand the risks involved.

I was born in South Africa, a country currently experiencing major grid instability. The adoption of micro-grids are increasing due to grid uncertainly and rising cost of energy. The reason for the grid issues are complex, regardless of this, the installation of micro-grids are a viable solution. By exploring the risks and techniques for a stable cost effective micro-grid, hopefully this transition will happen more swiftly and allow access for a larger group who are in need of a stable energy supply. South Africa is not alone in this, as this effects the majority of the world's population.

3.5 *Development of The Electrical Grid and the Impacts of Emissions*

The electrical grid has come a long way since its inception in the late 1800's, shown in Fig. 3.1, from a simple system of a few central generating plants to a complex network of generation and load sources spanning entire countries [19]. The recent advancement in micro-grid technology, such as generator- and inverter-based micro-grids, has helped to provide reliable power to remote or under-served areas and increased the overall resilience and reliability of the electrical grid. Inverter-based micro-grids that use renewable energy sources have the added advantage of reducing greenhouse gas emissions and decreasing reliance on fossil fuels [20]. But, with new technology comes new challenges.

Notes may appear in the right margin to provide a quick explanation for a point in the text or to draw attention to important information. Additionally, each chapter will begin with a brief summary to clarify the main points covered in the preceding chapter...

This project will focus on the effects of rapid load changes on isolated micro-grids. Micro-grids, due to them having lower grid strength or inertia in comparison to large electrical grids, are especially vulnerable to fast changing load conditions which result in EMC challenges. This is specifically aimed at the effects of changing load requirements in a very short time (micro- to millisecond-range), this could be inrush current due to load activation or ramp up current during the acceleration of a motor. There are many different aspects of EMC over large frequency ranges. In this case the focus will be on the low frequency range of generally below 150 kHz and mostly intermittent in nature. Meaning non-repetitive and not predictable. To expand on effects some conducted and radiated aspects will be highlighted in the next section.

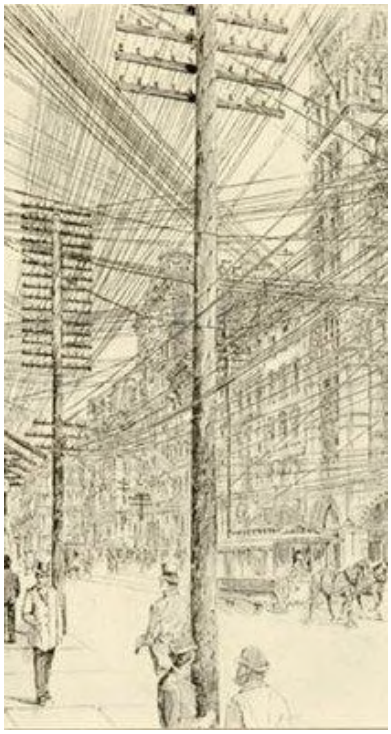


Figure 3.1: Image of grid infrastructure in New York City - late 1800's. [1]

3.5.1 *Conducted*

Increased implementation of renewable energy, distributed or isolated micro-grids, and technological advances create new EMC challenges [21; 22]. The growing prevalence of non-linear loads or SPECs loads, such as computers and other electronic equipment, poses a challenge to the EMC of the grid. Fast reacting control systems and non-linear (NL) loads are increasing EMC concerns on low-inertia power grids, especially since micro-grids lack the inherent strength of a large distribution power network. With a

weak grid, rapid load changes will result in volatile conditions on the local grid in terms of voltage and current. EMC problems are apparent even on large distribution networks, with an estimated financial damage of 22 billion USD in US industries reported in 1991 [23]. This is even more prevalent in weaker power systems due to the lack of inertia [24; 25]. Furthermore, different micro-grid power supply technologies have different characteristics and therefore exhibit different behaviors in response to power demands. Converter-type (or switching) power supplies are considered weaker with less stored energy (or inertia) in comparison to generators (rotating machines), which store energy in an inertial rotating mass. However, converter-type supplies generally have the ability to respond much faster to load changes due to the inherent capabilities of switching electronic components [26]. According to the findings of this thesis in simple terms, comparable converter type power supplies are able to handle a less severe power derivative compared to rotational type supplies, but can recover faster to steady state operation after a transient event. Low energy (short duration), high amplitude (impulse type) load change is better suited to a supply with a higher stored energy - generator. A high energy (longer duration), lower amplitude (step or ramp) is better suited to a converter type supply.

Generator-based power supplies are known to have a superior response to millisecond impulse loads compared to inverter-based systems due to the stored energy in the rotational mass being much larger than the energy required by the load capacitors or inductors to charge up. However, this advantage can be compromised if the intensity of the transient current activates protection systems or is sufficient to move the generator out of its characterized frequency operating range. For instance, multiple instances of generator failure were observed in tests for hospital emergency backup systems in the Netherlands, where numerous non-linear devices activated simultaneously [27]. On the other hand, inverter-based power supplies have minimal energy buffer to supply impulse-type loads, leading to excessive currents flowing through active components and a sharp decline in voltage output, which can result in instability, supply deactivation, and

even damage in some cases [3]. In the modern environment, impulsive-type loads are to be expected, and it is crucial for design engineers to carefully consider power supply characteristics and their compatibility with load current behavior. Overlooking these factors results in suboptimal system performance, premature end-of-life, and reduced adoption of the technology, leading to decreased energy access for those in need. Note that grid-forming (GFM) inverters are design to counteract these mentioned effects and stabilise the grid, although these devices are generally higher capacity systems with reaction times in the order or seconds [28; 29].

3.5.2 *Radiated - Relevant in Terms of Isolated Micro-grids*

Increasing reliance on uninterrupted operation of electrical and electronic systems worldwide underscores the importance of good EMC practice. Therefore, it is crucial to harden systems against continuous or intermittent high-impact events to ensure reliability, especially as systems become more complex and interdependent, presenting challenges for EMC engineers [30; 31; 32; 33; 34; 35]. Lack of immunity to radiated emissions can result in temporary device malfunction, permanent system damage, and infrastructure outages with significant implications in terms of productivity, financial loss, and even loss of life. As mentioned in the margin note, the focus of this thesis does not include radiated phenomenon. However a snippit of radiated EMC concerns are included to for a comprehensive view, especially of lightning which is a significant contribution to downtime and damage.



Figure 3.2: Lightning strike near electrical infrastructure [2]

In addition to natural occurrences such as lightning [36; 37] and geomagnetic storms [38; 39] as sources of electromagnetic interference (EMI), radiated emissions from man-made sources, such as electronic devices, also need to be considered. These emissions complement the radiated interference effects from nature and nuclear detonations, adding to the complexity of EMC considerations. A lightning strike near electrical infrastructure is shown in Fig. 3.2.

Lightning strikes, for example, account for an estimated 30% of power outages in the United States, costing around USD 1 billion, including damage to equipment, loss of production, and even loss of life [34]. In the case of a High-Altitude Electromagnetic Pulse (HEMP) attack, the consequences could be catastrophic, resulting in a long-term, complete blackout of almost all key infrastructure. As a result, the Electric Power Research Institute (EPRI) initiated a research project in 2016 to provide unclassified, technically based research into various aspects of intentional electromagnetic interference (IEMI) resulting from a HEMP attack, including threat characterization, mitigation and hardening, impacts, and vulnerability investigations [40]. During a measurement campaign at Baron Technopark - Indonesia, the original inverters installed in

Note: The scope of this project is not to delve into radiated interference, but it is mentioned to add a more comprehensive view of potential EMC concerns. Since isolated micro-grids are often in remote locations, inherently the site would be difficult to access, service and maintain. A stable energy supply with maximum up-time requires a multi-faceted design approach. Also, a direct lightning strike is a conducted phenomenon. In this sense the radiated is referring to the resulting electric and magnetic fields generated by the strike current.

the remote micro-grid were both destroyed by lightning. More on this microgrid system later.

Chapter 4: Compliance Standards, Measurement Techniques and Simulations

- Review of compliance standards for conducted emissions
- Analysis of the limitations of current measurement techniques
- Presentation of simulation models of inverters and loads

In this chapter, compliance standards will be reviewed and evaluate their effectiveness in regulating conducted emissions in isolated micro-grids. Additionally, an analysis into the current measurement techniques used to monitor these emissions and highlight any limitations or challenges that need to be addressed. The simulations and models of inverters and loads that were developed to illustrate the effects of impulse currents and their propagation through the internal components of systems are discussed. The reason being that the magnitude and nature of the current flowing in active components are not common knowledge. The limits of components and potential damage as a result, is important when formulating solutions.

Chapter 5: Measurement System Development

- Description of the measurement system development process
- Explanation of the PicoScope USB oscilloscopes and RaspberryPi micro-computers used
- Demonstration of the system's capabilities through a single command line argument

In this chapter, the development of a measurement system will be described that provides an enhanced characterization of conducted emissions in an isolated micro-grid. According to the investigation of the previous chapter, it was crucial to design a system which provides multiple measurement data points. As a micro-grid with multiple connected loads and branches can reach high complexity, the source of a disturbance (especially if intermittent) and the effect on the other loads in the local

environment and power supply must be measured with a comprehensive dataset. The system utilizes PicoScope USB oscilloscopes with multiple channels and measurement sample rate of up to 80 MSps. The oscilloscopes are interfaced with RaspberryPi micro-computers, which receive commands via a master PC over LAN. This allows for easy configuration of the measurement system, including measurement duration, the number of measurement nodes, sample rate, test name for tracking purposes, and active channels. The measurement system is designed to measure in multiple locations and channels to provide a full snapshot of the system and to detect intermittent events that could affect the grid environment for loads and supply.

Chapter 6: Analysis of Real-World Systems

- Presentation of lab measurements with a Cinergia grid emulator and various load types
- Analysis of the effects of individual load switching and failure rates of the inverter
- Description of practical measurements on an isolated microgrid in Indonesia and a military mobile radar system
- Identification of the main effect on the system and the ability of the power supply to compensate for transients

This chapter will present the results of conducting measurements using the developed measurement system from Chapter. 3.5 on real-world systems. The lab measurements were performed using a Cinergia grid emulator and a variety of different load types, including banks of LED lights and other non-linear loads with known bad inrush currents, as well as resistive and inductive loads. The effects on the grid of each load and the effect on the grid of switching individual loads were analyzed, and failure rates of the inverter and EMC concerns were identified. Practical measurements were also conducted on an inverter-based isolated micro-grid in Indonesia, with a variety of loads and a reverse osmosis (RO) water filtration system as the main large load. Measurements were also performed on a military mobile radar system powered by a diesel generator. The main identified effect is shown to be the fast change in power or $\frac{dP}{dt}$ and the inherent inertia

of the power supply. Also the ability of the power supply to react fast enough to compensate for the transient as a result of rapid load changes.

Chapter 7: Active EMI Filter Design

- Design and simulation of an active EMI filter using FPGA architecture in Matlab Simulink with HDL coder
- Demonstration of the system's ability to decrease the rising edge of a step function of a rapid load change
- Construction of the active filter using Cree SiC fast switching transistor half bridges
- Evaluation of the system's performance and reactivity to respond within 10 microseconds

In this chapter, the design and simulation of an active EMI filter using FPGA architecture is described. The system was simulated and designed in Matlab Simulink with HDL coder. The purpose of the system is to decrease the rising edge of a step function of a rapid load change. The active EMI filter was constructed using Cree SiC fast switching transistor half bridges to form a fast reactive inverter, which is intended to be added to an existing isolated micro-grid to improve the EMC of the system and reduce the risk of interference between loads. The system proved to be reactive enough to respond within 10 microseconds, and simulations showed fast reactivity to rapid load changes. This chapter will also discuss the potential benefits and limitations of the active EMI filter in reducing the intensity of transient responses in the system.

Chapter 8: Conclusion and Future Work

- Summary of the research findings and contributions
- Discussion of the limitations and future directions for research
- Implications and recommendations for the industry and regulatory agencies

4 Compliance Standards, Measurement Techniques and Grid Characteristics



4.1 Rational and Significance

When considering isolated micro-grids and EMC, the impact of rapid current fluctuations, such as variations in load conditions and inrush currents, significantly undermines the operational stability and safety of these systems [16]. This challenge forms the core focus of the investigation in this chapter, where the aim is to clarify the multifaceted issues that disrupt the harmonious coexistence of loads and system environments within isolated micro-grids. Especially with the current trend of accelerated technological advances [10], which at times leave standards outdated [5]. An inquiry sheds light on a series of interconnected concerns that collectively influence EMC and compliance standards. These concerns encompass the inadequacies of existing standards in addressing the proliferation of power electronic devices, which have become integral components of these micro-grids. Furthermore, analysis delves into the obsolescence of standardized grid impedance values, which no longer accurately reflect the dynamic realities of modern power systems. The matter of low grid inertia, a consequence of the shift toward renewable energy sources, also commands attention, as it has ramifications for system stability and EMC. [41] reports that the majority of micro-grid failures are caused by inverter failures, specifically the failure of insulated-gate bipolar transistor (IGBT)s, metal-oxide semiconductor field-effect transistor (MOSFET)s and capacitors which are the components most sensitive to over-current condi-

Note: this chapter explains some groundwork done which forms a basis for future chapters. The current state of EMC compliance standards, the measurement system developed and some early findings are discussed here, as well as the reasoning for the future conclusions.

tions. Additionally, an elaboration on the limitations of current measurement techniques, which fail to capture the intricacies of rapid load fluctuations within complex grid environments. By exploring these interconnected issues, the objective is not only to delineate their deficiencies but also to pave the way for potential mitigation strategies and improved measurement methodologies, all with the overarching aim of advancing the field of isolated micro-grids and EMC in response to evolving energy landscapes.

4.2 *EMC Compliance Standards*

EMC refers to the ability of a system to function properly in the presence of its intended environment, while not interfering with other systems used in their intended environment. EMI can be caused by a variety of factors. Conducted EMI is EMI that is transmitted through electrical conductors, such as power cables and signal lines. This type of EMI can be caused by switching transients, fault conditions, harmonics and other disturbances in the electrical system. Commercial devices in Europe are governed by regulation and must comply to the relevant standards for the intended use and environment.

The European EMC Directive is a regulatory framework established by the EU to ensure that electrical and electronic devices do not generate, or are not affected by, EMI that could hinder the functioning of other devices or pose risks to health, safety, or the environment. The directive sets standards for electromagnetic emissions and immunity of devices, and it applies to a wide range of products including consumer electronics, industrial equipment, medical devices, and more.

The EMC Directive requires that products comply with specific technical requirements related to their EMC, such as emission limits, immunity levels, and other performance criteria. Manufacturers must ensure that their products meet these requirements before they can be placed on the market within the European Economic Area (EEA), which includes the 28 EU member states, as well as Norway, Iceland, and Liechtenstein.

Compliance with the EMC Directive typically involves conducting appropriate testing and maintaining documentation or relevant certifications or markings. For certain products CE compliance can be self certified [42; 43], although it is important to follow proper procedures, to ensure the product complies with standards and is able to prove due diligence or accept legal risks. Failure to comply with the EMC Directive can result in products being withdrawn from the market, fines, or other legal consequences. It is important for manufacturers, importers, and distributors of electrical and electronic products to be aware of and comply with the EMC Directive to ensure their products can be legally sold in the European market and to ensure that they do not cause harmful interference to other devices or compromise safety. A list in Section 4.2.1 indicates standards which govern the compliance of the effects of rapid load changes and inrush current applicable to this project.

4.2.1 *Relevant EMC Compliance Standards*

1. **EN 61000-3-3:** Limits for voltage changes, voltage fluctuations, and flicker caused by rapid load changes in electrical power systems for equipment with a rated input current $\leq$ 16 A per phase.
2. **EN 61000-3-11:** Limits for voltage changes, voltage fluctuations, and flicker caused by rapid load changes in electrical power systems for equipment with a rated input current $>$ 16 A per phase or $>$ 75 A per phase for equipment intended for connection to low-voltage systems.
3. **EN 61000-3-14:** Guidelines and limits for voltage fluctuations, flicker, and harmonic emissions for equipment connected to public low-voltage distribution systems with a rated current $>$ 16 A and up to 75 A per phase.
4. **EN 61000-3-12:** Guidance on assessing the impact of voltage fluctuations and flicker caused by rapid load changes in electrical power systems, especially for equipment with a rated input current $>$ 75 A per phase.
5. **EN 61000-4-11:** Test methods and limits for voltage

dips, short interruptions, and voltage variations on public low-voltage power supply systems, addressing the immunity of equipment to rapid changes in the power supply voltage.

6. **EN 61000-4-29:** Guidance on measuring and assessing voltage dips, short interruptions, and voltage variations in public low-voltage power supply systems, especially for equipment with a rated input current > 16 A per phase.
7. **EN 61000-4-30:** Defines methods for measuring power quality parameters in electrical power systems to ensure consistency and accuracy in data collection and analysis.
8. **MIL-STD-461:** Requirements for the control of electromagnetic interference characteristics of subsystems and equipment used by the military.
9. **MIL-STD-1399:** Interface standard for shipboard systems to ensure compatibility of electrical power systems on naval ships.

4.3 *Factors of Concern*

4.3.1 *Inrush Current of SPECs Devices*

Large inrush current or the superposition of multiple small inrush currents can result in EMI that generates large net peak currents with high crest factor (CF), sometimes surpassing the impact of single higher power devices. Unfortunately, existing standards do not adequately limit this effect. For instance, according to MIL-STD-1399-300-2 [44], the maximum peak inrush is capped at 10 times the device's maximum rated current, with exemptions for devices rated under 1 kW. Meanwhile, IEC61000-3-3 [4] specifies that the maximum inrush current over a 10 ms half-cycle between zero crossings should not exceed 20 A RMS.

However, inrush currents typically manifest as short-duration, high-intensity pulses, which can result in a misleadingly low RMS current value, particularly for single low-power devices. In other words, adhering to all standards does not guarantee a reliable micro-grid installation when connecting a specific profile of loads, as the cumulative effect of multiple low-power devices switching on

simultaneously is shown to cause severe currents.

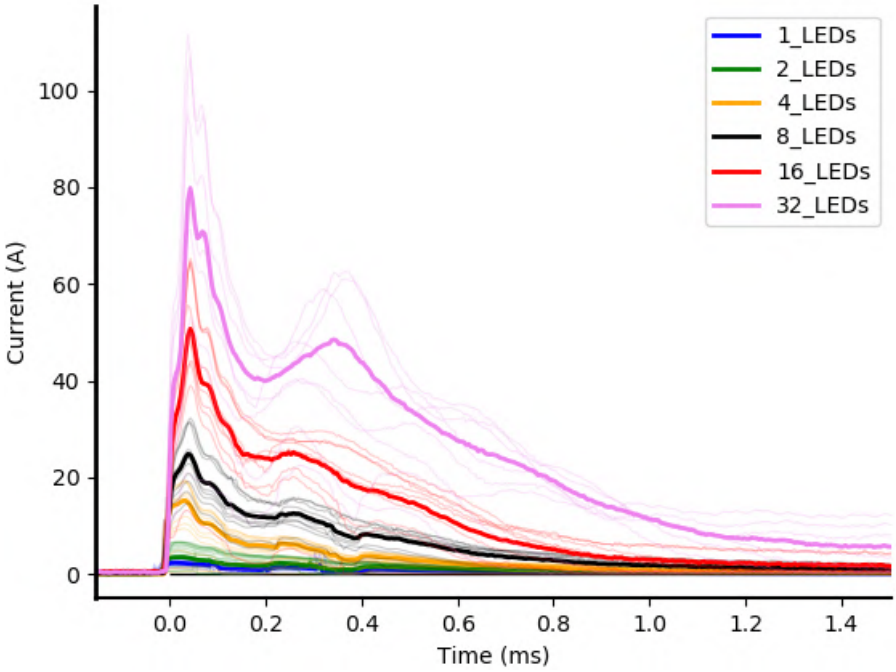


Figure 4.1: LED inrush current plots. Opaque plot for all data and solid line for average values per cluster [3]

To mitigate the impact of transient switching events, current strategies include incorporating negative temperature coefficient (NTC) devices [45], pre-charge capacitors or resistors [46], as well as low-pass filters or control strategies for electrical drives [47]. Additionally, it is common practice to switch portions of large inrush loads in stages after the initial activation of the supply source, but this is only feasible when the switching events are expected or known [27]. Micro-grid design typically involves calculating the rated load and applying a safety factor to overstate the supply, along with an additional buffer to account for the estimated power factor. However, this approach often leads to significant over-sizing of the supply source [12] in relation to the power rating requirements, and may not guarantee reliable operation.

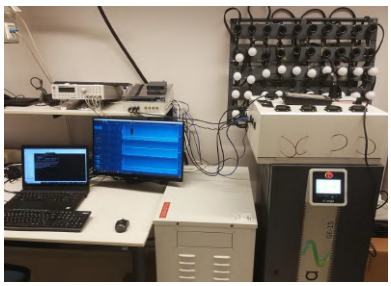


Figure 4.2: Measurement setup for LED inrush current.

In Fig. 4.1, a large measurement-set is plotted of the inrush current of common low power 2 W LED devices. The measurement setup is shown in Fig. 4.2. The equipment includes a Cinergia GE, custom made software controlled

relay system, measurement system developed in [10], and LED array. It is clear the inrush currents of these devices are very large, and increasing the amount of devices increases the effect dramatically. In Tab. 4.1, the RMS current calculated over a 10 ms period and CF of the rated LED RMS current vs peak inrush current [3].

LEDs Units	Power (W)	I_{rated} (mA)	Current Measured (Max)		
			$I_{InrushPeak}$ (A)	RMS (A) (t=10 ms)	CF
1	2	9	4.5	0.47	500
2	4	17	7.3	0.95	429
4	8	35	20	2.1	571
8	16	70	33	4.1	471
16	32	140	65	8.2	464
32	64	280	110	14.7	393

Table 4.1: Number of LED vs inrush current characteristics

If the cluster of LED devices were to be considered a single device or compliant in the sense that the activation current superposition comply with the current standards, Fig. 4.3 indicates the amount of devices where the limit would be exceeded [3]. Considering the peak currents shown in Fig. 4.1, this is a concerning factor that should be better addressed in EMC standards. This concern is difficult to address. Increased awareness of the extent and risks for design engineers would be a good start, with guidelines and best practices specifically targeted at low inertia micro-grids. Alternatively, standards require a revision or additional scope to include rapid load changes and activation currents of non-linear loads for low inertia micro-grids.

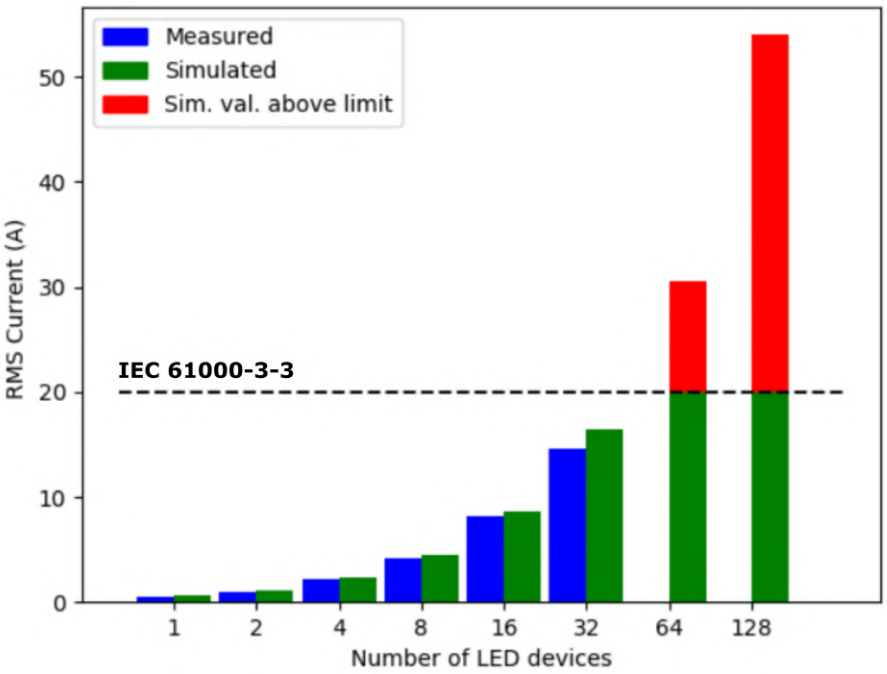


Figure 4.3: Measured and simulated inrush current RMS value over a 10 ms period [3]

4.3.2 *Standardized Grid Impedance Assumptions - Z_{ref}*

Much research has been dedicated to developing innovative techniques for measuring grid impedance in order to establish this parameter [48; 49]. Standards have also been established to regulate emissions from electrical and electronic products, which manufacturers are required to adhere to. However, with the wide range of possible grid impedance values and constantly changing grid conditions, the applicability of these standards to the modern grid environment is a topic of discussion. Studies have indicated that the standard reference impedance may not accurately reflect the real-world conditions [15], as the last comprehensive study that established these reference values was conducted in 1981 [50]. Since then, the power grid conditions and implementation of new technological advancements have undergone significant changes (especially the implementation of micro-grids, non-linear loads [3], and fast reactive control systems [51]). Factors such as load demand variations, high switching currents for load activation, and "steady state" operation can all contribute to EMC problems in a system that is fully compliant with

standards [5].

Let's take a look at a few measurements performed with different source impedance characteristics. A simulation of the reference impedance (Z_{ref}) specified in IEC61000-3-3 [4] is included for comparison. The load comprises of 32x2 W LED devices, for which inrush current is measured. Firstly, a Cinergia GE [14] is used with an output impedance of less than $0.4 + j0.25\,\Omega$ at 50 Hz, as indicated by the manufacturer, which aligns with the reference impedance (Z_{ref}) specified in [4]. The Z_{ref} is defined as a resistance and single inductor in series at 50 Hz, but it is not defined for higher frequencies. Additionally, the mains from the laboratory building at the University of Twente is employed, with and without the combination of two different line impedance stabilization network (LISN)s with isolation transformers. A Matlab Simulink simulation of the LED load with a standard impedance (Z_{ref}) as specified in [50] is also conducted[5].

From a total sample set of 103 inrush current tests, Fig. 4.4 presents the maximum inrush current values for different supply configurations. The results clearly demonstrate a distinction between the test scenario powered by lab power and GE, and the other test setups with LISN configurations, as well as the simulation with a standard supply impedance value of Z_{ref}. The rising edge of the inrush current, as well as the amplitude, are significantly higher for lab power and inverter supplies, exceeding 100 A, which is extreme compared to the steady state peak of 4 A shown in Fig. 4.5, highlighting the destructiveness of inrush currents (steady state in this case refers to the power draw during steady state operation).

Measurement clarification

- **Cin :** Cinergia GE (R-value - Variable internal resistance)
- **LISN1 :** PMM L2-25
- **LISN2 :** TEKBOX TBLC08
- **Lab :** Wall power outlet at the EEMCS power lab in the Carre building of the University of Twente
- **Sim_Z_{ref} :** Standardised impedance according to IEC61000-3-3

The maximum current peaks measured from the LISN supplied test scenarios were in the range of 57 A to 71 A, and for the simulated scenario, a peak of 75 A. To further analyze the characteristics, Tab. 4.2 presents a table of the differences in current behavior between the test configurations. As mentioned before, the peak currents are

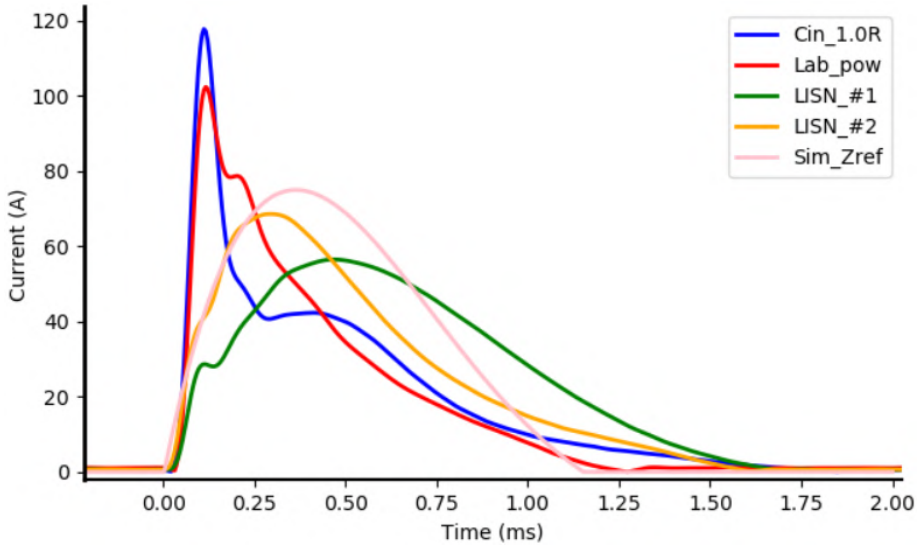


Figure 4.4: Inrush current waveforms for Cinergia GE, Laboratory power, LISN 1, LISN 2 as well as Z_{ref} impedance in accordance with [4] [5].

tabulated for each impedance configuration, as well as the fundamental components of the rate of change of the currents.

The lab power and GE unit are distinctly different from the LISN or simulation of standard reference impedance (Z_{ref}) configurations. Notably, the currents have a higher amplitude for the lab power and GE connection. The current rise time between 10 % and 90 % of the current peak shows a rising edge rise time in the 50 μs range for lab power and GE, while the other configurations have a rise time of hundreds of μs. Comparing the rate of change of current or slew rate, the difference is evident. The lab power and GE current rate of change is in the range of 1.6-1.8 A/μs, while the LISN and simulated Z_{ref} configurations have values in the range of 0.165-0.36 A/μs, significantly lower. It should be noted that this is merely a nominal power load of 64 W producing these large inrush currents. The delivered currents are able to activate protection systems or cause damage to switchgear or internal components of inverter systems. Inherently large currents will also affect the voltage waveform and (in "weak" grids) interfere with system operation or other devices [5].

Inrush currents behave differently to steady state cur-

Source	Peak I (A)	IΔ (A)	$T_{rise}(\mu s)$	Slew rate $(A/\mu s)$
GE	118	93	51	1.8
Lab	104	82	50	1.6
LISN_1	57	46	276	0.165
LISN_2	71	54	158	0.359
Sim_Z_{ref}	75	60	217	0.276

Table 4.2: Current characteristics per source impedance [5]

Source	Peak I (A)	Δ V (V)	Δ V (%)	Pulse Width (ms)
GE	118	85	28	1.8
Lab Power	104	123	38	1.3
LISN_1	57	282	87	1.7
LISN_2	71	260	80	1.6
Sim_Z_{ref}	75	238	76	1.2

Table 4.3: Current peak and voltage sag during inrush.

rents? Using the same setup with 32x2 W LED devices only for extra data points the internal resistance of the Cinergia GE is varied between 0 - 1 Ω. The same trends are visible as in Fig. 4.4, the GE and laboratory power outlet leading in rise time of the current waveform in addition to higher peaks. Interesting to note is the noise produced the switching frequency of the GE power electronics components.

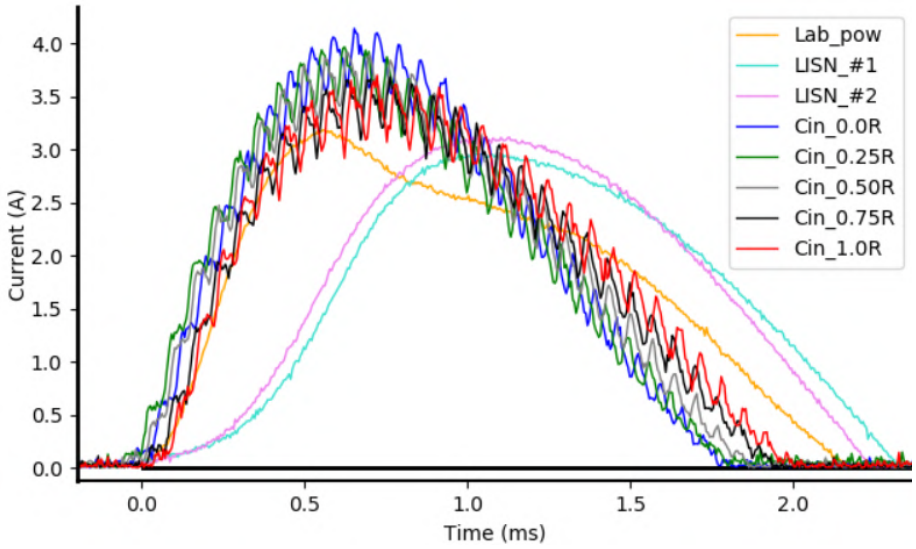


Figure 4.5: Steady state current wave-forms for Cinergia GE R_{int}[0 - 1 Ω], Laboratory power, LISN 1, LISN 2 .

4.3.3 *Grid Inertia*

The differences in inertia between large electrical grids and micro-grids can be complex and multifaceted, with implications for the stability, reliability, and performance of these power systems. Large electrical grids, characterized by extensive interconnections of power generation sources, often exhibit higher inertia due to the presence of large rotating generators powered by fossil fuels or nuclear energy, example shown in Fig. 4.6. These generators store significant amounts of energy in their inertial masses, resulting in slower response times to load changes. This higher inertia can provide stability to the system, as the stored energy in the rotating masses acts as a buffer against sudden changes in load demand or other operational factors [52].

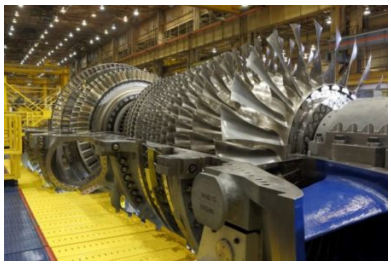


Figure 4.6: Chubu Electric Nishi-Nagoya power plant Block-1 in Japan — powered by GE's 7HA gas turbine [6]

The equation to calculate mechanical energy contained in a rotating mass is shown in Eq. 4.1. Where I is the rotational inertia of the generator rotor and mechanically connected components. This can be provided by the manufacturer or calculated [53]. The calculation may involve complexities as this involves geometrical attributes rotating around the reference axis , as shown in Fig. 4.7.

$$E_{rotor} = \frac{1}{2} I_{rotor} \omega^2 \tag{4.1}$$

On the other hand, micro-grids, which are localized power systems that often rely on renewable energy sources and energy storage, may have lower inertia due to the absence of large rotating generators. Instead, micro-grids may utilize converter-type inverters, which do not have significant inertial masses and therefore respond more quickly to load changes. Some energy is stored in electrical components which provides electrical (or virtual) inertia [52]. The energy stored in the output filter can be calculated as in Eq. 4.3 and Eq. 4.2. Some energy is fed through from the DC link or storage source through switching devices, but this is limited by the control signals. The lower inertia can result in faster response times to changes in load demand, which can be advantageous in some situations where quick adjustments are needed to match variable renewable energy generation or rapidly changing load demands.

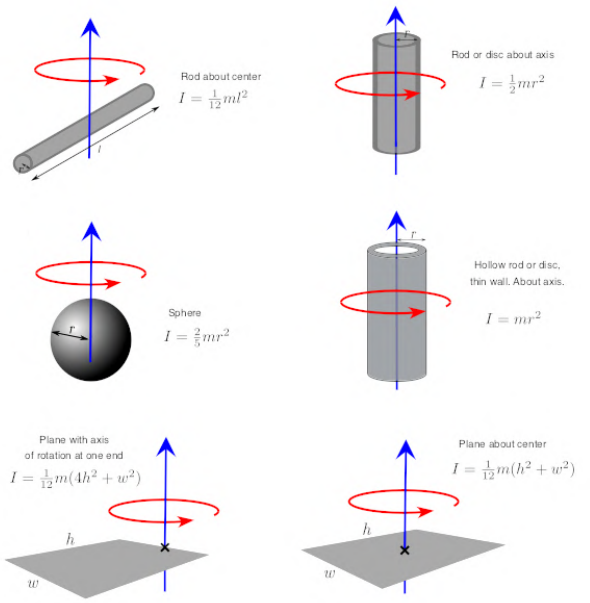


Figure 4.7: Equations for the rotational inertia of some simple shapes under rotation [7]

$$E_{capacitor} = \frac{1}{2}CV^2 \tag{4.2}$$

$$E_{inductor} = \frac{1}{2}LI^2 \tag{4.3}$$

However, the lower inertia of micro-grids can also pose challenges. For example, converter-type inverters may have limited energy buffer to supply impulse-type loads, such as those caused by transient or impulse currents from electrical and electronic equipment loads. This can result in excessive currents flowing through the active and passive components of the inverter with potential damage to the inverter-based power supplies [3]. Due to interaction with the inductor in the low-pass output filter, this leads to a sharp decline in voltage output, instability, or even supply deactivation. Moreover, the increasing installed capacity and technological changes to loads, such as SPECs, motor loads, and control units, have resulted in higher load fluctuations during activation, leading to EMC issues in micro-grids [26].

Additionally, failures and damage to both generator-based and inverter-based power supplies have been reported in micro-grids. For instance, in emergency backup

systems, such as those in hospitals, the intensity of transient currents from impulse loads may activate protection systems or move the generator out of its characterized frequency operating range, compromising its performance [27]. Inverter-based power supplies suffer from failures due to impulse-type currents, such as inrush or pulsed continuous currents from electrical and electronic equipment loads [16; 54]. More information is provided in Chapter 6, as well as examples and analysis of the difference between mechanical and electrical inertia.

4.4 *Measurement Techniques*

There are various monitoring systems available for transients and grid stability, such as is proposed by Rietveld et al. [55], Xaunwei et al. [56], and products like PQube [57]. Another widely used method is synchrophasors or PMU [58; 59; 60], which are commonly used for large grid stability, power flow analysis, and fault diagnosis. These methods are commonly used in large complex systems to identify the source of major inrush current and the propagation of voltage disturbances such as rapid voltage changes (RVC).



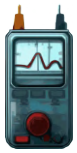
Figure 4.8: Siemens PMU [8]

However, when it comes to short duration intermittent or transient events, devices designed for long-duration measurements may not have the bandwidth to capture critical information, and in some cases, the emissions may even interfere with the measurement devices themselves, such as energy meters [27; 61]. Diagnosing problems related to conducted EMI can be challenging due to the intermittent nature of transient events, the difficulty in identifying their sources, and the need for specialized equipment. While devices like PMU, shown in Fig. 4.8, or synchrophasors [58; 60] are effective for large grid stability, power flow analysis, and fault diagnosis, smaller devices like the Iskra iMC770 [62] or the PowerQube [57; 63] with synchronization capabilities are also available. However, these devices are not optimized to provide the sample rates required for microsecond-level analysis or come at a high expensive.

It is important to note that fast Fourier transform (FFT) type EMI receivers, which are commonly used for EMI

measurements, may not be suitable for intermittent measurements due to the lack of time-domain information in FFT results. FFT type EMI receivers provide frequency-domain information, which may not fully capture the characteristics of transient events. Therefore, when dealing with intermittent transient events, it is crucial to carefully choose appropriate measurement systems that can capture the necessary time-domain information for accurate analysis and diagnosis of the conducted EMI problems. Further research and development in this area can help improve the availability and affordability of suitable measurement systems for conducting transient analysis in diverse power systems and grid environments.

5 Measurement System Development



5.1 State Of The Art

Time domain measurements offer crucial advantages for EMI assessments, particularly when evaluating lower frequency ranges from DC to 150 kHz and addressing intermittent, unpredictable events. Time domain measurements excel in capturing transient disturbances and anomalies, providing a comprehensive understanding of the dynamic electromagnetic environment. These measurements facilitate the identification of intermittent EMI sources, aiding in the detection of irregular patterns that might otherwise remain undetected in frequency domain analyses. By enabling the visualization of temporal variations, time domain measurements allow for a more nuanced analysis of the behavior of intermittent events, leading to the development of targeted mitigation strategies and the implementation of effective shielding solutions.



Figure 5.1: PowerQube power analyzer [9]

Devices for long duration measurement generally do not have the bandwidth to show the critical information, due to the massive datasets created for high sample rate measurement over time. Therefore, measurement systems are usually optimized for either long term low sample rate data acquisition (system monitoring), with or without averaging, or short duration high sample rates (such as oscilloscopes) Emissions may even result in interference with measurement devices, such as static energy meters [27; 61]. With electrical systems complexity, the diagnoses of problems of this nature can be very difficult. The often intermittent nature of transient EMI events and the requirement of specialized equipment can pose

challenges. Some devices exist which can partially satisfy the requirement, such as PMU or synchrophasors [58; 60] which are used for large grid stability or power flow analysis and fault diagnosis. Smaller products, such the Iskra model iMC770 [62] and PowerQube [57; 63] are effective, especially the PowerQube which has synchronisation capabilities in the latest models. Although, these devices may be very expensive and with limited sample frequencies which limits the applications. Alternatively, systems which utilise high-speed processing devices and global positioning system (GPS) or precision Time Protocol (PTP) can achieve synchronization between analogue signals in the range of 10-12.5 ns [64; 65].

That being said, considering the practical aspects of measurements performed in micro-grid environments, some other criteria need to be taken into consideration. Since a criteria of EMC is use in intended environment, and often EMI problems occur outside of lab environment in the field. In a shielded enclosure, GPS lock may not be available. The rugged unforgiving environments in which micro-grids operate (to be shown in Sec. 6), need to be considered when developing useful measurement tools.

5.2 *Design Parameter Wish List*

- Time synchronization: The measurement system must be able to synchronize data between multiple channels, devices, and measurement points to link events with a location and time.
- Mobility: The measurement devices must be mobile and lightweight, with a battery-powered option, to enable easy movement around the micro-grid and to minimize disruptions to the system.
- Time-domain data: The measurement system must be able to capture and store time domain data, which allows for analysis of transient events and waveform distortions that can affect the performance of the micro-grid.
- Data processing: The measurement system should be designed to provide easy-to-process data, preferably in a standardized format, to enable quick and efficient analysis of the results.

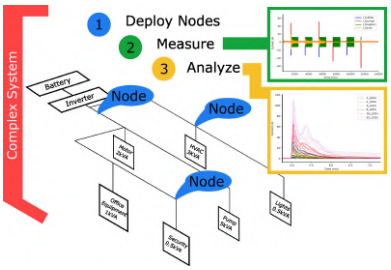


Figure 5.2: Node deployment in multi-zone micro-grid [10]

- Ruggedness: The measurement devices should be able to function in harsh and rugged environments, with a high degree of resistance to environmental factors such as humidity, temperature fluctuations, and dust.
- Scalability: The measurement system should be scalable, to enable the addition of new measurement points and devices as needed, without requiring significant changes to the system architecture.
- Real-time monitoring: The measurement system should provide real-time monitoring of the microgrid, to enable operators to respond quickly to any changes or anomalies in the system.
- Compatibility: The measurement system should be compatible with a wide range of communication protocols and standards, to enable seamless integration with existing systems and devices.

5·3 *Components and Interfaces*

The developed system utilizes any version of Picoscope universal serial bus (USB) oscilloscope compatible with PicoTech application programming interface (API). Picoscope devices are versatile USB oscilloscopes, the selected version is the 4824A with specifications shown in Tab. 5.1. The 8 channels of the Picoscope has BNC connectors for connection of any compatible external probe for measurement of current, voltage or other. One channel is used for trigger signal in order to synchronise devices when multiple nodes are deployed in one measurement setup.

Picoscope 4824A Specifications	unit or type
Device Interface	USB
Number of Channels	8
Bandwidth	20 MHz
Sample rate	80 MSps
Input voltage range	±10 mV to ±50 V

Table 5.1: Picoscope specifications

Additional components, listed in Tab. 5.2, include a Raspbery Pi 4 (RP4) linux based mini computer as a node-local

control device. The RP4 devices have Picotech API libraries installed which interface with Python software written to relay commands, setup configurations and extract data from the Picoscope. The code is available on Github [66].

No.	Component
1	Raspberry Pi model 4
2	Picoscope 4824A
3	Battery bank (5V, 20Ah)
4	Network jack
5	USB connector

Table 5.2: Measurement node components

The nodes can be used in isolation or interconnected between multiple units. In case of multiple deployment, an external network switch is required. An image of two measurement nodes is shown in Fig. 5.3. The units are placed in opposite orientation to show all internal and external components and interfaces. The network switch is connected to a node for convenience, and is powered by the on board battery bank. The only connection required to connect and operate the nodes is an ethernet cable from the master personal computer (PC) to the network switch. And inter-connecting ethernet cables between all the nodes and the switch.

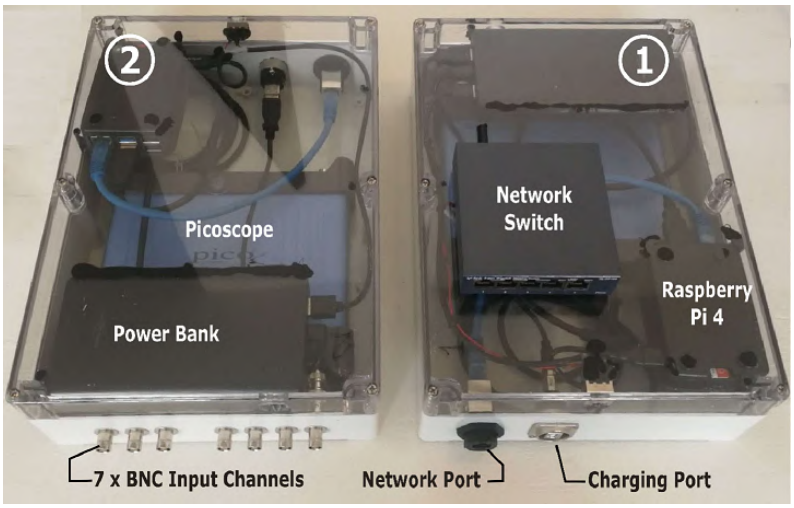


Figure 5.3: Measurement node components and interfaces [10]

The internal connections are shown in Fig. 5.4. The 5 V dc

power supply powered all internal devices and network switch. 7 BNC connectors are available on the enclosure for connection to measurement inputs.

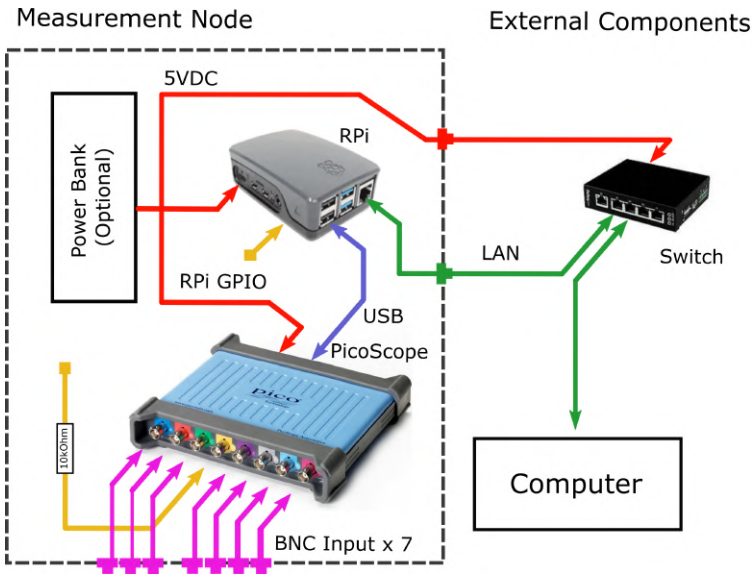


Figure 5.4: Measurement node components and interfaces [10]

The system has many possible configurations, including use of WiFi interface if preferred. Fig. 5.5 shows a connection diagram for coupling of multiple nodes during a standard measurement. The main node, whether used in isolated or in a cluster is connected to a network switch to simplify IP routing. Additional nodes used are connected to the same network switch. One network cable is the routed to a PC which orchestrates all the commands, data collection and pre-processing before data is stored or viewed in plots.

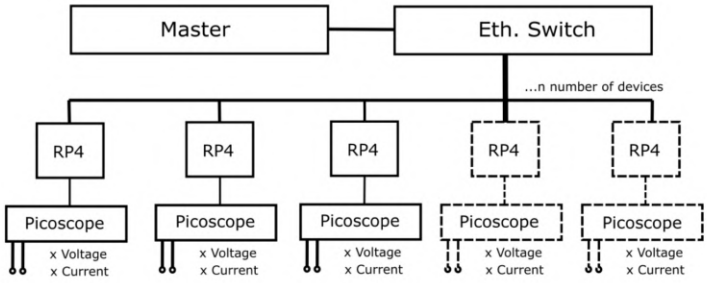


Figure 5.5: Measurement node components and interfaces [11]

5.4 *Software*

The basic control of software commands and data flow is shown in Fig. 5.6. The measurement requirements are configured by a command line with arguments. The commands are sent over local area network (LAN) using SSH to the RP4. This includes sample rate, channels enabled and duration etc. The RP4 initializes the Picoscope with correct channel and measurement configuration. The Picoscope begins streaming data and returns preset packets of data back to the RP4 for temporary storage for the duration of the measurement window.

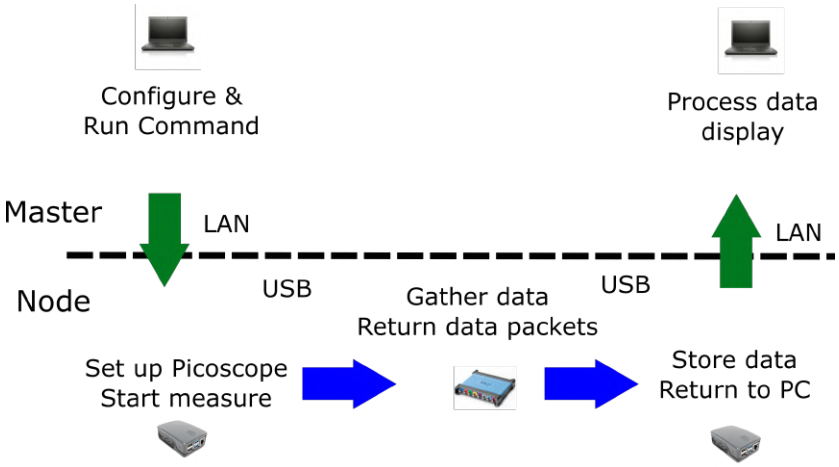


Figure 5.6: Software command paths

If multiple measurement nodes are in operation and data streaming has successfully started on all nodes the PC will send a command to the RP4 to trigger a pulse on a general purpose input output (GPIO) pin which is connected to channel 4 of the Picoscope. The Picoscope indexes the rising edge of the pulse and returns this index in the data structure. The RP4 returns feedback to the PC during the whole process to acknowledge successful operation. Once the preset measurement window is complete, the RP4 copies all measurement data back to the PC. The data structure of a single or multiple nodes are then processed and combined. The trigger index in the data is used to synchronise all the data between multiple nodes.

The raw data is stored on the PC in a single binary file.

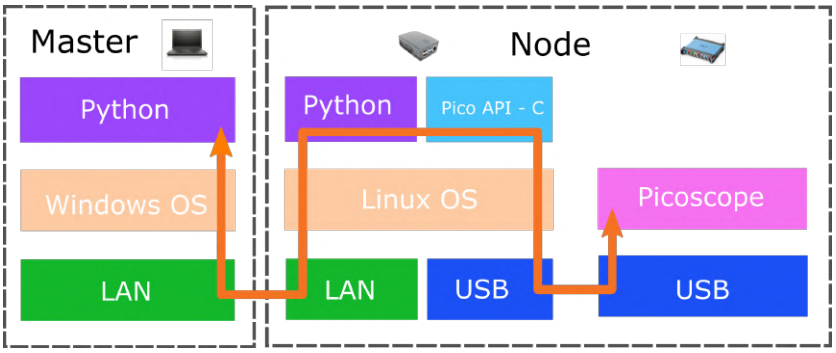


Figure 5.7: Basic software stack

This data can be processed or displayed. A representation of the software stack interaction is illustrated in Fig. 5.7. The scripting language used on the master (PC) is Python, which controls all the functionality, communications as well as data processing and visualization of the data plots. The main Python script communicates with the RP4 via LAN which in turn also runs Python scripts locally. The Python scripts running on the RP4 interfaces with the Picoscope API libraries which are coded in C. The Picoscope API libraries then issues commands and returns data to and from the Picoscope via USB. The measurement nodes have BNC connectors on the enclosure directly connected with the Picoscope allowing measurement of 7 channels, configurable to any compatible input (voltage probes, current clamps etc.). The API interaction and stream buffer of the Picoscope follows the process shown in a flow diagram in Fig. 5.8. Once measurement is initiated, a data buffer is filled , in accordance with the size of the buffer specified in the callback time argument, a specific data buffer is returned to the RP4. This continues until all data is transfered.

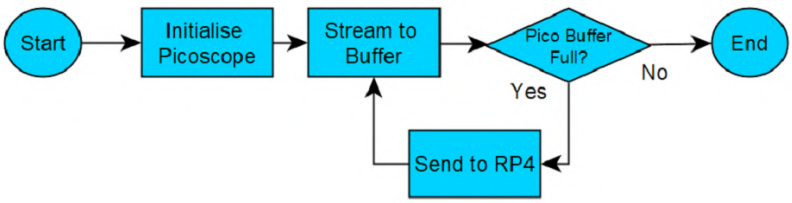


Figure 5.8: Basic software stack

5.4.1 *Software Event Timeline*

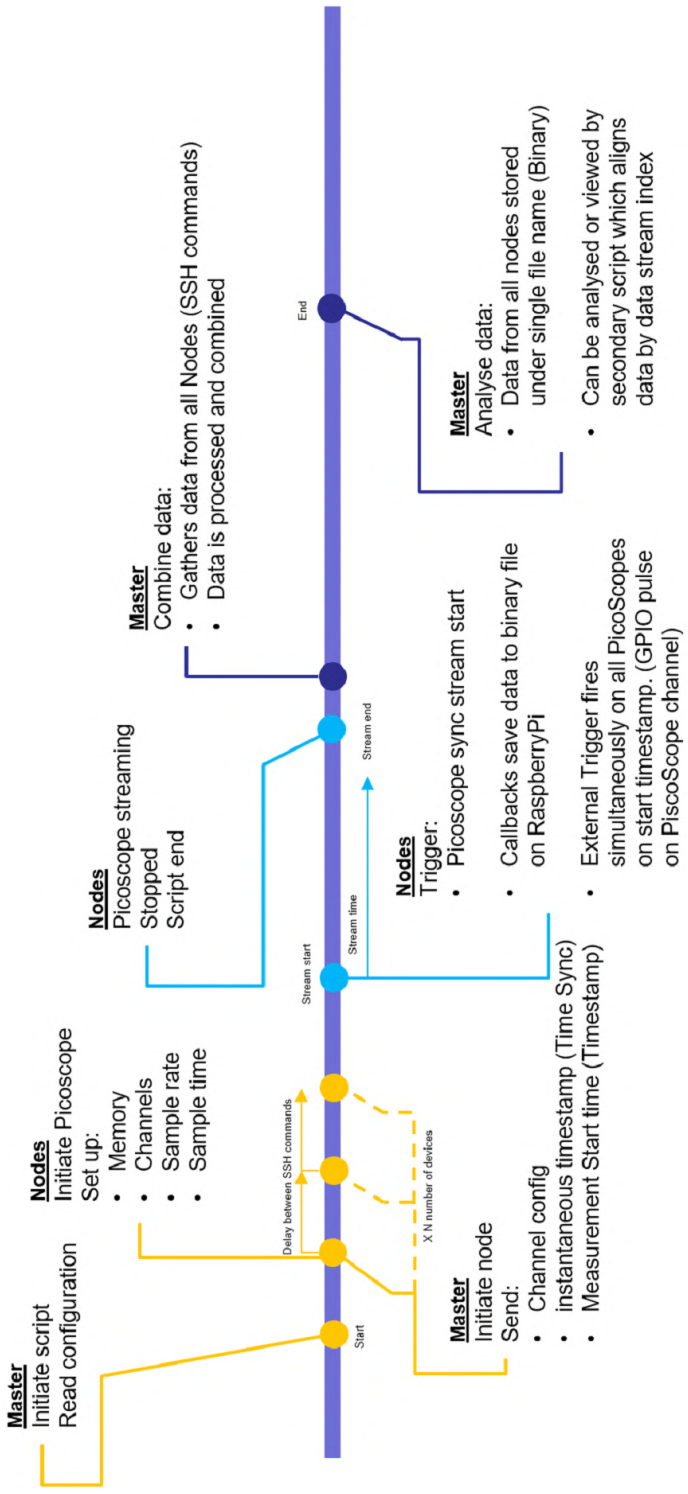


Figure 5.9: Time line of software events during measurement

5.5 *Applications*

5.5.1 *Case Study - Carré Building (UT) - Long Duration Monitoring*

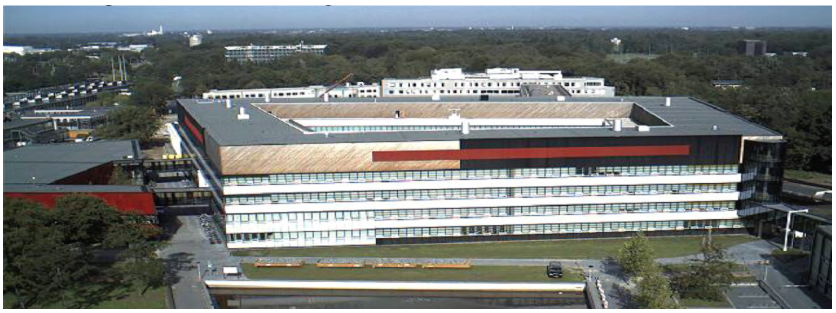


Figure 5.10: Carré Building - The University of Twente, Enschede, The Netherlands

Major power quality (PQ) issues were experienced in a modern building, Carré building of the University of Twente in Fig. 5.10, directly after commissioning. Initially, a capacity of 4.0 MVA of power capacity was installed and determined to be sufficient, according to the consultants. Once energized, as the building is equipped with LED luminaires and many other power electronic devices, significant distortion of the supply occurred. LEDs and other lighting were found to especially contribute to overall total harmonic distortion (THD) [54].

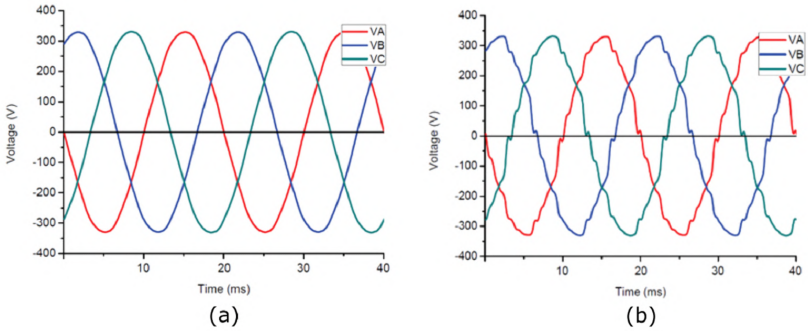


Figure 5.11: Phase voltage input waveforms a) No load, b) Energized

Fig. 5.11(a) shows the supply voltage of the point of connection (POC) with no load connected to the transformer feeding the building. Fig. 5.11(b) shows the same phase voltages with the load connected, which exhibit heavy

distortion. After investigation it was determined that the most cost effective action for correction was to increase capacity to the power supply to the building (more transformers). Important to note, the standard solution for addressing EMI concerns is over-dimensioning, whether in micro-grids or in this specific case. Another two transformers of 1.6 MVA each were thus installed. The total deliverable capacity was increased to 7.2 MVA for a real power load of less than 3 MW. This can be considered a large, and arguably unnecessary, expense to the developer. Especially since, during the design phase, technicians raised concerns that this was a possibility. More details of the negative effects of this distorted supply voltage are discussed in [54]. The measurements discussed in this case study are performed once more after the corrective measures were implemented. Measurement points are shown in a single line diagram in Fig. 5.12. Measurement point A is located at the POC with the building and the grid in the main control room (MCR) on the low voltage (LV) terminal of the transformer. The second point is measured in a lab located at a significant distance from the MCR, and is referred to as point B.

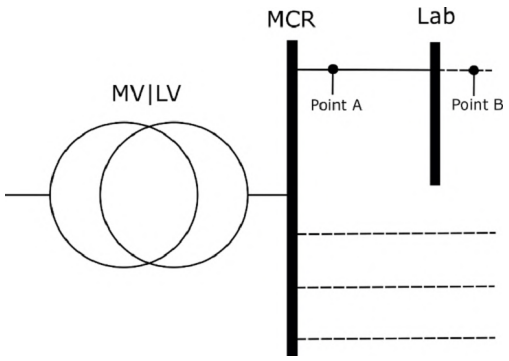


Figure 5.12: Single line diagram of Measurement points

The latest measurement results shown in Fig. 5.13(a) and (b) illustrate an improvement in the harmonic content of the voltage waveforms. The total harmonic distortion compared to the fundamental frequency is shown in Tab. 5.3

By visual inspection of the measurement and values of THD, the situation looks clearly improved, but is far from ideal. The improved PQ of phase C can be attributed to many factors. Most likely this is due to the changes im-

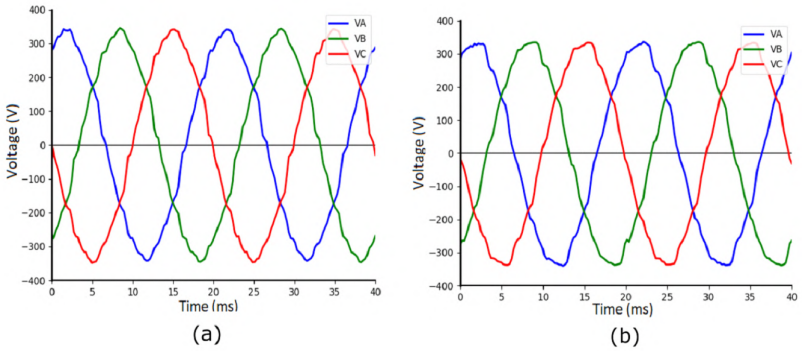


Figure 5.13: Phase voltage input waveforms a) No load, b) Energized

plemented with the corrective measures taken initially with the addition of extra supply capacity. What can be concluded from this data is that the interference sources causing the THD of the voltage wave-forms are closer to point A than point B. With more measurement points a clearer picture can be formed of the sources of disturbance in a large complex system.

Using the equation shown in Eq. 5.1, the measured voltage THD is calculated at approximately 4.4 % for the POC location and 3.3 % at the laboratory. A reduction from the previously measured 5.4 % and 6.7 % respectively [11]. The previous measurements from [54] were performed at different time intervals. Measuring EMI in different zones synchronously improves the quality of the data, which this measurement system provides.

$$THD_V = \sqrt{\sum_{n=2}^{40} \left(\frac{V_n}{V_1}\right)^2} \tag{5.1}$$

THD_V **(% Fundamental Freq.)**						
	Point A			Point B		
Phase Voltage	VA	VB	VC	VA	VB	VC
Previous [54]	5.4	5.4	5.7	-	-	6.7
Recent	4.5	4.4	4.4	3.4	3.3	3.3

Table 5.3: THD_V measurement results

5.5.2 *Short Duration High Band-Width (1 MSps) Measurements*

A test setup, shown in Fig. 5.14, evaluate different load profiles and their interaction with power sources by emulating a weak grid in the test environment. The setup is a simplified example of a use case, with a measurement point placed at the output of a four quadrant amplifier as the weak supply (weak in the sense of low-inertia in comparison to a electrical grid, stored energy « electrical grid). Two additional measurement points are used to measure sub-branches: one with a purely linear load and the other with a non-linear load consisting of an LED light bank and dimmer circuit connected to a water pump, both of which are known to cause high harmonic current distortion [54; 67; 68]. Current and voltage wave-forms are measured at these three nodes. Although the setup has only three measurement nodes, many more may be measured should it be required. The proposed system is scale-able and can be adapted to practical constraints via a local network configuration and processing capabilities of the measurement node (MN).

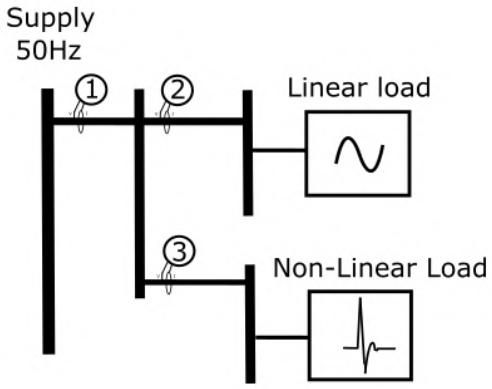


Figure 5.14: Single line diagram of test setup and measurement points 1, 2 and 3

The connected loads are listed in Table. 5.4. A Pacific Power Smart Source 140-TMX AC power supply, which is a 4-quadrant amplifier, acts as the weak or low inertia source. Differential voltage probes and current clamps measure voltage and current and are connected to the relevant nodes for each.

The linear load is expected to draw sinusoidal current

Branch Number	Description	Nominal Power (W)
(2)	Resistor bank	900
(3)	LED light bank (32x2W)	64
(3)	Water pump with dimmer	40

Table 5.4: Branch loads

with minimal distortion of the supply, whereas the non-linear load is expected to cause significant transients and stress on the supply. The test procedure involves starting the voltage supply (four quadrant amplifier) at zero load. This four quadrant amplifier emulates a renewable energy source or a local small generator (the emulation criteria being that the supply closely resembles the stored energy and control system of an inverter power supply). Once the measurement devices are activated and triggered, the load in branch (2) is activated. As shown in Fig. 5.15, the current drawn from this load is linear and has a constant sinusoidal amplitude. Subsequently, the non-linear load or branch (3) is energised, as indicated in Fig. 5.15.

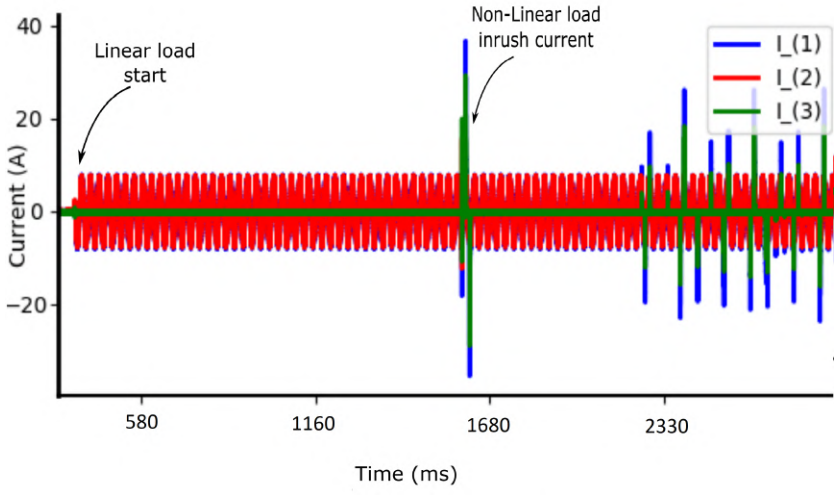


Figure 5.15: Current profile vs measurement window time for branch (1), (2) and (3) [12]

The energise time offset between branches clearly show the difference in response between sub-systems. This results in inrush current pulse, which exceeds 17 A caused by capacitors in the power electronics shown at approximately 1653 ms on the sampling time window scale in Fig. 5.16. Shortly after, near the peak of the next half cy-

cle at 1660 ms, another larger current pulse is present and repeated in the next negative half-cycle. Likely caused by current draw in the power electronics circuitry to enable operation of the systems. The peak of this transient current pulse is approximately 31 A. After a settling period whereby the LED bank luminaires activate and the pump begins operation more current pulses are apparent but not in regular amplitudes nor precisely periodic.

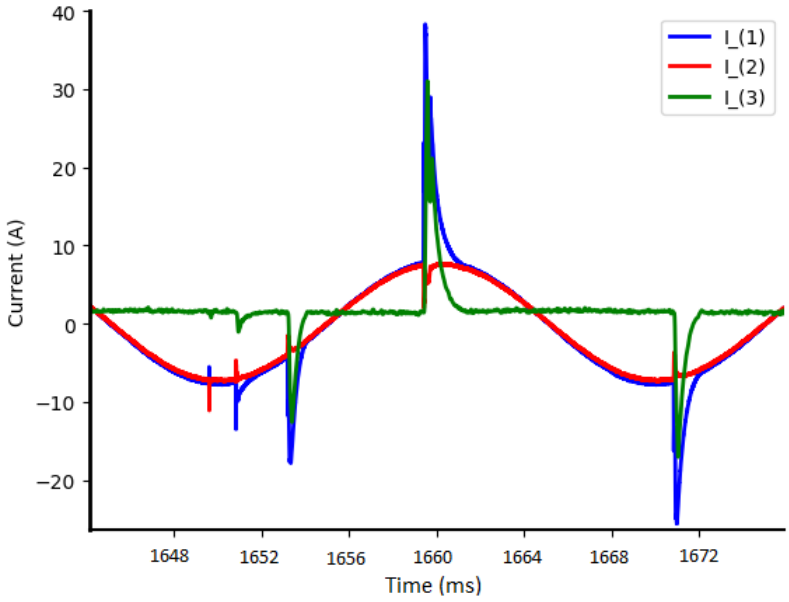


Figure 5.16: Current plot in branch (1), (2) and (3) [12]

Note: The peak currents listed in Table 5.5 are individual maxima and thus branches (2),(3) do not sum to branch (1). There are instances where current flows between branches (2) and (3) due to charge stored in components (capacitors) connected to respected branches.

The total current draw of the supply is the sum of the currents in the two branches. Hence, the current measured at branch (1) with inrush and secondary peak of (3) is 17.8 A and 38.3 A, respectively. Table 5.5 shows the measured RMS currents and power, peak current, and inrush current for each branch. Additionally, a ratio ($Ratio_I$) is shown for the peak measured current in each branch compared to the RMS current per branch.

No.	I_{RMS} (A)	I_{pk}(A)	P_{RMS} (W)	I_{inrush}(A)	$Ratio_I$
(1)	5.65	38.3	846	17.8	6.7
(2)	4.89	11.2	713	11.2	2.3
(3)	0.39	31	58	12.5	78

Table 5.5: Current and power values per branch [12]

In a more complex system with such transient current pulses present, it will further challenge the behavior and reliability of the power supply. Fig.5.17 shows the voltage deviation caused by the transient current pulse. Fig.5.18 provides a closer look at the effects on the supply and system voltage at various points. The plot displays the positive half-cycle deviation of the measurement time between 1658.4 ms and 1660.8 ms at measurement points (1), (2), and (3).

Note: The voltage plots have small time offsets due to non-perfect triggering between MNs. This is shown in some of the plots deliberately to display the data in a more legible format. The time offsets between channels are generally in the order of 0.5 - 1 ms due to network latency and time between trigger commands on the RP4. During post processing this is/can be compensated by aligning zero crossings of the a selected voltage waveform channel. This is done by calculating the phase shift, represented in radians, between two input signals, x_1 and x_2, utilizing their dot product and normalized vector lengths.

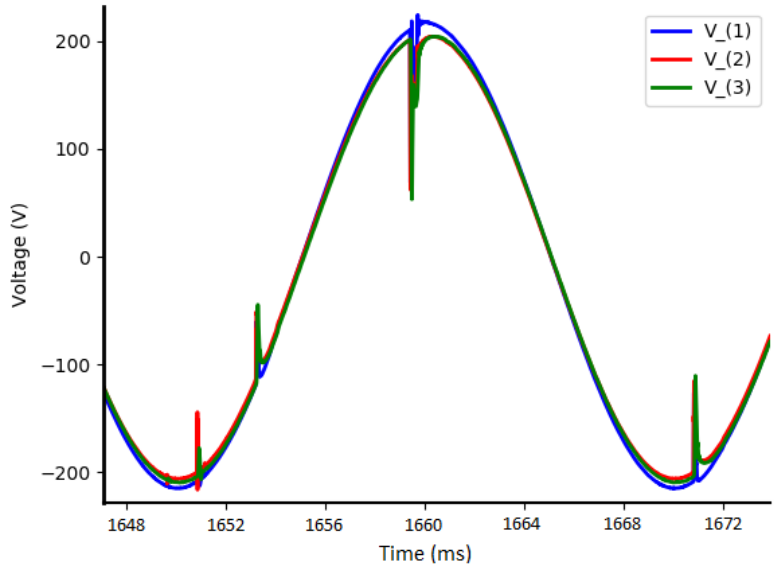


Figure 5.17: Inrush period voltage - branch (1), (2), (3) [12]

At a specific moment in the voltage cycle, the voltage dip seen by all branches reaches as low as 54 V from an initial voltage of 187 V and lasts for 0.344 ms in branch (3). This corresponds to a voltage draw down of over 71 %, which exceeds the 20 % limit of the STANAG 1008 standard. The voltage recovery in branch (2) takes longer due to the resistive and inductive components of the load. As a result of the disturbance in branch (3), the voltage and current of branch (2) are also affected. Although the operation of the load in branch (2) is unaffected in this case, more sensitive loads may experience a high probability of malfunction. Additionally, the significant stress placed on the supply could cause disruptions or overloads, leading to a total system shutdown. The significant voltage dip in

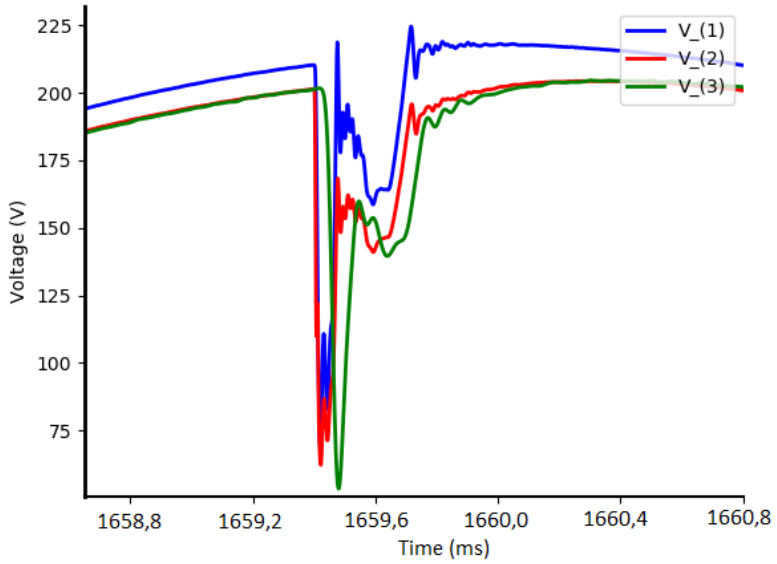


Figure 5.18: Sub-millisecond window inrush voltage deviation [12]

the system can be attributed to the low inertia or weak nature of the source [69; 70]. In the case of a strong grid, the deviations in voltage may be less severe[25].

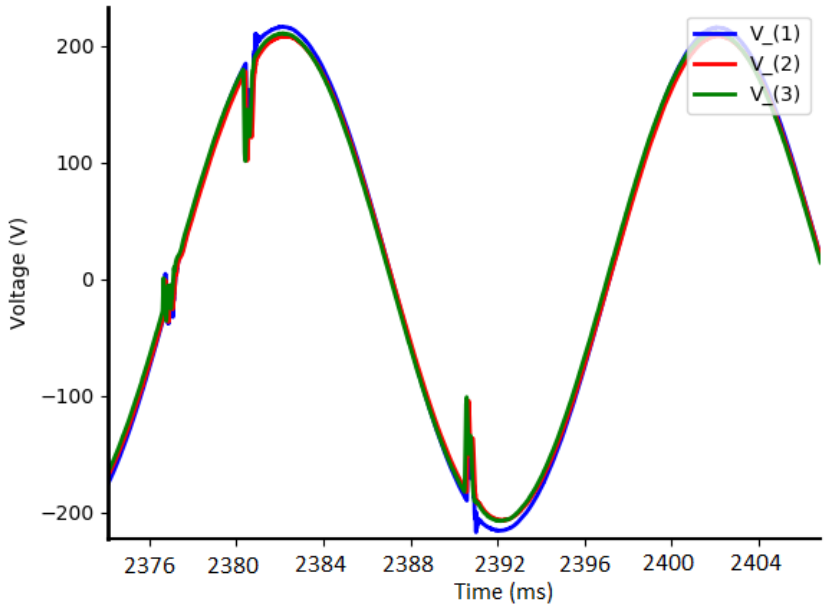


Figure 5.19: Voltages in branch (1), (2) and (3) with additional capacitor on (3) [12]

As a common engineering practice (old fashioned and applicable to linear systems), capacitors are often added to power systems for power factor correction [71], which may inadvertently result in an increase in transients. To

study the impact on system behavior, a 17.7 µF capacitor is introduced to branch (3), known (measured in this case) for generating large current activation pulses. The voltage wave-forms for all branches during the same inrush current phase, as shown in Fig.5.19, are depicted in Fig.5.18. During this specific phase, the inrush current coincides with the zero crossing in the voltage cycle, leading to a notable reduction in the inrush current magnitude, as illustrated in Fig. 5.20. Additionally, the voltage dip at all measurement points exhibits a less pronounced drawdown, particularly on the supply line, which records a voltage drop from 184V to 111V. However, when the branch is energized at the peak of the voltage waveform rather than at or near the zero crossing, the outcomes differ significantly. Despite the capacitor's role in mitigating the impact of periodic current transients on voltage deviation following the inrush current, the initial inrush current pulse experiences a significant increase.

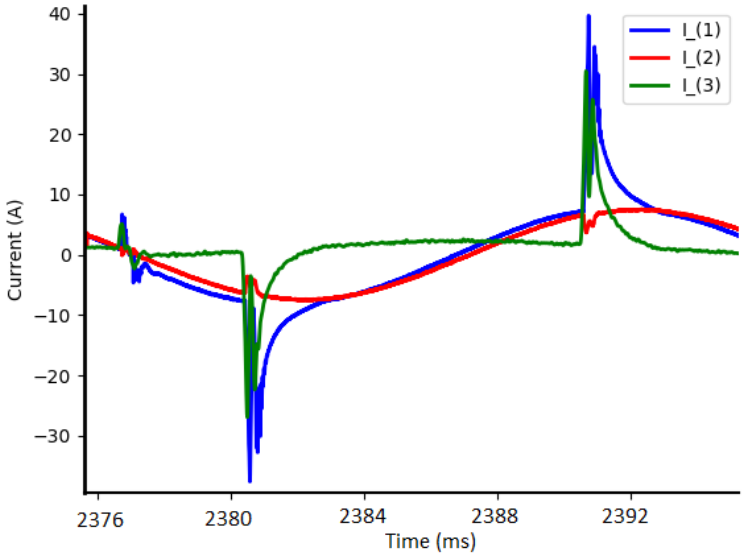


Figure 5.20: Current in branch (1), (2) and (3) [12]

The voltage at all measured nodes is depicted in Fig.5.21. At measurement point (1), the voltage dips to zero (and even to negative) for a very short time interval, most likely due to reactive currents flowing through the system during a phase of source instability. The inrush current is illustrated in Fig.5.22. The peak of the pulse is measured at 54.9A in branch (1), representing a 44 % increase over the previous highest peak current. Table 5.6 shows

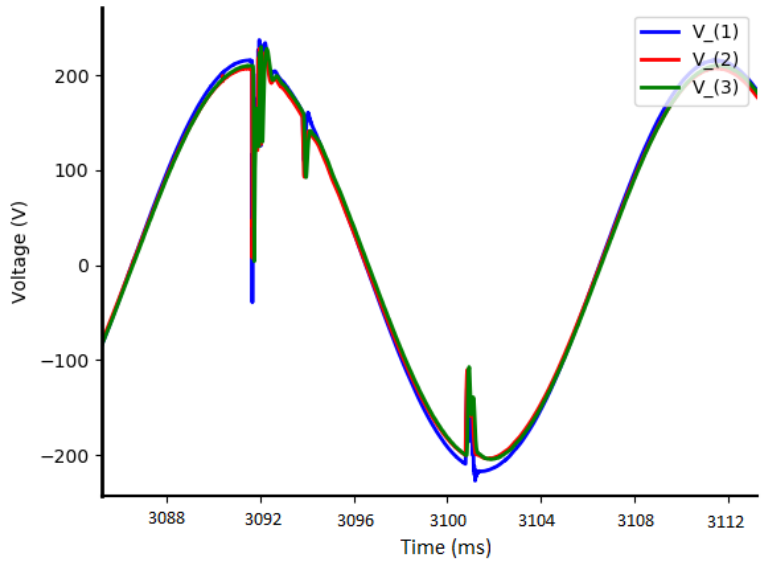


Figure 5.21: voltages in branch (1), (2) and (3) with inrush on peak of voltage cycle [12]

the RMS currents and power, peak current, and inrush current for the measured changes after the capacitor was added.

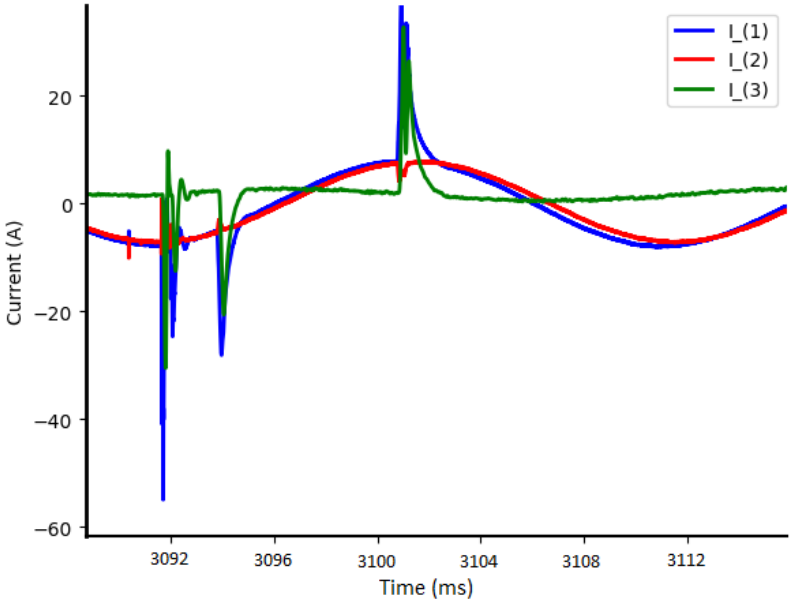


Figure 5.22: Sub-cycle voltage vs time in branch (1), (2) and (3) [12]

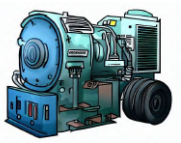
The peak currents observed in the operational current of the non-linear load show a slight increase, reaching 32.8 A. When implementing capacitors for power factor

correction (or other measures), it is essential to consider various criteria within the electrical environment. Especially when dealing with large inrush currents that can interfere with system operation [71]. The measurement validation setup employed has furnished crucial insights into the power conditions of both the supply and the subsystem.

No.	I_{RMS} (A)	I_{pk} (A)	P_{RMS} (W)	$I_{in-rush}$	$Ratio_I$
(1)	5.75	54.9	876	54.9	9.6
(2)	4.91	10.2	715	10.2	2.0
(3)	0.49	32.8	72	32.8	475.1

Table 5.6: Current and power values per branch with added capacitor on branch (3)

6 Analysis of Real-World Systems - Effects of Rapid Load Changes



6.1 Reasoning and History

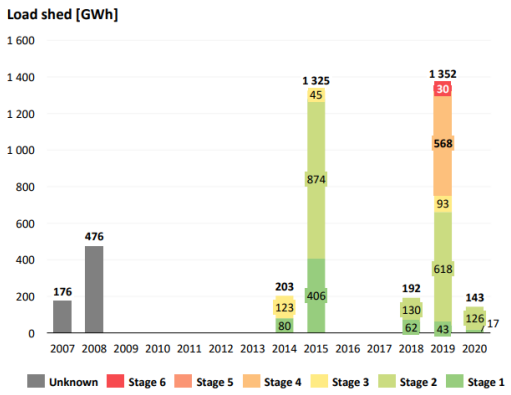
In recent years, there has been a global shift towards renewable power sources and distributed generation, driving the energy transition. Distributed energy and microgrids have emerged as alternatives to traditional transmission lines in cases where they are impractical or when islanded configurations provide protection from external influences. However, the system environment can become complex [72], primarily due to the unique characteristics of power sources in these setups. These power sources can be classified into two categories: inertial sources (such as rotating generators) and static sources (such as inverters). Inertial sources, like generators, possess higher inertia due to the stored energy in the rotating inertial mass of the alternator itself or the energy source. As a result, they exhibit a slower response to load changes. On the other hand, inverters have lower inertia but can respond quickly [26].

The implementation of both supply configurations has encountered increased EMC issues in recent years. This is partly due to the growing installed capacity and technological changes in loads, including SPECs, motor loads, and control units. These changes result in rapid load fluctuations during activation or connection/disconnection of loads are well as fast operational mode changes. Several concerns have emerged from these developments. For example, there have been instances of generator failures, particularly critical situations involving backup emergency generators in hospitals in The Netherlands [27]. Similar issues have been observed with inverter

power supplies, where failure and damage occur due to impulse-type currents like inrush or pulsed continuous currents from SPECs loads [16; 54].

In developed nations, grid instability is not a common occurrence. Energy security plays a vital role in maintaining a high quality of life, and any compromise in this security can have severe impacts [73]. These impacts include substantial financial losses to businesses [74; 34], disruption in communications, loss of life [35] or even increase in electricity theft [73]. A prominent example of the consequences of compromised energy security can be observed in South Africa, where load shedding has been implemented [75] (Please see side margin note for a description of load shedding). In response to these challenges, there has been a growing emphasis on the development of independent micro-grids, which represents a positive step forward [74; 76]. In Fig. 6.1, the histogram shows amounts of load shed in South Africa from the initial occurrence in 2007 until 2020. Multiple factors resulted in the failure of ESKOM, which is increasing in intensity. The highest levels of load shedding being recorded in 2023 thus far.

Note: Load shedding is a strategy implemented by the national electrical grid operator of South Africa (ESKOM) for load reduction due to lack of generating capacity. Stages of load shedding start at 1, and escalate to higher levels which reflect more load reduction. The implications are rolling blackouts which (at stage 1) reflect periods of no electricity for certain designated areas of 2 hours, daily or every second day and reaching many hours and longer blackout duration for higher stages. Currently, this is also the case in Ukraine, due to the reduced electrical system capacity of war damaged infrastructure.



Year	Duration of outages (hours)	Energy shed (GWh)	Est. econ. Impact (ZAR-bln)
2007	-	176	8-15
2008	-	476	21-42
....	....	....	....
2014	121	203	9-18
2015	852	1325	58-116
2016	-	-	-
2017	-	-	-
2018	127	192	8-17
2019	530	1352	59-118
2020 (YTD)	80	143	6-12

Figure 6.1: ESKOM load shedding histogram from 2007 - 2020 [13]

Note to Fig. 6.1: Load shedding assumed to have taken place for the full hours in which it was implemented. Practically, load shedding (and the Stage) may occassionally change/ end during a particular hour; Total GWh calculated assuming Stage 1 = 1 000 MW, Stage 2 = 2 000 MW, Stage 3 = 3 000 MW, Stage 4 = 4 000 MW, Stage 5 = 5 000 MW, Stage 6 = 6 000 MW; Cost to the economy of load shedding is estimated using COUE (cost of unserved energy) = 87.50 R/kWh Sources: Eskom Twitter account; Eskom se Push (mobile app); Nersa; CSIR analysis [13]

Indonesia is another country where the roll-out of micro-grids is on the rise [77; 78]. However, many instances of micro-grid failures have been reported [79; 41; 80], emphasizing the need to understand the underlying causes, including EMC issues. It is crucial to investigate real-world examples to gain insights into these challenges.

Consequently, measurements were performed on micro-grids, including an isolated inverter-based remote grid in Indonesia that powers multiple loads, including a large RO water filtration system (used for desalination of sea water).

The analysis of rapid load changes on isolated micro-grids is a topic of significant interest in the research community, especially considering the increasing utilization of micro-grids in off-grid and rural areas. However, the existing literature on this subject lacks practical real-world measurements, which hampers progress towards more reliable and efficient micro-grids. Recent studies have primarily relied on simulation-based methods to examine the effects of rapid load changes on generator and inverter-based systems [81; 82; 83]. While these studies provide useful insights, they do not fully capture the complexities of real-world scenarios, such as the impact of environmental factors on micro-grid performance in their specific environment or the aim of the studies focus on either system specifically or a combination of (hybrid) interfaced systems [84]. Therefore, there is a need for empirical studies that integrate laboratory experiments with real-world measurements to obtain a comprehensive understanding of the effects of rapid load changes on isolated micro-grids.

Generators inherently exhibit a superior response to millisecond impulse load currents compared to inverters, thanks to the stored energy in their rotational mass. However, there are instances where the intensity of the transient current can activate protection systems or push the generator beyond its designed frequency operating range. In scenarios with high-profile SPECs loads, such as during emergency backup system tests in hospitals [27], numerous uncharged capacitors are connected simultaneously. This leads to a large inrush current, which the generator control electronics perceive as a short circuit, triggering disconnection. On the other hand, inverters have limited energy buffers to supply impulse-type loads. Consequently, a high-rising edge and amplitude of current impulses result in excessive currents flowing through the active components and a sharp decline in voltage output. This instability can lead to supply deactivation and,

in some cases, equipment damage [3].

6.2 *Case studies*

6.2.1 *Indonesia - Rapid Load Changes on Static Power Supplies (Inductive Dominant Load)*



Figure 6.2: Solar PV panels at Baron TechnoPark, Indonesia (BRIN)

In this case study, a test facility for micro-grid research is investigated located at Baron Techno Park in Indonesia, this was in collaboration with the ANRGI project. The micro-grid system at this facility operates completely off the grid and in a remote area. One specific installation within the facility incorporates several SMA inverters along with a large battery bank, as depicted in Fig. 6.3. This installation is powered by solar panels and stores it in a substantial 96.2 kWh battery. While the micro-grid supplies electricity to various small local loads within the MCR, a major load in the system is the water desalination system which includes a raw water pump, which pumps sea water from the nearby ocean to the treatment facility.



Figure 6.3: SMA micro-grid Baron Technopark (BRIN) Indonesia

The water desalination system plays a crucial role in the facility, as it pumps raw seawater to a shed for RO purification. The specific RO system utilized is the MTech 300M, with a power rating of 3.7 kW. The nominal power ratings of the SMA inverters amount to 23.4 kW, with a maximum power rating of 36,2 kW for a duration of 3 s (short duration over-capacity power). According to the manufacturer's device specifications, each Sunny Island battery inverter has a peak current draw of 120 A

for 60 ms. The impact of the desalination pump's current draw on the output voltage of the system is illustrated in Fig. 6.4.

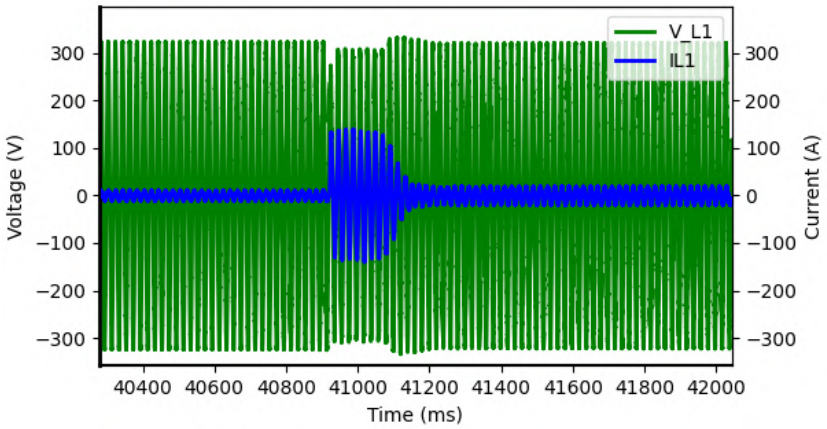


Figure 6.4: Micro-grid output voltage and current

Measurements were conducted using the system described in Section 5 [10], which utilized differential voltage probes and current clamps to measure the output voltage and current at the electrical distribution box of the micro-grid. Although the nominal power levels of this inverter-based micro-grid are lower than those of the generator discussed in Section 6.2.2, the power differential during load draw is comparable in both cases.

Fig. 6.5 illustrates the power profile, showing an initial power draw of 1.5 kVA that escalates to 22 kVA at the peak of the load power draw. The impact on the output voltage is shown in Fig. 6.5, where a sharp voltage dip is observed, dropping from 230 V to 184 V_{RMS}, accompanied by a frequency deviation from 50 Hz to 45 Hz. This corresponds to an RMS voltage drop of 20 %, surpassing the (10 %) limit specified by the EN 50160 standard. The EN 50160 standard is a European guideline that outlines the requirements for the quality of electrical power supplied by public distribution networks. It is important to note the difference in the time to recover (compared to a generator based system discussed in Section 6.2.2). The frequency and voltage stabilize back to their nominal levels within 200 ms. The primary consideration lies in the interplay between the system inertia and the time constant of the control system, which will be compared with the response of a generator-based system in Section 6.2.2.

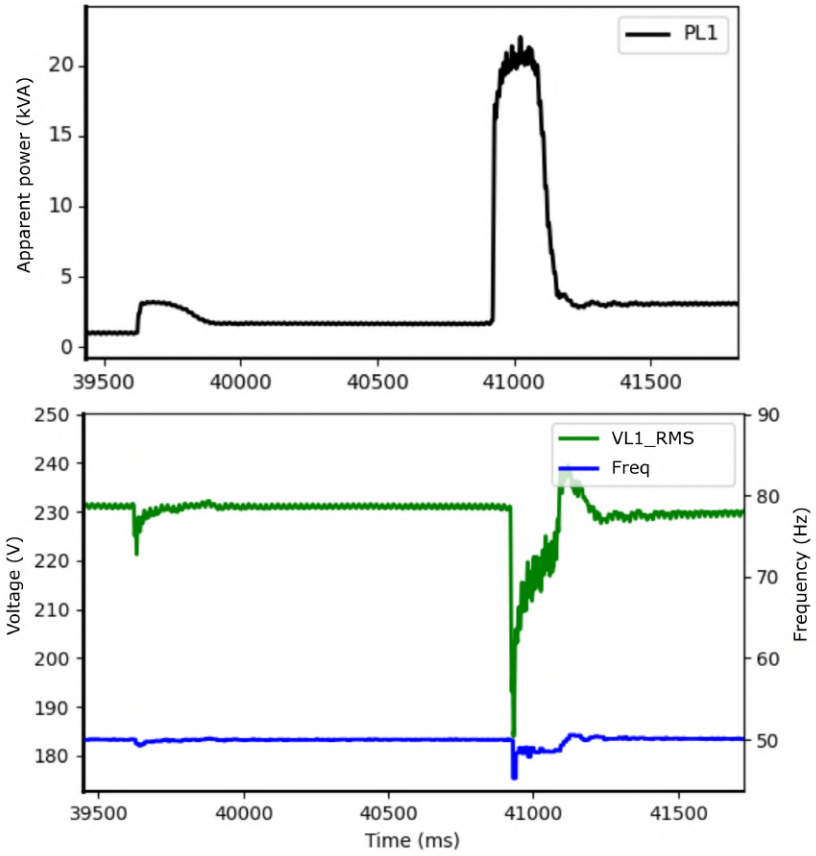


Figure 6.5: Apparent power (top) and RMS voltage with frequency (below)

6.2.2 *Netherlands - Rapid Load Changes on Diesel Generator (Inductive Dominant Load)*

Rapid load changes have significant impacts on generator-based power supplies. When there is a sudden increase in load, generators may struggle to meet the demand, resulting in voltage and frequency deviations. According to IEC 60034-1, diesel generators should operate within a range of ±2 % of the rated frequency (50 Hz or 60 Hz). If the frequency deviates beyond this range, protective devices, such as an ANSI 81 relay, are typically employed to shut down the generator and prevent equipment or system damage. The specific frequency range triggering the protective action may vary depending on the application and relevant standards, typically falling within ±0.2 to ±5 Hz of the nominal frequency.

Aternatively, sudden decreases in load can lead to overspeed conditions in generators, as the reduced load diminishes the driving force (see note in the margin). This can cause the generator to exceed its rated speed, potentially resulting in mechanical damage. Rapid load changes impose significant wear and tear on generator components, including the exciter and voltage regulator. Therefore, it is crucial for generators to possess sufficient capacity to handle rapid load changes, along with appropriate protective systems to mitigate their effects.

Note: To further examine the response of a generator, let us consider a step or impulse load change, if the energy drawn in a short time interval is a substantial amount in comparison to the energy contained in the rotational mass, this will alter the rotational velocity of the armature in the generator. This is if the control system is unable to respond fast enough to deliver the energy from the source. The initial energy is derived from the rotational momentum, leading to a decrease in rotor speed and output voltage. Subsequently, the control system attempts to compensate for the speed reduction by increasing the energy input, resulting in an increase in speed that often overshoots the set-point.

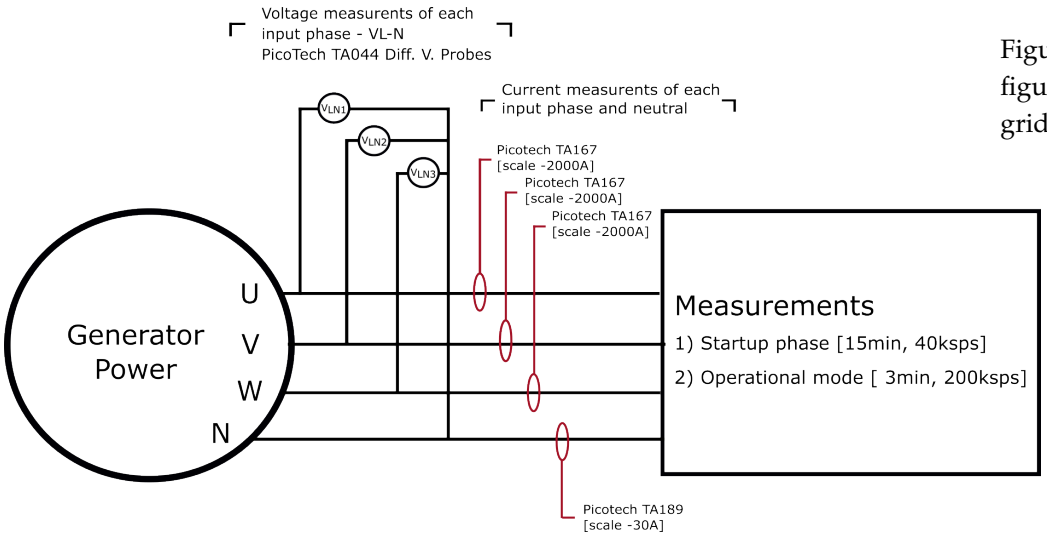


Figure 6.6: Measurement configuration on generator micro-grid

Measurements are performed on a 44 kVA diesel generator powering a mobile military vehicle platform [85]. In Fig. 6.6 the measurement configuration is shown. Three phase voltages and line currents are measured with differential voltage probes and current clamps, and the neutral current line.



Figure 6.7: Measurement setup on radar platform.

Note: Energy balance is extremely important. Over-dimensioning a generator results in a large generator, high mass, large volume, and low efficiency. This is factor is crucial, especially in mobile applications.

The platform, shown in Fig. 6.7, is equipped with sophisticated equipment, including computational processors, heating, ventilation, and air conditioning (HVAC) systems, variable frequency drives (VFDs), various communication devices, and high-power radar, all relying solely on the diesel generator for power. In Fig. 6.8, measurements are shown of the startup mode of the system, the duration is approximately 15 min. During the initial few

minutes some loads are activated, these are highlighted in the smaller time scale plots in Fig. 6.8.

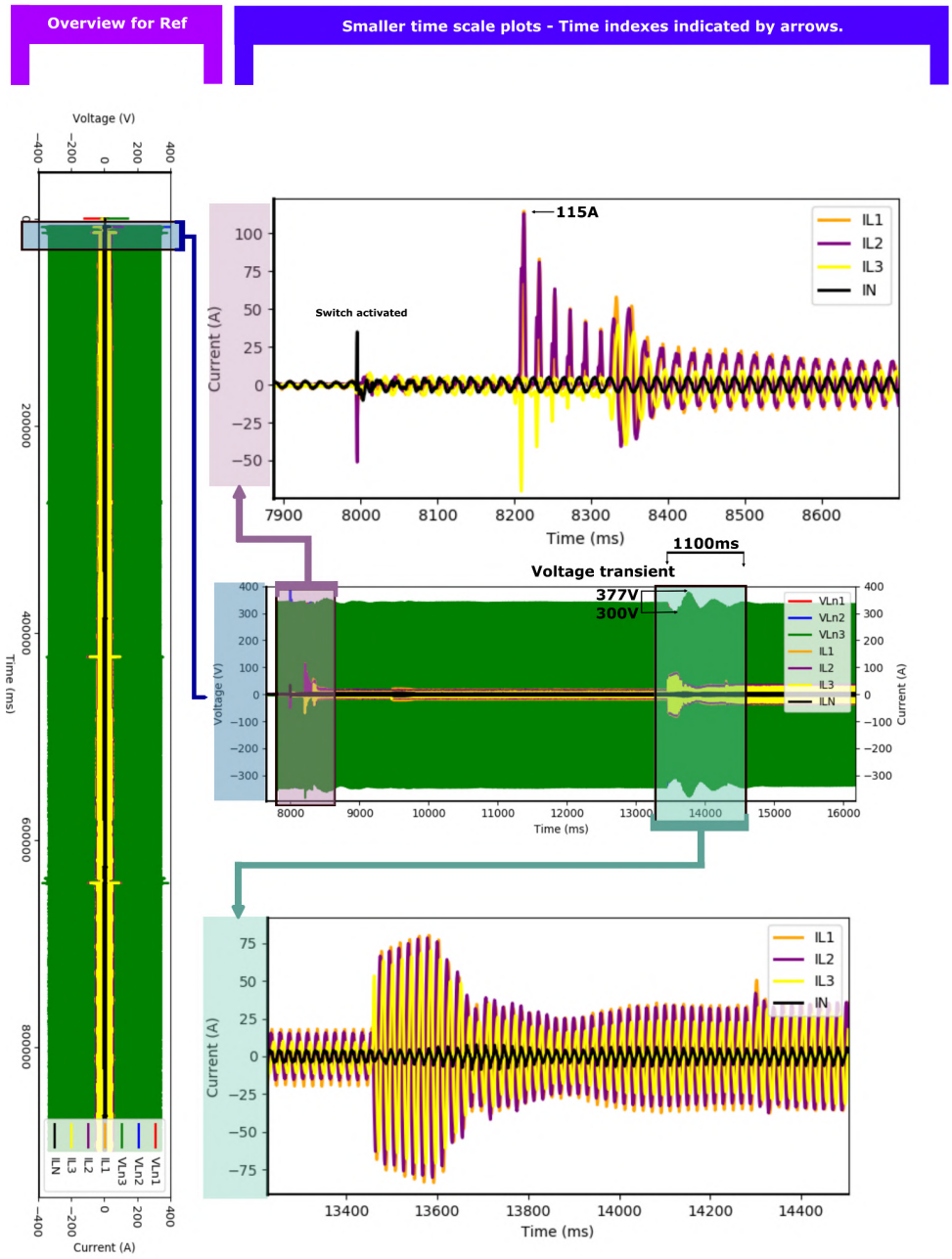


Figure 6.8: Startup sequence measurement of voltage and current, with multiple highlighted events

Note: In Fig. 6.8 a 15 min measurement window is shown which represents the startup sequence of the system. On the left, the entire measurement for current and voltage is shown. The plots on the right highlight points of interest, such as a startup of a system or and event which resulted in temporary instability or voltage and frequency fluctuation.

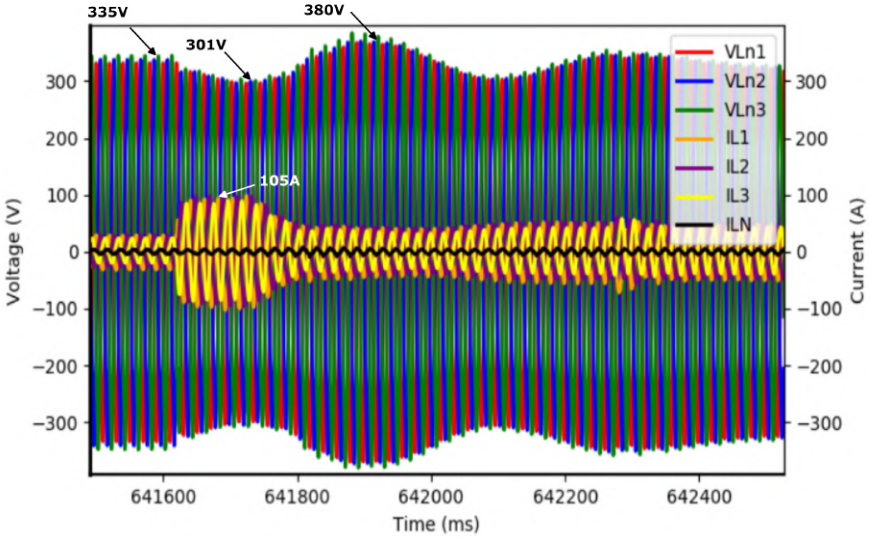


Figure 6.9: 3 Phase voltage and current during transient event on diesel generator

In Fig. 6.9 a transient is shown resulting from a large power pulse draw. The RMS voltage, apparent power and frequency is shown in Fig. 6.10. A voltage dip of 16 % (202 V) and a peak surge of 12 % (262 V) from the base value of 234 V.

These voltage fluctuations lie outside the specified range defined in EN 50160 [86]. Furthermore, the frequency of the power supply deviates from the baseline of 50 Hz, reaching a minimum of 46.5 Hz and a maximum of 52 Hz. A more detailed view of this event on a smaller time scale is presented in Figure 6.11, where the rising edge of the power draw corresponds to a rapid reduction in frequency from 50 Hz to 48 Hz within the rising edge of the load pulse with a duration (0 - 10 ms). This impulse initiates a response observed in Fig. 6.10 for approximately 2 s.

If another load change occurred before the system has recovered, the two transient events would superimpose, potentially increasing the intensity of the effect. In Fig. 6.12 the power conditions during operation mode of the system is shown. Fast toggle activation of certain sub-systems result in very volatile load power draw. This reflects on the stability of the system as shown in Fig. 6.13, with an unstable frequency and voltage during operational mode.

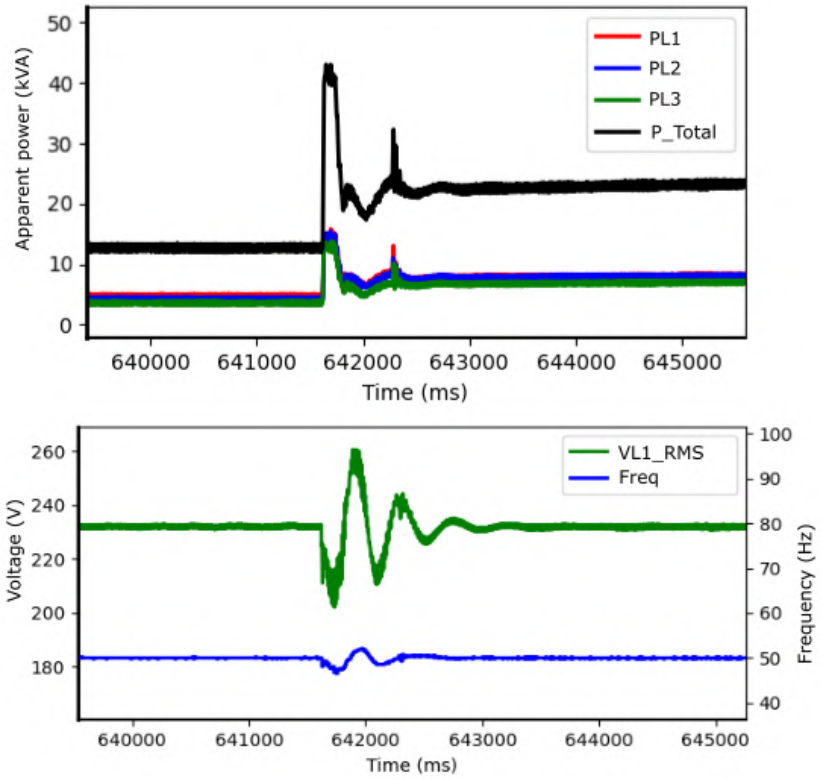


Figure 6.10: Apparent power (top) and RMS voltage and frequency (below)

The observed transient event demonstrates the impact of rapid load changes on the voltage and frequency stability of the micro-grid. The deviations from the specified voltage and frequency ranges highlight the need for further investigation and potential mitigation measures to ensure the stability and reliability of the micro-grid system. Section 6.2.4, will delve into the response characteristics of the generator-based system and compare them with the inverter-based micro-grid.

Note: As mentioned before, energy balance is extremely important. This can be very difficult to estimate in the design phase due to a variety of factors.

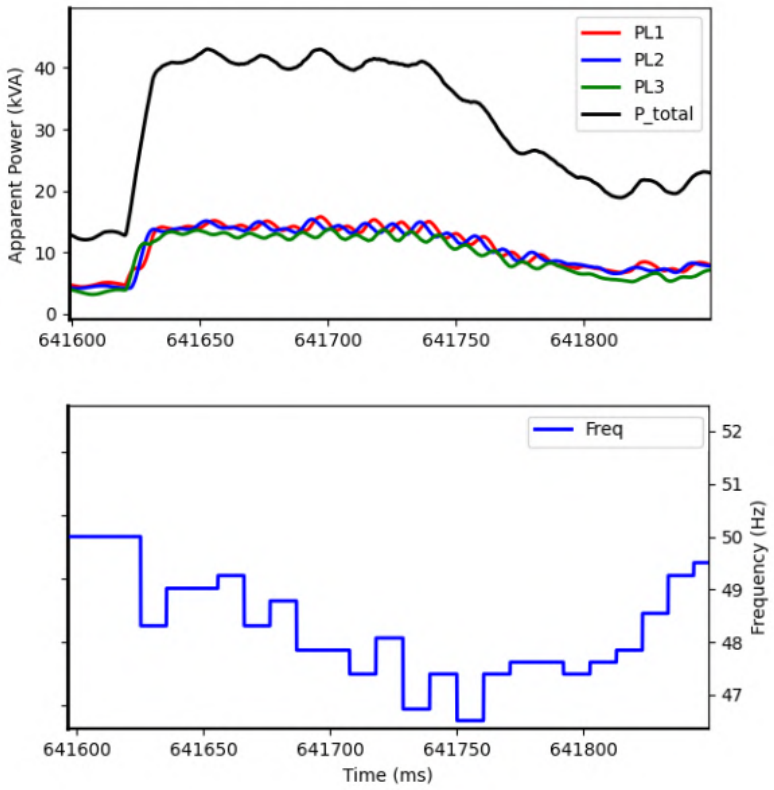


Figure 6.11: Apparent power (top) and frequency (below)

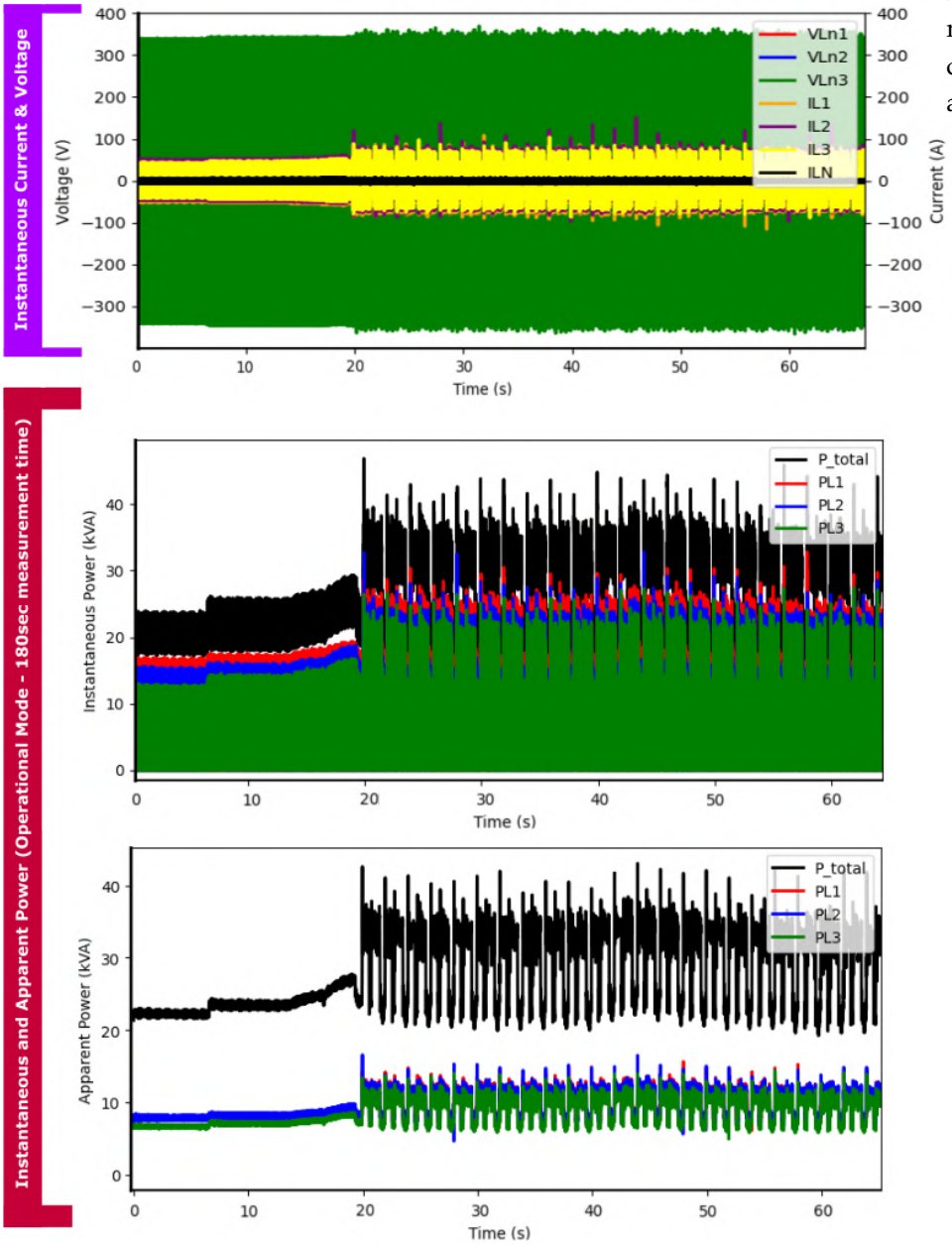


Figure 6.12: Operational mode measurement of voltage and current, with instantaneous and apparent power as labeled.

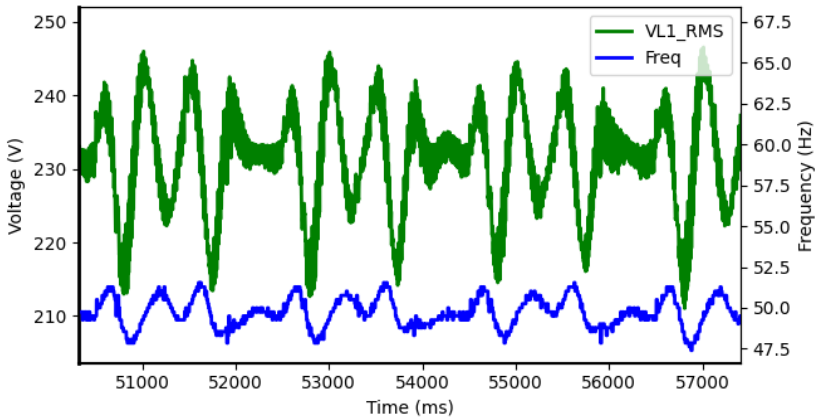


Figure 6.13: RMS voltage and frequency for operational mode

6.2.3 *Static Power Supplies and Capacitive Dominant Loads - Lab Micro-grid Emulation*

To ensure the performance and safety of power supplies, various standards, such as the IEC 61000 series, have been established. These standards define test procedures for evaluating the immunity of power supplies to electrical disturbances and transient currents. They also set limits on electromagnetic emissions to prevent interference with other electronic equipment in the micro-grid environment. Transient currents pose a risk of damaging power electronic components in converter-based power supplies, including diodes, transistors, and capacitors, leading to reduced reliability and shortened lifespan. Moreover, transients result in voltage and current spikes that can cause EMI issues when interacting with other local electronic equipment. Typically, inverters employ LC low-pass filters on the output to mitigate switching noise entering the local grid environment. Such as an (LC) low-pass filter can be viewed as a second-order system [87].



Figure 6.14: Cinergia GE, 15 kVA (5 kVA per phase) [14]

The standard reference impedance used for testing in-rush current in electrical devices, as specified in the IEC 61000-3-3 and IEC 61000-3-11 standards, is often regarded as too high [15]. This is because the last comprehensive study of worldwide grid impedance was conducted in the 1980s, and since then, power distribution characteristics have evolved significantly, particularly in micro-grids or isolated grids. Consequently, there is a need for updated standards that reflect these changes and provide

Note: This is also applicable to situations such as what is mentioned in Section. 5.5.1. Because of EMI issues (large THD), the installers were forced to increase the capacity of the transformers (and need for thicker and more expensive cables), which ultimately resulted in a vary low mains impedance.

more accurate reference impedances for inrush current testing. In Fig. 6.15 measurement results are shown of grid impedance from 4 different countries with 81 total locations (performed at typical public LV networks).

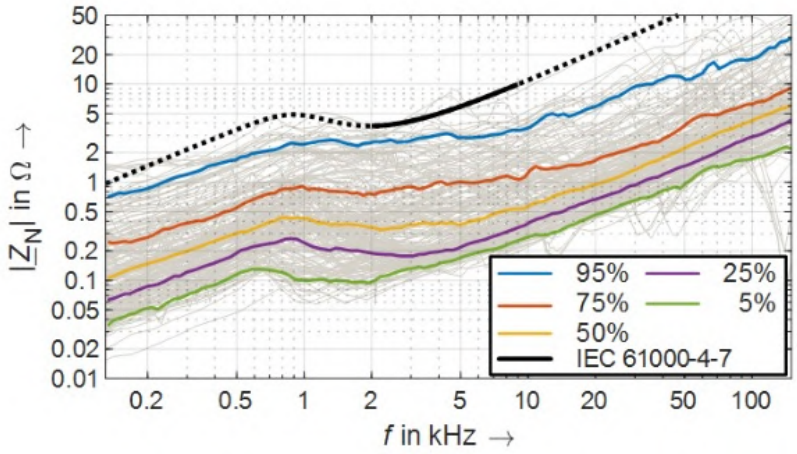


Figure 6.15: All measured loop impedances compared with IEC 61000-4-7 and percentile curves [15]

Fig. 6.16 illustrates a pulse current with a high rising edge and short duration. Considering the system as a Linear Time-Invariant (LTI) and the short pulse current drawn by the non-linear load as an impulse, the behavior follows a classic impulse response. The intensity and waveform shape of current pulses are significantly influenced by the system impedance, as mentioned in [5]. The tests were conducted using a Cinergia GE with an output impedance of less than $0.4 + j0.25\,\Omega$ at 50 Hz, according to the manufacturer, which aligns with the reference impedance (Z_{ref}) specified in IEC 61000-3-3.

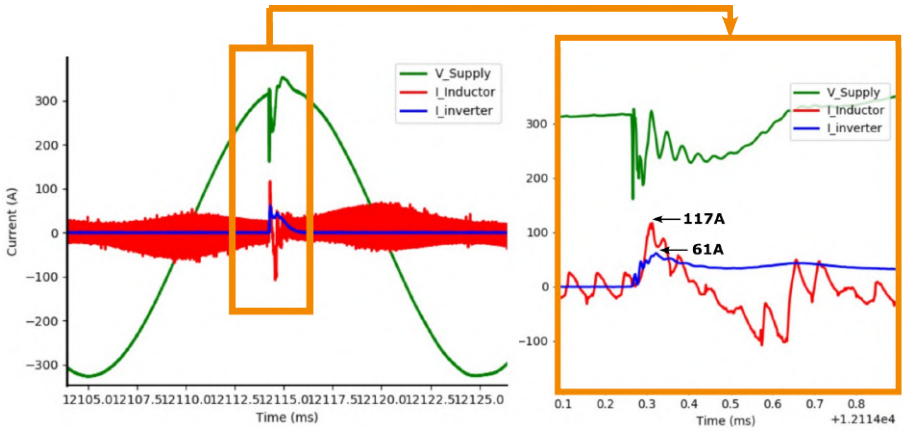


Figure 6.16: Impulse response of converter type power supply

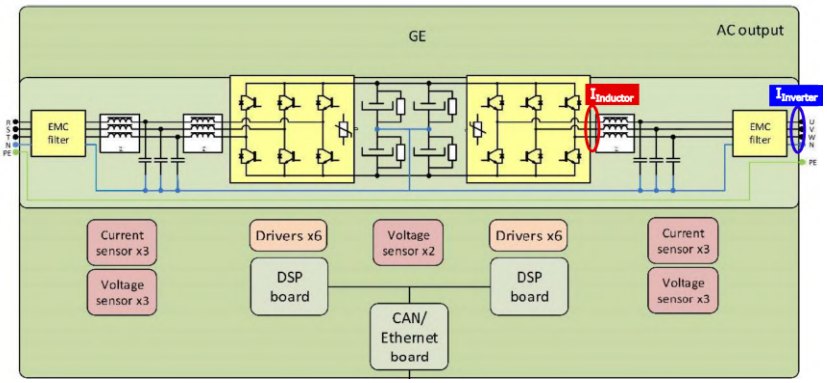


Figure 6.17: Measurement points for internal (inductor current) and output current. Internal components of Cinergia GE obtained from datasheet [14]

It is crucial to differentiate between currents supplied by the inverter and those experienced by the internal components, as well as to understand how these events interact with internal components. In Fig. 6.16, a non-linear load (only 125 W) is connected to a Cinergia GE (50 Hz, 240 V, 5 kVA per phase), and the current and voltage at the output are measured. The current is measured after the low-pass filter (i.e., the current flowing through the active switching components), as shown in Fig. 6.17. Notably, the peak inrush current drawn by the load reaches 61 A. However, due to the interaction between the current waveform (impulse) and the second-order system (low-pass filter), the impulse response of the system results in a decaying transient response with a peak of 117 A. This current directly flows through the active switching components of the inverter. The duration of the transient response is approximately 0.6 ms.

6.2.4 *Factors of Impulsive load Changes and Micro-grids*

Micro-grid power supplies, whether based on inverters or generators, have the ability to exceed their rated power capacity for short durations. Some manufacturers provide this information in data sheets or sales documents, specifying the acceptable overload conditions the device can sustain. For example, the SMA Sunny Island series SI8.0H can supply 120 A for 60 ms, which exceeds the continuous rated current of 20 A or 2.5 times the rated current for 3 s [88]. However, switching events occur during which the power demand far exceeds the rated or even overload capacity, albeit for milliseconds, when SPECs loads are

switched [16].

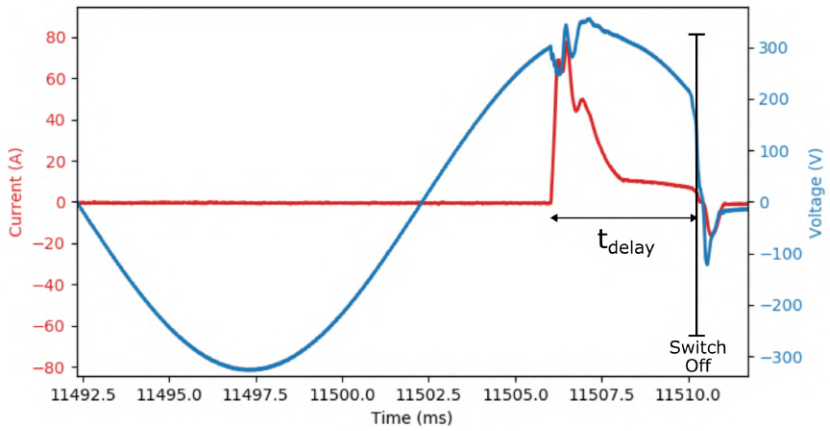


Figure 6.18: Inrush current resulting in inverter failure [16] (t_{delay} = time from switch event to inverter deactivation)

In the case of inductive loads, such events typically last longer, ranging from milliseconds to a few seconds, during motor startup, acceleration, or transformer magnetization. These events may not cause severe interference issues in robust power distribution networks, but they pose significant challenges in low-inertia micro-grids with limited capacity. Fig. 6.18 illustrates the failure of a 5 kVA inverter connected to 250 W of non-linear loads during an inrush event, despite the load being approximately 5 % of the inverter's rated capability [16].

6.2.5 *Active Switching Component SOA*

An Insulated Gate Bipolar Transistor (IGBT) is a commonly used in power electronic device in micro-grids due to its fast switching speed, high efficiency, and low conduction losses. MOSFET devices also belong to this category. The IGBT has a Safe Operating Area (SOA) that defines the operational limits in terms of voltage, current, and temperature. Similarly, MOSFET SOA is typically characterized by its maximum drain-source voltage V_{DS}, maximum drain current (I_D), and maximum junction temperature (T_j). It is crucial to operate the switching device within these limits to ensure its reliable and safe operation. Fig. 6.19 depicts the SOA of a PSMN1R5-30BLE MOSFET.

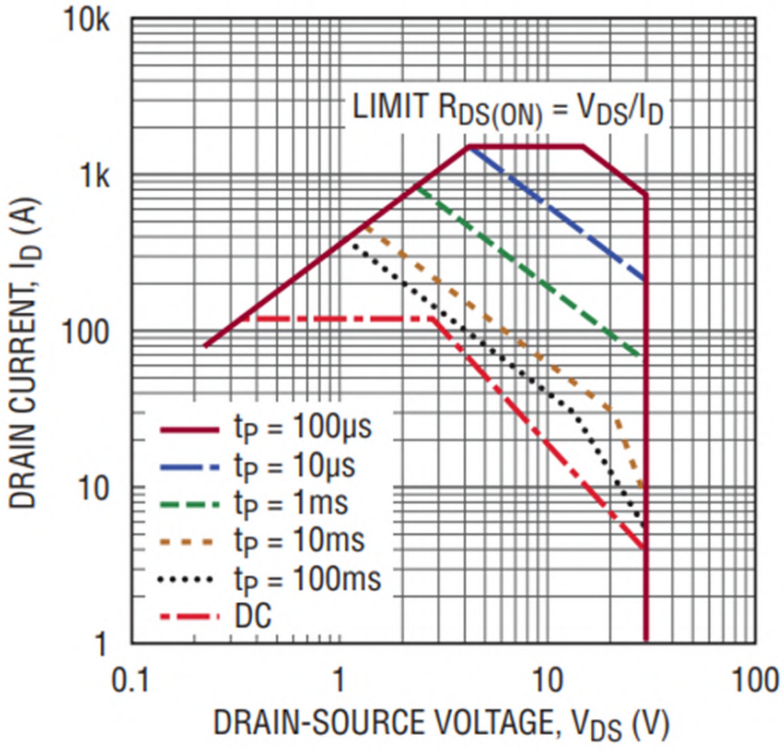


Figure 6.19: Safe operating area for a MOSFET [17]

The graph clearly illustrates that the pulse duration (t_p) and peak current (I_D) are bounded. In other words, a higher amplitude pulse can be conducted but for a shorter duration and under a certain drain-source voltage, compared to a lower amplitude pulse.

6.3 *Inrush Current*

6.3.1 *Emulated Grid Setup and Measurements*

Inrush currents occur when loads are initially powered on. The characteristics of this phenomenon depend on the components present in the load. SPECs type loads incorporate an input filter that includes a capacitor, which charges up during startup and results in a significant current draw [3]. In contrast, motor loads or transformers, being inductive type loads, can draw large currents, often exceeding their rated value. However, the equation that characterizes the current response for these loads is related to the change in current rather than the change in voltage across the load. Resistive type loads generally do not exhibit inrush currents. However, in the case of a resistive heater, where the heating element consists of a positive temperature coefficient material and the resistance changes from low to high during the heating phase, a low-intensity inrush current effect may occur in terms of crest factor and time. In generators or inverter-based microgrids, inrush currents can lead to EMI when the load is activated. This may trigger the over-current protection of the power supply or cause damage to internal components [53; 89], as illustrated in Fig. 6.20.

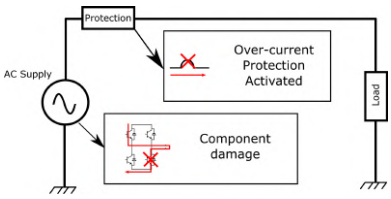


Figure 6.20: Potential failure points diagram [5]

Inrush currents can reach levels that trigger protection systems, damage power supply components [90], and cause significant voltage sags, swells, or flickering effects that compromise EMC [91]. It has been demonstrated in [16] that connecting a small SPECs load component, which is only a fraction of an inverter's nominal capacity, can disable the system, requiring a restart and severely impacting system reliability.

EMI caused by the superposition of numerous small inrush currents can result in large net peak currents with high crest factors, which may equal or exceed the effects of single higher-power devices. Unfortunately, existing standards (at least for typical consumer or military applications) do not sufficiently limit this effect. For instance, [44] limits the maximum peak inrush current to 10 times the maximum rated current of a device, excluding devices

rated under 1 kW. According to [4], the maximum inrush current over a 10 ms half-cycle between zero crossings should not exceed 20 A RMS. However, inrush currents are typically high-intensity pulses of short duration, so the RMS current value during an inrush event may appear low, especially for a single low-power device. In essence, it is possible to comply with all standards while connecting a specific load profile to a micro-grid installation, only to render the system unusable. Although low-power devices may not seem likely to cause interference in isolation, a large number of low-power devices switched on simultaneously can result in severe currents.

Current strategies to mitigate the effects of transient switching events include incorporating NTC devices [45], using pre-charge capacitors or resistors [46], implementing low-pass filters or control strategies for electrical drives [47], and employing staged switching of portions of large inrush loads after the initial activation of the power source, if high currents are expected [27]. In micro-grid design, the common practice is to calculate the rated load, multiply it by a safety factor to overstate the supply, and add an additional buffer to account for the estimated power factor. However, this approach often leads to significant over-sizing of the power source [11] relative to the actual power requirements, and it may still not be sufficient to maintain reliability.



Figure 6.21: Cinergia GE with LED light bank

An inrush emulation system composed of a bank of 2 W LED lights are connected and switched by a software-controlled relay, with a diagram shown in Fig. 6.22. The relay switches the LED light bank on and off in specified intervals, with each activation lasting for 2 s. This switching pattern initiates an inrush current event, which is measured using a system developed in [11]. The system logs the data to a personal computer using differential voltage probes and current clamps. The power supply used is a Cinergia GE with a capacity of 5 kVA. It operates in single-phase independent mode, as depicted in Fig. 6.21. After the LED light bank is activated, the relay deactivates it for 2 s to allow the discharge of filter components in the devices, along with a 1 ms offset to enable the inrush event to trigger at different phase firing angles on the voltage waveform. This test procedure is repeated

for 10 cycles, with different numbers of LED devices connected, ranging from 1 to a maximum of 32.

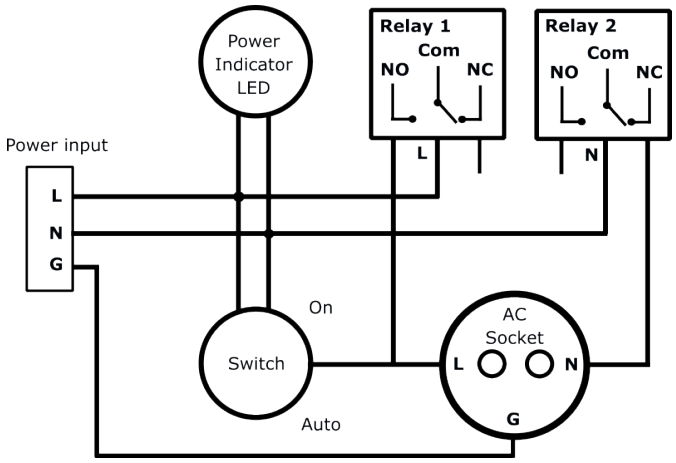


Figure 6.22: Software controlled relay diagram (1 channel out of 8 total)

Fig. 6.23 depicts large inrush currents for various quantities of activated 2 W LED clusters. The crest factor of these specific measurements varies between 500 and 393 for a single LED and 32 LEDs, respectively. It's worth noting that the current pulse time-frame falls within the range of 1-2 ms, with a rising edge to peak duration of 50-100 μs. The characteristics of the waveform are highly dependent on the system impedance [5].

LEDs Units	Power (W)	I_{rated} (mA)	Current Measured (Max)		
			$I_{Inrush_{peak}}$ (A)	RMS (t=10 ms)	CF
1	2	9	4.5	0.47	500
2	4	17	7.3	0.95	429
4	8	35	20	2.1	571
8	16	70	33	4.1	471
16	32	140	65	8.2	464
32	64	280	110	14.7	393

Table 6.1: Number of LED vs inrush current characteristics

6.3.2 *Simulation of Setup*

To simulate the effect of a common SPECs device (in this case, an LED light) and investigate the impact of stacking multiple devices, a Simulink model was developed. The model allows for the inclusion of any number of computationally feasible LED devices within a model block.

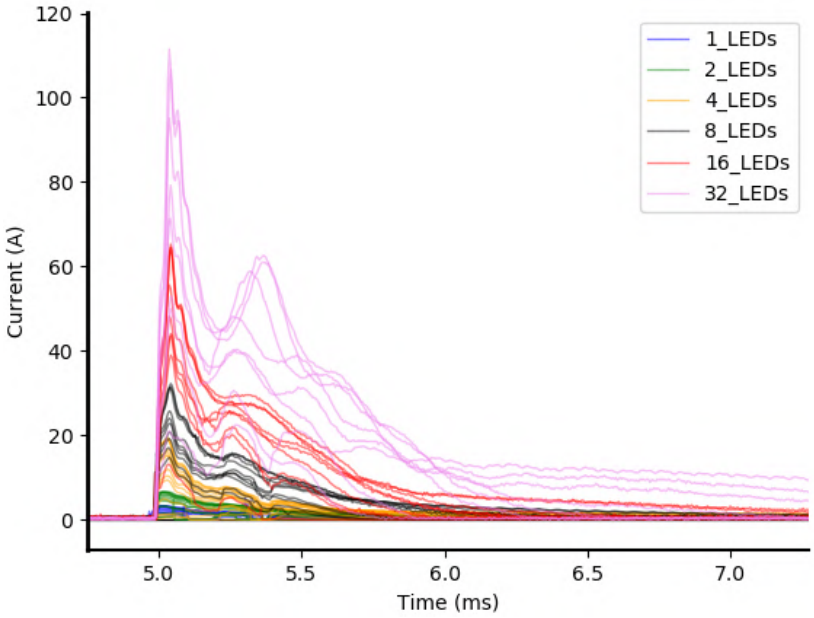


Figure 6.23: Inrush currents for clusters of 2W LED lamps [3]

Fig. 6.24 illustrates a schematic of a single LED unit in the model, specifying component values ($R1 = 10\,\Omega$, $L1 = 470\,\mu H$ with $R_{\text{internal}} = 65\,\Omega$, $C1 = 4.7\,\mu F$, $C2 = 330\,\mu F$). This LED unit model is integrated into a larger simulation of an inverter, which shares similar characteristics with the Cinergia inverter used to supply the devices under test. Fig. 6.25 presents an example of the high-level system simulation, including a switch that controls the timing of the LED cluster activation. The firing angle of the voltage can be adjusted in the simulation to replicate real-world switching conditions and produce comparable results. Values in simulation ($L2$ $R_{int} = 0.05\,\Omega$, $L2 = 50\,nH$, $L1$ $R_{int} = 0.05\,\Omega$, $L1 = 0.5\,H$ and $C1 = 68\,\mu H$)

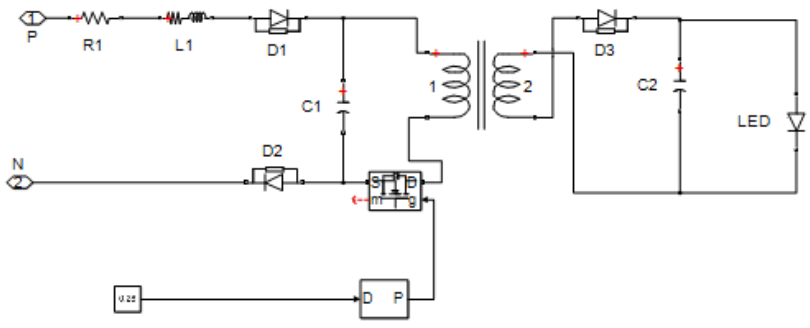


Figure 6.24: 2W LED simulation model - Simulink

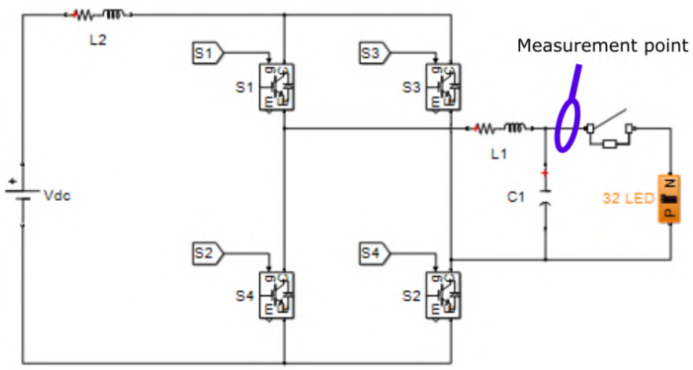


Figure 6.25: Simulink model of an H-bridge inverter

The simulation results align well with the measured data. Fig. 6.26 shows the currents for different combinations of activated LEDs, with solid lines representing the measurement data (maximum peak measured inrush) and dashed lines representing the simulation data. The colors correspond to the same number of activated LEDs. The comparison between measured and simulated current characteristics is summarized in Tab.6.2. The maximum peak current difference between the measured and simulated values falls within 20 %. The rising edge of the inrush event is also compared between the measurement and simulation, with a maximum difference of 22 % observed in this sample case. The rising edge current slope, ranging from 10 % to 90 %, increases with larger numbers of devices, varying from 1.01 A/μS for the 4-LED simulation case to 4.15 A/μS for 128 devices.

LEDs	I_{PkSim}	I_{PkMeas}	I_{Sim} $(\frac{dI}{dt})$	I_{Meas} $(\frac{dI}{dt})$	Diff.
Units	(A)	(A)	(A/μS)	(A/μS)	(%)
4	17	20	1.01	0.888	14
8	33	33	1.35	1.49	10
16	60	65	2.04	1.67	22
32	105	110	2.85	2.37	20
64	165	-	3.44	-	-
128	236	-	4.15	-	-

Table 6.2: Current deviation and time derivatives

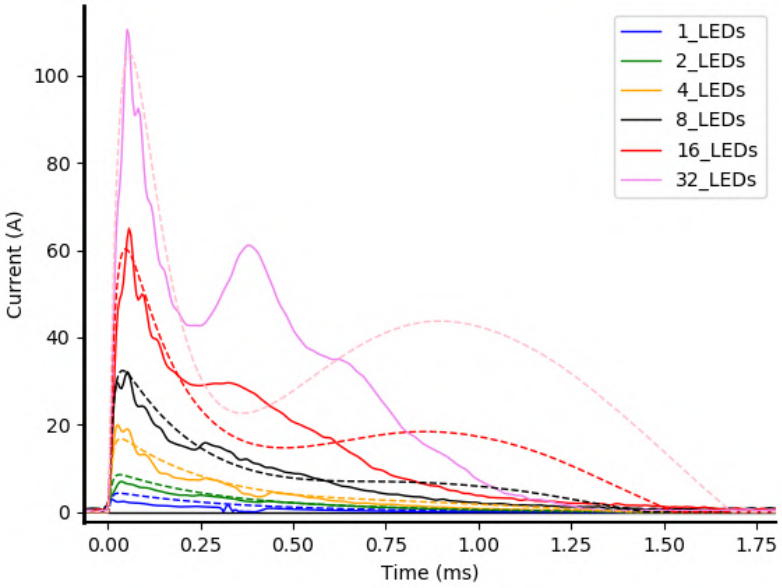


Figure 6.26: Simulation vs measured data of LED devices. Measured data with solid lines, simulation data with dashed lines.

The current waveform characteristics are influenced by factors such as the strength of the power supply and the system impedance. A system with lower impedance or higher inertia can result in more intense current waveforms during switching events. Increasing the number of devices in the simulation further intensifies the current waveforms. Eventually, a point is reached where the stored energy in the output filter of the supply, along with the supplied energy, is no longer sufficient to sustain the voltage levels. This leads to a voltage sag that prevents the devices from charging within a single cycle, as the voltage is inadequate to charge the input filters of the LED devices. The convergence trend of the current is illustrated in Fig. 6.27, while the effect of the inrush currents on the voltage supply is shown in Fig. 6.28.

The accumulation of a large number of low-power SPECs devices can act as a significant source of disturbance when switched simultaneously. This factor may be underestimated when considering EMC standards for small, low-power devices. However, in environments where many SPECs devices are used and switched in clusters, such as after a power failure or during testing/use of backup power systems, this issue becomes more prominent. According to the standard [4], equipment is deemed compliant if the maximum RMS input current (includ-

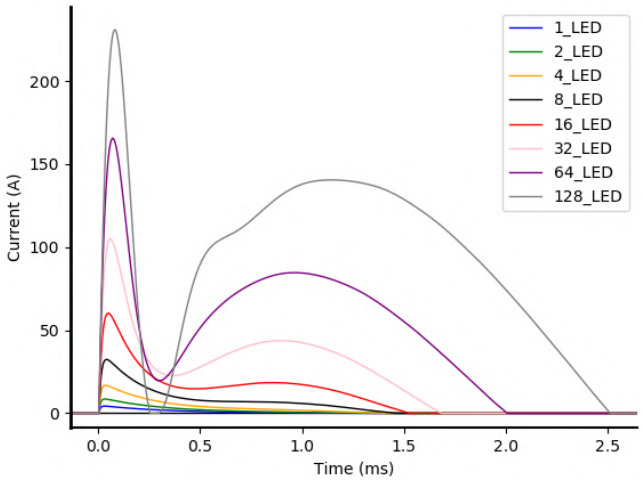


Figure 6.27: Simulation of inrush current trend with increased number of LED devices

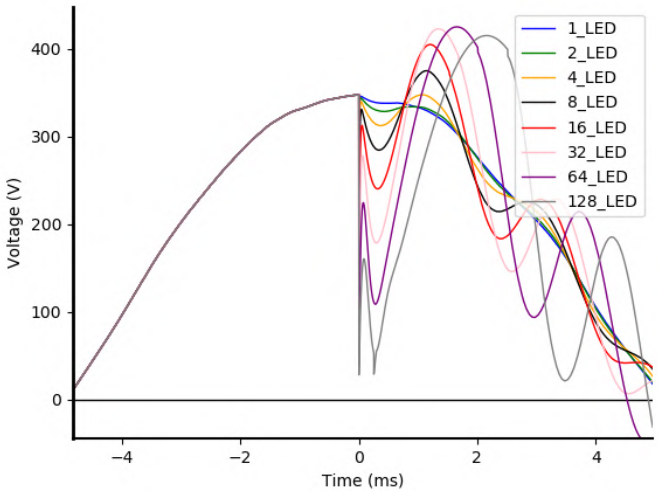


Figure 6.28: Voltage deviations caused by switching currents for different amount of LED lights (Simulated)

ing inrush current) evaluated over each 10 ms half-period between zero-crossings does not exceed 20 A for voltage changes caused by manual switching. This criterion can be easily met with a single low-power device. However, as the trend moves toward a higher number of lower-power SPECs devices, the cumulative effect of stacking these devices should be considered. Fig. 6.29 shows that the simulations predict RMS levels of 20A over 10 ms with only 64 devices switched, resulting in a total power of 128 W. The deviation between measured and simulated results can be attributed to various variables.

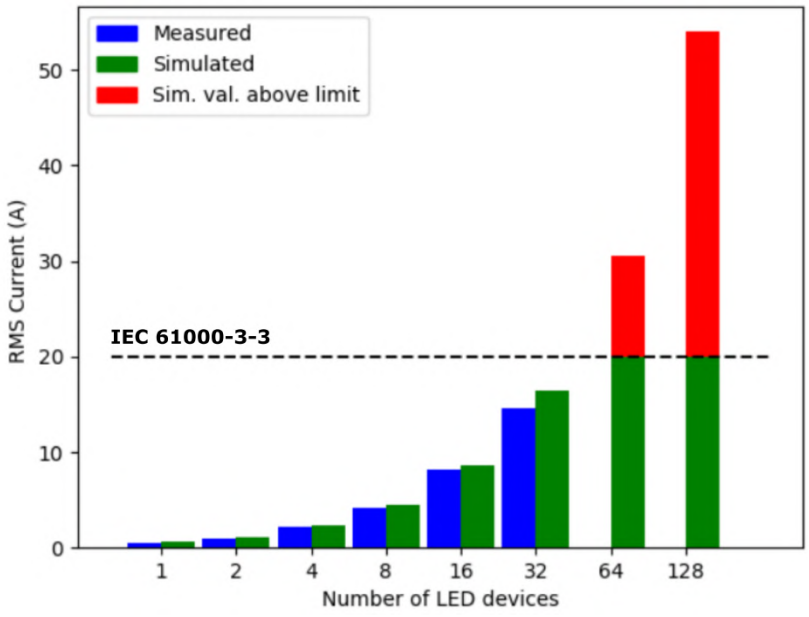


Figure 6.29: Measured and simulated inrush current RMS value over a 10 ms period

The most significant factor is likely the response of the GE's control system. Other factors include differences in output filter characteristics between the simulation and practical implementation, the absence of parasitic components in the simulation, and deviations in the voltage firing angle triggering the inrush current.

This level of SPECs power accumulation is commonly encountered in public, commercial, or industrial grid zones. Adding high-power low-pass filters to sensitive power zones may mitigate the situation, although it could be expensive or impractical. In addition to the mitigating factors mentioned in Section 6.1, providing a rating

on devices indicating the peak or form of the maximum inrush current could increase awareness among system designers. This would enable them to make informed judgments or compensate for very high transient-prone loads.

6.3.3 *Recap: Understanding the Implications of Rapid Load Changes on Micro-Grids and Generators*

The recent global shift towards renewable energy sources and distributed generation has highlighted the complexities arising from rapid load changes, particularly in the context of micro-grids and generators. Micro-grids, especially those reliant on inverters, face significant challenges due to their limited energy buffers, leading to voltage instability and potential equipment damage. Similarly, generators, while exhibiting superior response to transient load currents, are not immune to the impacts of rapid load changes, such as voltage and frequency deviations beyond specified limits, and potential mechanical damage in cases of over-speed conditions. As such, the stability and reliability of both micro-grids and generator-based systems become crucial aspects to consider in the context of the broader energy transition landscape. Important to note, the problems are not isolated to "socio-economically challenged" regions but global and current.

This chapter and the preceding emphasizes the importance of high-bandwidth measurements that encompass multiple locations and synchronized data, providing a comprehensive snapshot of the behavior of complex energy systems like micro-grids. Such measurements allow for a deeper analysis of the interplay between system inertia, control system time constants, and the impacts of rapid load changes on the voltage and frequency stability of these systems. Additionally, the data underscores the consequences of rapid load changes and the main factor contributing to the intensity of a occurrence. In Chapter 7 a solution is presented to mitigating transients through the use of an active filters. A fast responding active filter, especially response to non-predictable or intermittent events, is shown to be capable in stabilizing voltage and frequency fluctuations, ensuring the robustness of the

overall energy distribution system.

By combining empirical research with laboratory experiments and real-world measurements, this study contributes to the ongoing efforts to develop reliable and efficient micro-grid systems. The findings highlight the need for advancing the understanding of the dynamic behavior of both inverter-based and generator-based systems, enabling the implementation of targeted mitigation strategies to enhance their resilience and performance within the evolving energy landscape.

7 Active EMI Filter Design



7.1 Overview

An AEF is a type of electronic filter that employs active components such as transistors or operational amplifiers to suppress or mitigate unwanted electromagnetic emissions generated by electronic devices or systems. It operates by actively sensing, analyzing, and processing the incoming electrical signals, and then generating counteracting signals to minimize the impact of EMI on sensitive components or neighboring systems. The primary function of an AEF is to reduce the interference caused by electromagnetic emissions, ensuring the smooth operation and reliable performance of electronic equipment while adhering to EMC standards.

An AEF implemented on an FPGA platform with an H-bridge inverter offers a dynamic and efficient solution for EMI management in power electronic systems. By utilizing the programmable capabilities of the FPGA, the AEF can adaptively analyze and process complex signals, allowing for real-time adjustment and optimization of the filter parameters based on the specific EMI characteristics. The integration of an H-bridge inverter enables precise control over the power delivery to the load, further enhancing the filter's effectiveness. This digital approach is superior to an analog approach due to its greater adaptability, allowing for easy reconfiguration to meet changing EMI requirements. Additionally, the digital platform facilitates more precise signal processing, leading to enhanced EMI mitigation without compromising the efficiency and performance of the power system. The FPGA architecture, specifically the Xilinx Arty 35 FPGA chosen in this case, offers the necessary resources and flexibility to implement the control algorithms and signal process-

ing required for the system.

The computational core is designed using Matlab Simulink with HDL coder, enabling efficient modeling and synthesis of the control algorithms. By utilizing this approach, the system can achieve fast reaction times in the low μs range, which is crucial for effectively counteracting transient effects caused by rapid load changes and impulse-like load currents. The FPGA control block in the system includes various sub-blocks, ADCs for current sensing, and the control algorithm implementation. One essential component is a Butterworth bandpass filter core block, which processes the grid signal and isolates the measured grid signal from unwanted frequency components. The filter is optimised for a 50 Hz supply frequency.

7.2 *Architecture*

The Silicium-Carbide (SiC) full bridge inverter is employed for the power conversion in the system due to its superior characteristics compared to traditional silicon-based devices. A SiC MOSFET, unlike traditional silicon-based MOSFETs, offer superior thermal conductivity, faster switching speeds, higher efficiency, and improved performance at elevated temperatures, making them ideal for high-power, high-frequency applications requiring enhanced power density and reliability in demanding environments. The high-switching frequency capability of the SiC MOSFET allows for rapid and precise control of the inverter output, aligning with the requirements of the AEF system to counteract high-frequency current components. This fast response capability contributes to reducing the rise time and peak of the load current waveform, effectively mitigating the transient effects caused by rapid load changes.

Note: The higher switching speed of SiC MOSFETs enables faster response times in control systems, as the increased frequency allows for quicker reactions to changes in the system. In an active filter, for instance, the ability to respond is limited to the time of a switching cycle, a faster switching frequency = shorter switching period = faster response.

Furthermore, the reduced power losses in SiC devices lead to higher power conversion efficiency, minimizing energy wastage and improving the overall performance of the micro-grid system. By leveraging the FPGA's computational power and the SiC MOSFET's high-switching frequency capabilities, the proposed system design achieves enhanced performance in terms of response time, accuracy, and efficiency. These technological choices play a

pivotal role in realizing the objectives of the system to enhance electromagnetic compatibility EMC in low-inertia micro-grids, particularly in the presence of rapid load transitions and non-linear loads.

7.3 *FPGA Design and Implementation*

A Xilinx Artix-7 is chosen for this project. The advantages of the Xilinx system are explained; The Arty A7 board, built around the Artix-7 FPGA, boasts developer-friendly features, such as an onboard analogue to digital converter (ADC) that enables direct interfacing with analog signals. Additionally, the board provides an large number of general-purpose I/O (GPIO) pins, facilitating flexible connectivity with external devices and peripherals. Xilinx's documentation, reference designs, and community support are positives for the Artix-7 FPGA. This is marketing jargon provided by the company, the real reason is that coding in VHDL is torture, and HDL coder environment which is used in Matlab Simulink provides a user friendly platform that can take you from simulations to implementation relatively easily. The reference designs are very helpful and besides some troublesome non-obvious issues, the workflow is survivable.

7.3.1 *Design Process for Programming Xilinx Artix-7 FPGA*

Note: Using the Xilinx Vivado software coupled with the Matlab Simulink HDL coder environment is not easy, but the integration of the Simulink simulation environment to final implementation can be done on the two coupled platforms. This enables a start to end linked environment from concept to implementation. The list in this section describes the main steps in the process.

1. Conception and System Design:
   - Clearly define the requirements and functionality of the FPGA design.
   - Identify the input and output interfaces, data processing algorithms, and any additional components needed.
2. MATLAB Simulink Modeling:
   - Use MATLAB Simulink to create a high-level system model design.
   - Implement the desired functionality using Simulink blocks and subsystems.
   - Configure parameters, inputs, and outputs of the model.
3. Algorithm Design and Verification:
   - Develop and validate algorithms within Simulink.

- Use MATLAB's simulation capabilities to test and verify the behavior the design, this may include simulated external components.
- Ensure that the system meets the specified requirements and functions correctly.

4. HDL Code Generation:
   - Use HDL Coder in MATLAB to automatically generate synthesizable VHDL or Verilog code from the Simulink model.
   - Configure HDL code generation options according to the target FPGA device (Artix-7 in this case).
   - HDL Coder translates the Simulink model into a hardware description language (HDL) representation.
5. Vivado Project Creation:
   - Launch Xilinx Vivado and create a new project.
   - Select the appropriate FPGA device (Artix-7) and specify the project settings (such as project name, location, and language).
6. IP Integration and Customization:
   - Import the generated HDL code into a Vivado project.
   - If required, add Xilinx Intellectual Property (IP) cores to the project, such as memory controllers, communication interfaces, or pre-designed modules.
   - Customize and configure the IP cores and HDL code as necessary.
7. Synthesis and Implementation:
   - Perform synthesis to convert the HDL code into a gate-level representation.
   - Run the place-and-route process to map the synthesized design onto the Artix-7 FPGA's resources.
   - Verify that the design meets timing constraints and resource utilization requirements.
8. Bitstream Generation:
   - Generate a bitstream file from the implemented design.
   - The bitstream file contains configuration data for programming the FPGA.
9. FPGA Programming:

- Connect the Artix-7 FPGA board to the computer running Vivado.
- Configure the FPGA using the generated bitstream file.
- Verify that the programming process is successful.

10. Testing and Validation:
    - After programming the FPGA, verify that the design functions correctly on the target hardware.
    - Test different input scenarios and ensure that the outputs match the expected results.
    - Perform any necessary debugging or optimization iterations.

7.3.2 *Simulation - Proof of Concept*

A simulation of an auxiliary source is incorporated in a larger group of previously verified simulations of a GE and bank of LED lights [16; 5], the activation currents of which result in large impulse like currents. The current characteristics of this specific load profile has shown to deactivate the power supply or create large voltage deviations [16]. Using measurement data of load current obtained from previous experiments, an AEF is simulated as shown in Fig. 7.1.

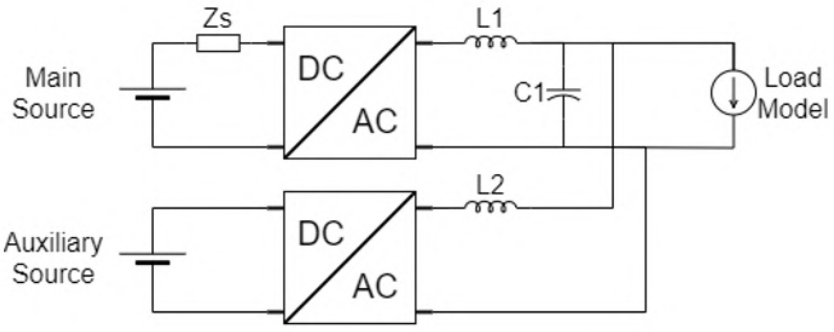


Figure 7.1: Diagram of AEF as auxiliary high frequency power source [18]

The concept is that the device is attached to an existing inverter or generator (or a specific load zone with the intention to stabilize it), the AEF supplies higher frequency

currents to the load. The original power supply is only required to, for the most part, supply 50 Hz current. In Fig. 7.2 the results of the simulation show a significant decrease in volatility due to the transient event by incorporating the AEF. The key is of course the reaction speed, the faster the device is able to react to a high-frequency current component (such as the rising edge of a power step function), the more effective the system [92; 93].

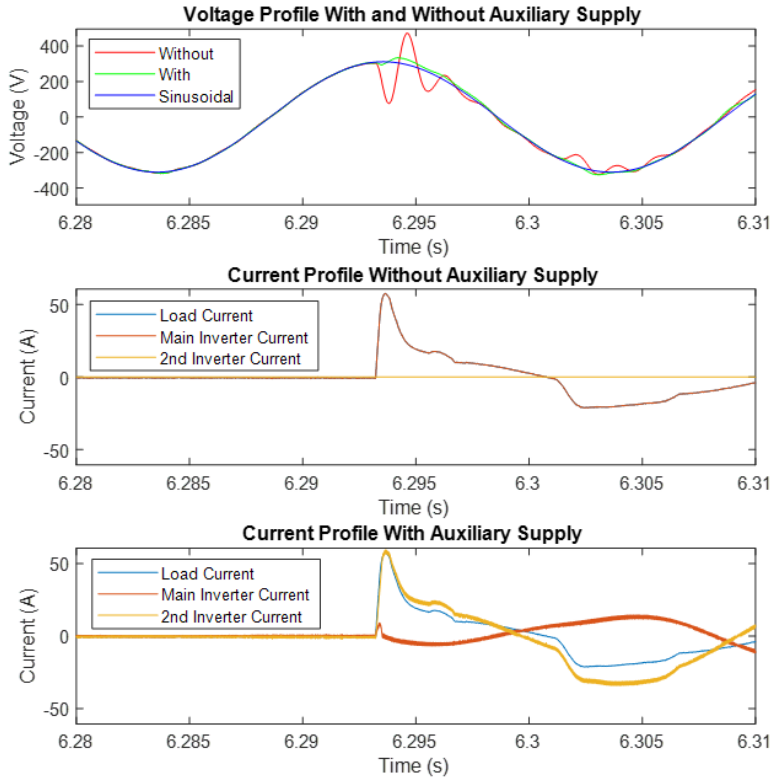


Figure 7.2: Effects on voltage stability and current delivery with and without compensation [18]

The power source, whether generator or converter based, contains an internal store of energy. A rapid load change will extract energy from the source, the rate of which energy is extracted as well as the value of energy with respect to the internal energy storage corresponds with the resulting transient. The control system of the power source will react at a certain speed (reaction time t_R). The rate and intensity of the reaction by the power supply control system defines the characteristics of the transient. The concept is to increase the reaction (speed and amplitude) of the control by use of a very-high-speed AEF or

auxiliary power supply.

The simulation is created in Matlab Simulink environment and is intended as a proof of concept. The simulation is not yet implemented or compatible with an FPGA device, the model blocks used are not available in the Simulink HDL coder environment for discrete format synthesis. Therefore the concept is ported and modified for feasible operation with Simulink HDL coder and FPGA synthesis, which is discussed in the following sections.

7.3.3 *System components and software*

The AEF control system operates on a Xilinx Arty 35 FPGA architecture, designed in Matlab Simulink with HDL coder and exported to a custom IP block. The block is imported in Vivado FPGA environment all necessary peripheral blocks as shown in Fig. 7.3. This includes definition of input output (I/O) ports, clock and reset configuration, ADC definitions, and AXI definitions which allow hardware in-the-loop (HIL) control and access to internal registers. Vivado generates a bitstream file which is loaded and run on the FPGA device.

The system layout composes of various components shown in Fig. 7.4. The AC supply and load are represented here and operate as usual. Firstly, an input circuit is designed using a resistor network with a voltage pull-up. The reason for this is the voltage range of the FPGA ADC input is from 0-3.3 V. The AC voltage must therefore be scaled down from the grid max level of $\pm$ 311 V to a valid range. Resistors are chosen to avoid any filtering of the input signal, the values are selected to match with the input impedance of the ADC I/O pins of the FPGA and PicoTech TA167 current clamps. The current clamp is used to accurately sense the load current and transduce this to a measurable and isolated voltage level.

The FPGA interfaces with a PC via USB for programming, parameter setting and power. For the SiC half-bridges, CREE half-bridge modules accept a PWM control signal from the FPGA. The half-bridge modules require a 12 VDC input for the control circuitry and a variable

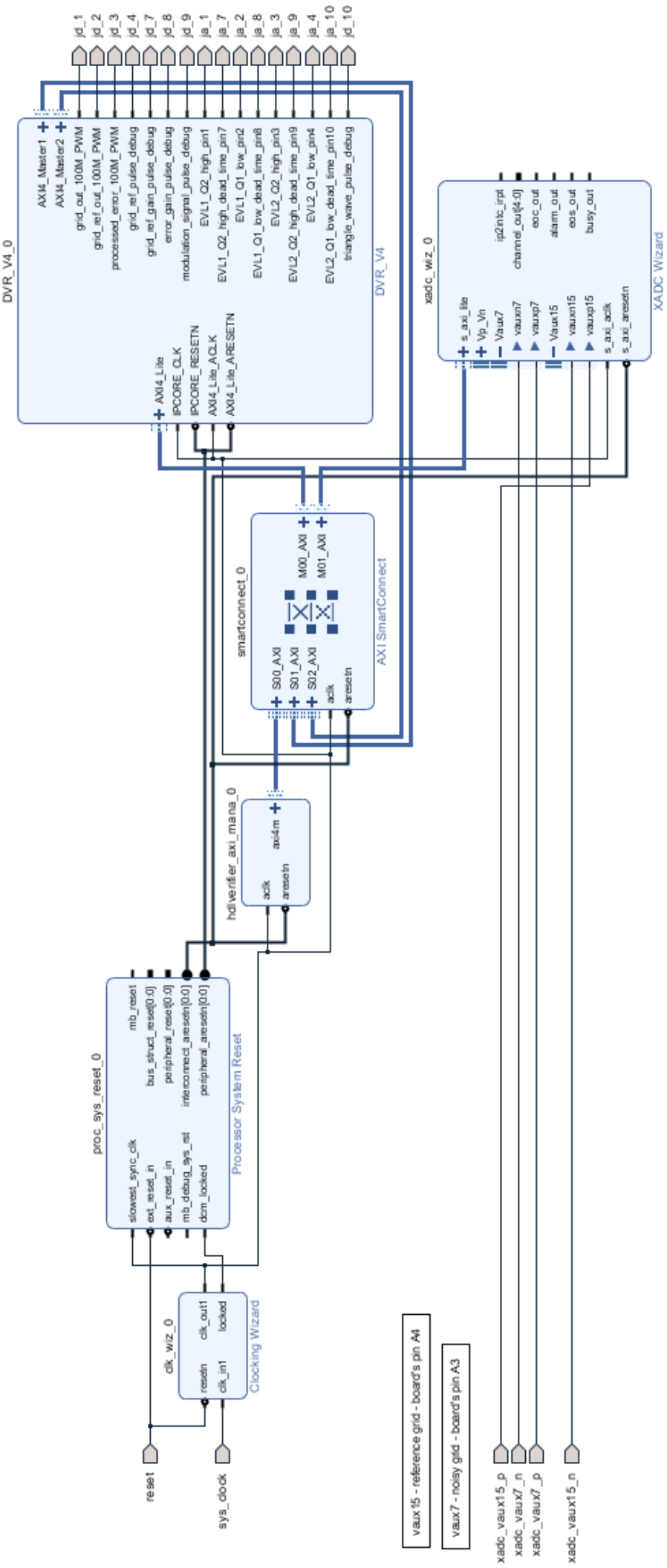


Figure 7.3: Vivado FPGA IP block design

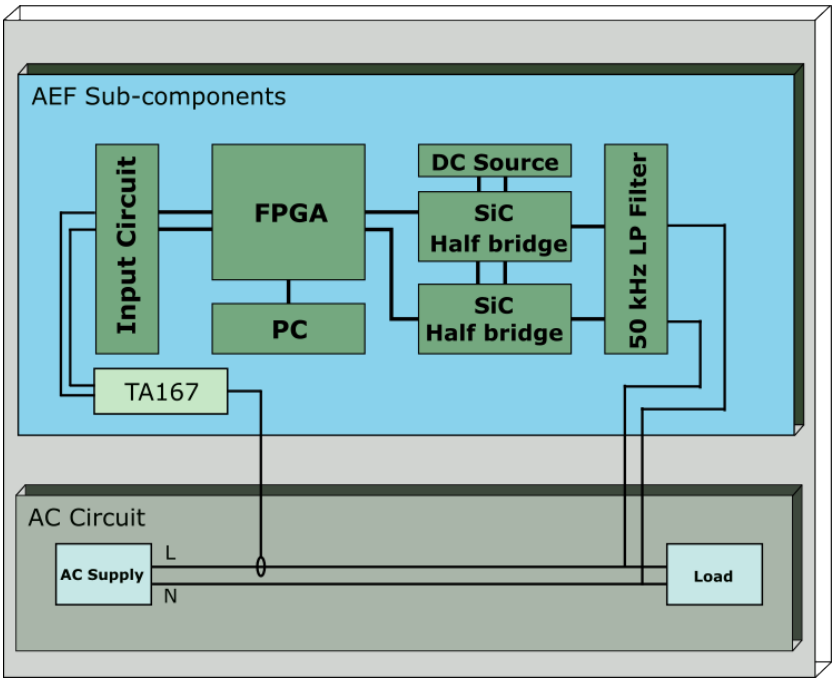


Figure 7.4: AEF system component diagram

DC-link voltage up to 600 V. Finally an output filter is designed to reduce the effect of the high frequency (1 MHz) switching noise from the SiC half-bridges. A 130 kHz power filter is designed which has the advantage of a small size footprint with sufficient attenuation. The targeted frequency range for power injection falls below the 50 kHz range and will therefore not interfere with the intended injected power with an inductance value of 1.7 μH and 1 μF capacitor. The intention is that the output of the filter is to be connected to the line between the AC power supply and load.

7.3.4 *Layout and Block Creation*

The Simulink design in organised into many sub subsystems which are connected with signal ports. In Fig. A.1 shows the main components of the main IP block to be exported to converted Verilog IP block used in the Vivado block design. The sub systems interact via defined ports. The main ports from the FPGA synthesized block diagram will interact with these set of sub-systems with signals such as I/O, reset, clock, RAM access and other internal signal lines.

7.3.5 *Control block design*

The control block consists of the input signal of the grid without any filtering or any elements which would effect the signal phase. This signal is filtered with a Butterworth band-pass filter with minimal phase shift with the notch of the filter designed to fall on 50 Hz. The signal is therefore fed straight through to the processed output but with all frequency components excluding the 50 Hz fundamental subtracted from it. This serves to provide a minimum phase shift between the grid reference signal and the measured grid signal for PWM output signal of the device. The flow diagram of this operation is illustrated in Fig. 7.5.

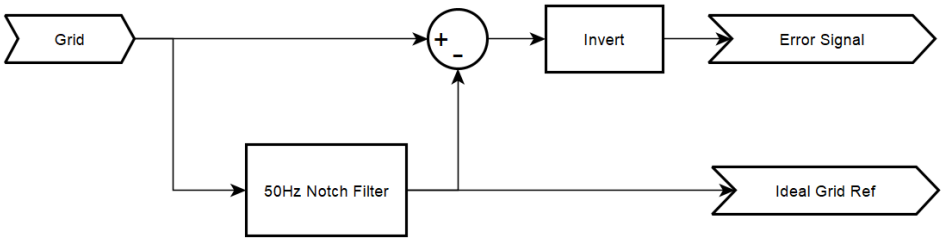
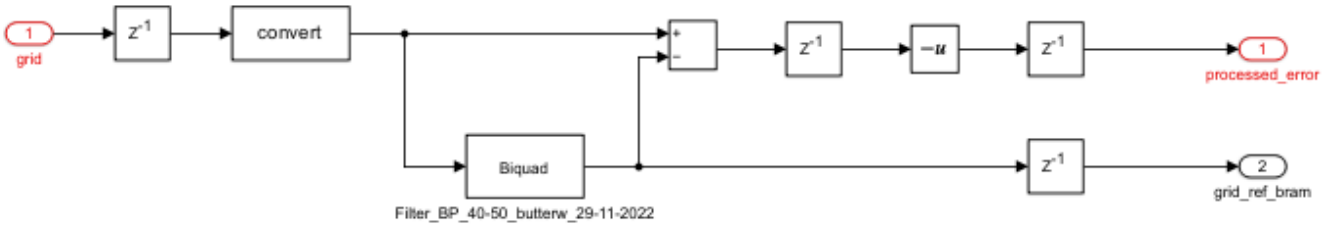


Figure 7.5: Control system flow diagram

The actual HDL coder model of this operation is shown in Fig. 7.6. The grid label is output from the XADC IP block to the control block. Signal post-processing includes further blocks which converts the signal error to the output PWM switching signal of the SiC Full bridge inverter configuration. The AXI smart connect IP block enables

access to on-board FPGA memory blocks which control signal gain, PWM switching frequency, switching dead-time.

Figure 7.6: Control system for grid deviation - Simulink



The FPGA implementation differs from the simulation performed in Section 7.3.2 due to differences in the block properties between continuous and HDL coder available block functions. The former utilizes a hysteresis-Fourier controller which extracts an errors signal from the grid and is processed in a hysteresis band, which provides the control signal for injection or absorption of current from the grid [18]. In future this may be optimized. The present objective is to observe the reaction time and waveform generating capabilities of the AEF with real-time grid measurement input.

7.3.6 *HDL Simulation/Emulation*

The same input waveforms from Section 7.3.2 is used as the input for the simulation. In Fig. 7.7 the input current waveform (I_{meas}) as well as the produced output voltage waveform (V_{out}) after filtering in sofware using a Butterworth filter with a 5 kHz cut-off frequency (to filter out the AEF switching noise and make the signal visible in the plot). The input voltage is biased around 1.65 V due

to the input requirements of the ADC. The output is normalized as this is the switching signal which will be injected into the gate driver of the SiC half-bridge modules.

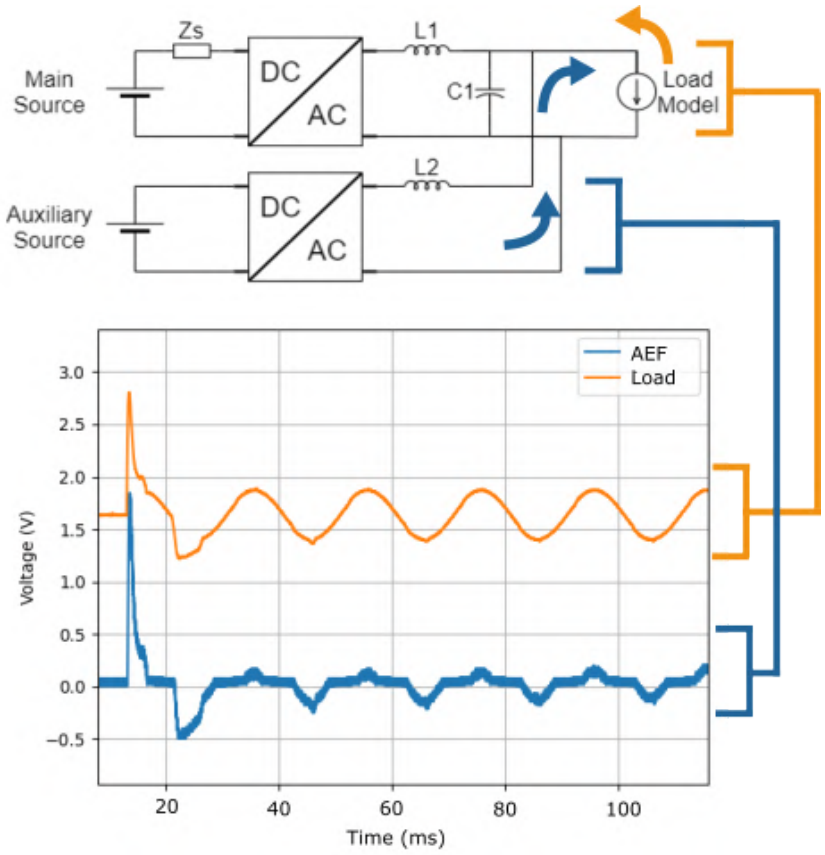


Figure 7.7: Simulation output voltage waveform with input measurement of current (Voltages scaled down for input voltage range of ADC and output voltage of control signal FPGA).

The response time of the AEF to a rapid load change in 13.2 μs according to the Simulink simulation and the waveform shape produced corresponds well with expectations. The connected loads in this specific profile are listed in Tab. 7.1. The Large initial activation current is a result of the LED light bank and Water pump with dimmer. A higher order harmonic component due to the inductive nature of the electric drill. The largest load component, which is a clean 50 Hz sinusoidal waveform is due to the resistive heater. The intention is to give a broad frequency spectrum of common load current implemented in practice, in order to evaluated the performance of the AEF. Although performing an FFT of a waveform gives insights on the frequency components contained in a signal, the main objective of the AEF is

compensating for intermittent or non-predictable rapid load changes. Performing an FFT of continously changing waveforms will not provide insights as an FFT by it's very nature averages out intermittent events. The AEF has the ability to compensate continuous waveforms, but it is not part of the scope of this project. When applying the FFT to a non-repetitive pulse signal, the length of the data sample or the segment used for the FFT calculation plays a significant role. If the length of the data sample is too short compared to the duration of the pulse, the pulse may not be fully captured in the frequency domain representation, leading to a loss of information.

Load	Description	Nominal Power (W)
1	Resistor bank (heater)	1800
2	LED light bank (32x2W + 5x11W)	119
3	Water pump with dimmer	130
4	Motor load (drill)	850

Table 7.1: Load profile of input current data

7.3.7 *Custom IP Block Creation and Porting*

Note: An IP block is a block representing custom code written for a function in FPGA design.

To create custom IP blocks for FPGA synthesis in Vivado from HDL Coder in MATLAB Simulink include several crucial steps. First, it is important to ensure proper clock timing synchronization between the MATLAB Simulink design and the FPGA. This can be achieved by using clocking wizard IP blocks in Vivado and configuring them according to the desired clock frequency and input/output requirements. HDL Code Advisor provides system checks on created IP blocks and settings, industry standards, model configuration, native floating point and ports and subsystems. The checks should be performed after every design iteration or modification of the code or model to prevent unnecessary errors. HDL Workflow Advisor assists with the next steps for settings and generation of the IP core specific to the target device (FPGA device) and synthesis tool. The serialization and version control is specified in this step. This file is then imported in the Vivado environment.

Templates and signal lines can be defined in MATLAB Simulink to represent the desired functionality of the cus-

tom IP block. These templates serve as a blueprint for generating the corresponding HDL code. By using MATLAB's HDL Coder, the Simulink model can be converted into synthesizable HDL code. To create reusable and configurable IP blocks, generic blocks can be incorporated into the Simulink model. These blocks allow for parameterization and customization, making the IP block more versatile. By exposing parameters as generics, users can easily modify the behavior of the IP block without making changes to the underlying HDL code.

For ADC connectivity, MATLAB Simulink provides built-in ADC blocks that can be utilized to interface with the FPGA. These blocks enable the capture of analog signals and their conversion to digital format for further processing within the custom IP block. Accessing custom random access memory (RAM) within the FPGA can be accomplished by defining appropriate interfaces and memory controllers in the MATLAB Simulink model. These interfaces enable the IP block to read from and write to custom RAM locations, facilitating efficient data storage and retrieval.

7.3.8 *Vivado Block, Synthesis, Implementation and Programming*

Vivado analyzes the design and generates a gate-level representation. During synthesis, the high-level code is transformed into a netlist that consists of logical gates and their interconnections. Vivado performs optimizations to improve performance, minimize area utilization, and enhance power efficiency. Once synthesis is complete, the design can be implemented on FPGA. Implementation involves mapping the gate-level netlist onto the physical resources of the FPGA, configuring the programmable interconnects and routing paths. Vivado optimizes placement and routing to ensure efficient utilization of resources and meet timing constraints, Fig. 7.10 shows the mapped logic gate configuration which will be implemented on the FPGA (gate matrix).

Note: In FPGA synthesis, negative slack refers to the amount of time by which a timing constraint is violated in a digital circuit design. It represents the difference between the required arrival time (specified by the constraints) and the actual arrival time of a signal at a particular point in the circuit.

The WNS is the largest negative value of slack within a design. It indicates the most critical timing violation in the circuit, meaning that the violated path has the least margin for error and is most likely to cause functional or performance issues.

When the WNS is too large, it suggests that the design is not meeting its timing requirements. This can lead to various problems, such as incorrect operation of the circuit, reduced performance, or even complete failure of the design.

FPGA synthesis tools aim to optimize the design by performing various transformations, such as logic restructuring, pipelining, and resource allocation, to improve the timing characteristics and reduce negative slack. Achieving positive slack (where the actual arrival time is earlier than the required arrival time) is desirable to ensure reliable and correct operation of the circuit.

Detailed reports are provided on timing analysis, highlighting any potential violations and suggesting corrective measures. The signal timing information is shown in Fig. 7.8. In addition to synthesis and implementation, Vivado also supports programming the FPGA with the generated bitstream file. The bitstream file configures the FPGA's internal connections and functionality, allowing it to operate as desired after programming. Power estimation, allowing designers to estimate the power consumption of their FPGA designs is provided.

Path Group	From Clock	To Clock	Edges (WNS)	WNS (ns)	TNS (ns)
async_default	clk_out1_design_active_fil...	clk_out1_design_active_fil...	rise - rise	8.529	0.000

Failing Endpoints (TNS)	Total Endpoints (TNS)	Edges (WHS)	WHS (ns)	THS (ns)	Failing Endpoints (THS)	Total Endpoints (THS)
0	4	rise - rise	0.402	0.000	0	4

Figure 7.8: Timing information and WNS

Design Timing Summary

Setup		Hold		Pulse Width	
Worst Negative Slack (WNS):	1,401 ns	Worst Hold Slack (WHS):	0,056 ns	Worst Pulse Width Slack (WPWS):	3,000 ns
Total Negative Slack (TNS):	0,000 ns	Total Hold Slack (THS):	0,000 ns	Total Pulse Width Negative Slack (TPWS):	0,000 ns
Number of Failing Endpoints:	0	Number of Failing Endpoints:	0	Number of Failing Endpoints:	0
Total Number of Endpoints:	16977	Total Number of Endpoints:	16977	Total Number of Endpoints:	6025

All user specified timing constraints are met.

Figure 7.9: Design timing summary

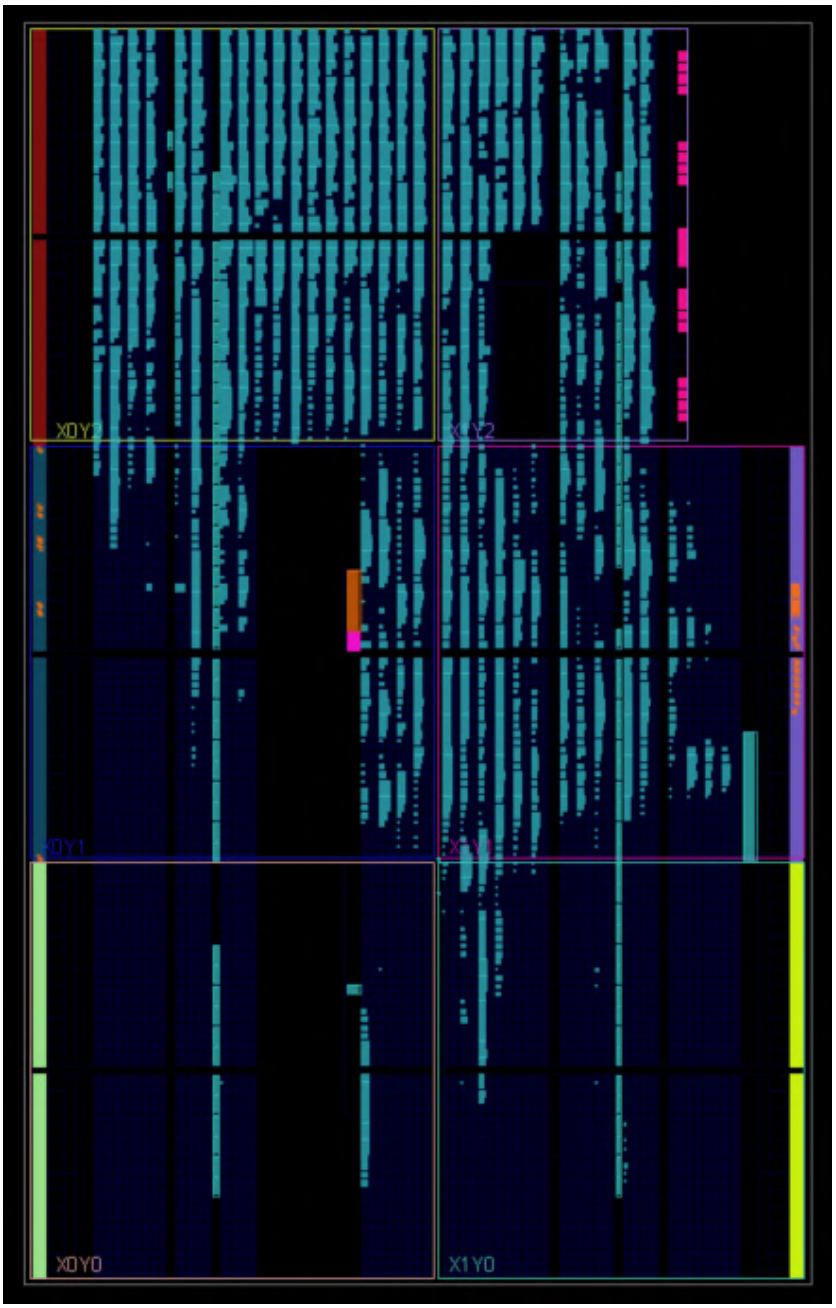


Figure 7.10: Vivado visualization of implemented FPGA logic elements

Note: This image shows the physical fabric of the FPGA. The highlighted blue sections indicate where the synthesized logic is utilised and how these logic elements interact.

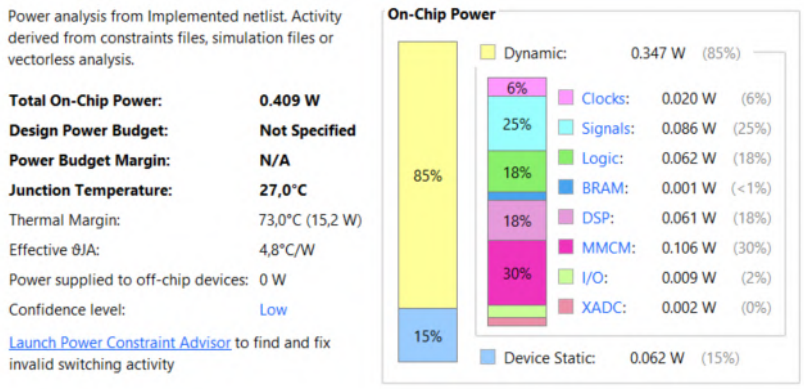


Figure 7.11: Estimates of power usage and system temperature

7.3.9 *Control signals*

The FPGA is programmed with the synthesized model. Using the same input current waveform (actual measurements used in different design steps for continuity) as the previous simulations, a Teledyne T3AFG120 AWG is configured with the wave-forms as output which is connected to the ADC sense input pins of the FPGA. A Matlab script is created which converts the waveform into the correct format and communicates with the AWG via USB. The wave-forms generated by the FPGA aligns well with the simulation and produces a response which aligns with the output of the simulations. The AWG does not produce a precise analogue waveform, therefore some discrete steps are noticeable and may effect the output generated by the AEF. The waveform is shown in Fig. 7.12.

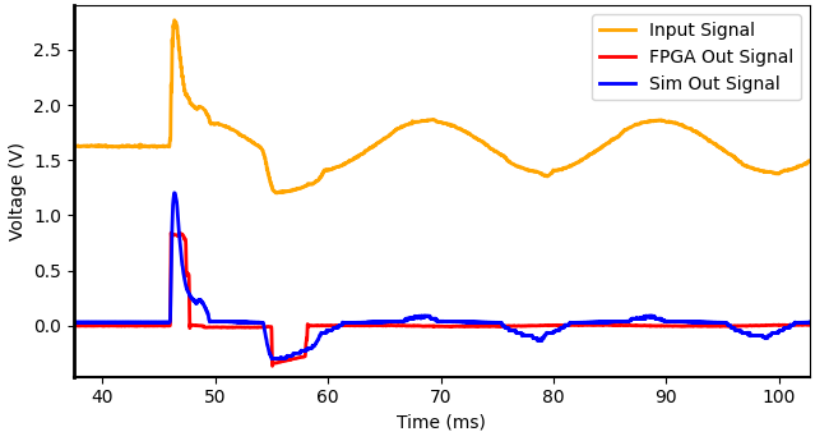


Figure 7.12: Physical AWG input (orange) and FPGA control output signal (red) measured, Simulink simulation output signal (blue)

The measurement of the input and output signals is performed with Picoscope and TA043 differential voltage probes. Fig. 7.12 is a capture of the Picoscope application with digital 5 kHz low pass filtering performed on the FPGA output (pulse width modulation (PWM) waveform) to produce the output waveform shown in red. The low pass filter is to show the signal waveform otherwise otherwise not clearly shown in the carrier PWM waveform. On a smaller time scale, the response time of the AEF become apparent, shown in Fig. 7.13. This is measured at 13.7 μs, which is in line with the expectations following

the simulation of the system.

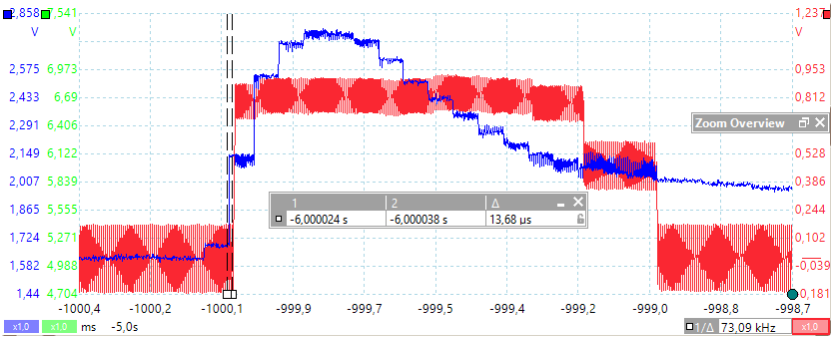


Figure 7.13: Response time of AEF to a rapid load change, small time window measurement (Error signal - Blue, AEF response - RED, time scale on x-axis in ms)

7.4 *Full System output measurement*

The system is tested while not connected to the grid (system diagram shown in Fig. 7.4). The purpose of the test is to verify the response time of the AEF as well as the waveform produced, which should be sufficient to reduce the rise time and peak of the un-compensated load current waveform. This to verify the previous simulations and have an operation system which has all active components included. The output of the AEF is loaded with 5 Ω. The grid consists of a Cinergia 5 kVA per phase, with the GE representing the isolated grid. A bank of 32x2 W LED lights are connected as the load which is activated to emulate a switching event which causes a large impulse-like current peak (a similar load profile which was used to produce the input current wave-forms used in the simulations).

A PicoTech TA167 current clamp measures the load current and is the input to the AEF as shown in Fig. 7.4. A measurement is shown in Fig. 7.14 of activation current from the LED load with a peak of 52 A, the total time of the rising slope is 544 μs. The grid voltage decreases rapidly from 323 V to 185 V.

A smaller time scale of the AEF output voltage and load current is shown in Fig. 7.15. The response time of the filter to a higher frequency current component is measured

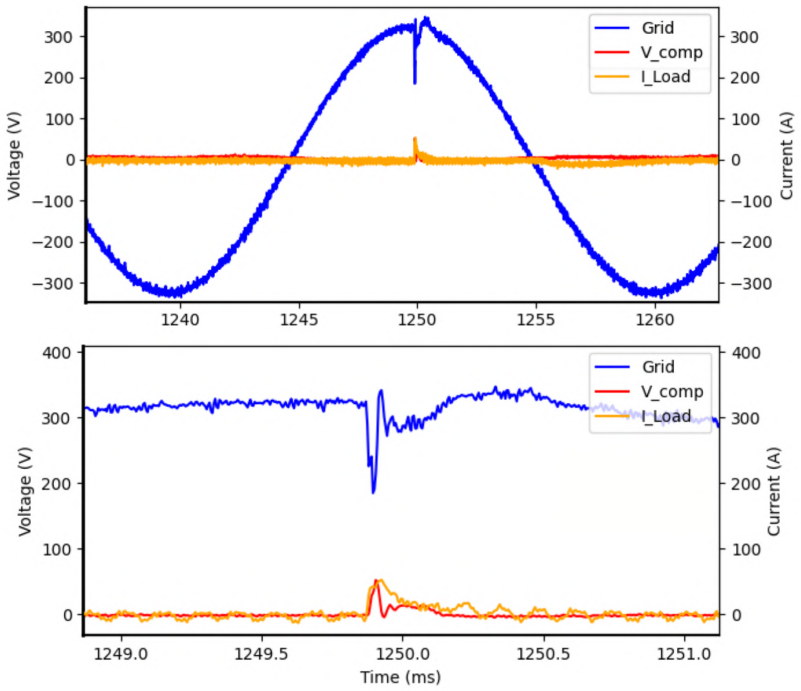


Figure 7.14: AEF response to pulse load current and grid voltage deviation

at 14 μs. If the system were connected directly to the grid, the wave-form shape would differ, due to the impedance differences in comparison to a 5 Ω resistive load.

7·5 *Results Discussion*

Experimental validation was conducted to assess the performance of the AEF control system in terms of pulse load effect mitigation and EMC. The initial proof of concept simulations performed in Section 7.3.2 show promising results and significant improvements in the transient response after a rapid load change event. Further simulations and practical measurements validate the concept of a very fast reactive AEF. The results demonstrate significant improvements in mitigating the high rising edge of load changes, maintaining power quality, and reducing the impact on the micro-grid's electromagnetic environment. High derivative of power being identified as the cause of adverse transient reactions in power supplies as shown in Chapter 6.

The system achieves reactive timing goals with expected

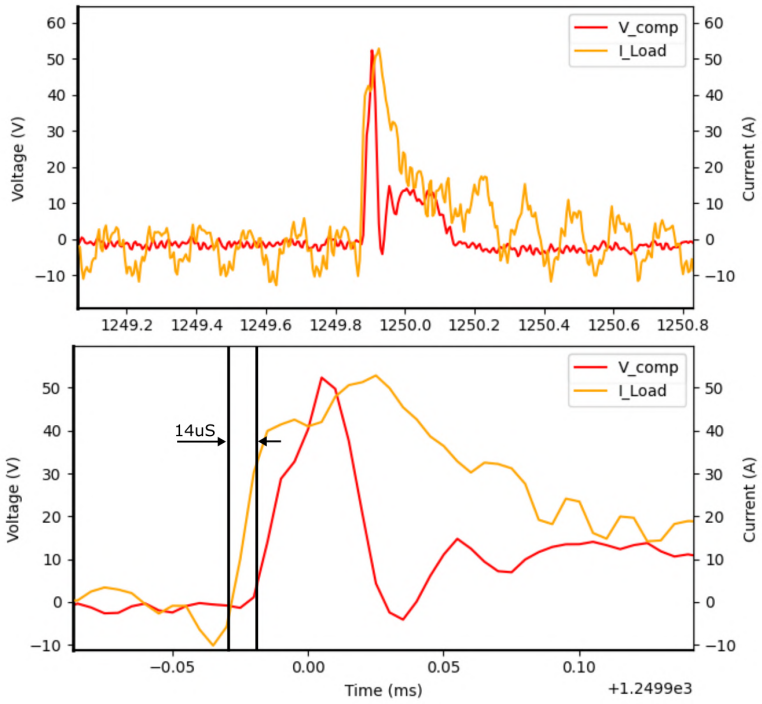


Figure 7.15: AEF response to pulse load current - small time scale

waveform outputs for power injection to counter the effects of high frequency current components arising from a rapid load change. The key factor being the response time in the low μs range, the achieved reaction time of 14 μs is therefore satisfactory with the information currently available. Further investigation is recommended to perform grid-tied tests, and analyzing the AEF performance under grid environment impedance and noise conditions. Adequate filtering of the 1 MHz switching noise of the AEF and harmonics thereof should be further investigated to avoid EMC in the higher frequency spectrum.

The results of the AEF design chapter demonstrate the successful implementation and performance of the proposed system. The FPGA-based control system, utilizing the Xilinx Arty 35 FPGA, allows for high-speed, real-time processing and control capabilities, enabling fast reaction times in the low microsecond range. The AEF system incorporates a MOSFET full bridge inverter and a Butterworth bandpass filter to process the grid signal. The use of SiC (silicon carbide) devices in the full bridge inverter offers superior characteristics compared to traditional

silicon-based devices, including higher switching frequencies and lower power losses. This enables faster response times and improved overall efficiency of the micro-grid system.

The AEF designed aligns with the simulations performed in the proof of concept models created. Although the output of the AEF is not physically connected to the grid, the output wave-forms from the AWG and current measurements directly input into the AEF show potential to effectively counteract transient effects caused by rapid load changes and impulse-like load currents. The simulation and emulation of the system show a significant decrease in volatility and improved voltage stability when the AEF is incorporated. The AEF successfully mitigates the transient effects, reducing the rise time and peak of the load current waveform in simulation. The future work recommended is to connect the AEF to the grid, test and verify the behavior of the system in use case scenarios.

The custom IP block design and implementation process using MATLAB Simulink and Vivado is described. The HDL code generation, synthesis, and implementation steps ensure the successful programming of the FPGA with the generated bitstream file. The timing analysis and power estimation provide valuable insights into the performance and resource utilization of the implemented design. The experimental results confirm the effectiveness of the AEF system. The response time of the AEF to rapid load changes is observed to be in the range of 13.4 μs, which aligns well with the simulation results. The AEF's ability to generate the desired output waveform, reducing the rise time and peak of the load current, is demonstrated through measurements and comparisons with simulation results.

As a final point, An FPGA was selected for implementation of the control system due to the signal processing capabilities and design flexibility as apposed to analog designs. FPGAs are superior in certain cases to general micro-controller designs due to timing certainties provided by the real time logic, and parallel processing capabilities.

8 Research Conclusion & Future Work



8.1 Overview

In this study, the focus revolved around investigating the impact of rapid load changes on the EMC of isolated micro-grids. The primary objective was to comprehensively analyze the extent of the problem and develop effective mitigation techniques to enhance the reliability of power supplies in these systems. To achieve this, a systematic approach was adopted, encompassing three key steps: the development of a measurement system, conducting measurements on real-world diesel generators and inverters, and proposing a viable solution.



Figure 8.1: Project path

The first step involved the development of a sophisticated measurement system capable of capturing and analyzing data with a sufficiently high bandwidth to accurately diagnose problems in complex micro-grids. This system was designed to facilitate comprehensive understanding of the EMC issues arising from rapid load changes. By employing this system, a comprehensive data-set was gathered, enabling detailed analysis and assessment of the extent of the problem.

With the measurement system in place, the second step focused on performing measurements on real-world diesel generators and inverters. The objective was to gain a deeper understanding of the nature and causes of the EMC challenges faced by isolated systems. Through these measurements, it was identified that isolated micro-grids, due to their inherent characteristics, exhibit weaknesses and lack the necessary inertia to effectively handle rapid load changes. This finding served as a crucial insight into the root causes of the EMC-related reliability issues faced by these systems.

Building upon the understanding gained from the measurements, the final step involved proposing a solution to address the identified problems. The proposed solution centered around the design and construction of a high-speed FPGA-controlled H-bridge inverter or an AEF. By employing such control technologies, the aim was to compensate for the fast-rising edge power changes and mitigate the adverse effects of rapid load fluctuations on the isolated micro-grid's power supply.

This research encompassed a comprehensive approach to studying the EMC challenges associated with rapid load changes in isolated micro-grids. Through the development of a measurement system, extensive measurements on real-world systems, and the proposal of a solution involving advanced control techniques, this study aims to contribute to the understanding and mitigating the impact of rapid load changes on the reliability of power supplies in isolated micro-grids. The findings and recommendations of this research provide valuable insights for improving the overall performance and EMC of isolated micro-grids in various applications.

8.1.1 *Development of Models, Measurement Tools and Software*

The research includes an exploration of measurement systems for conducting EMC measurements in the time domain. The advantages of time domain measurement devices over frequency domain devices are discussed, emphasizing their ability to capture transient phenomena and provide a more comprehensive characterization of electrical systems. The challenges associated with con-

ducted EMC issues are also highlighted, along with the limitations of existing measurement devices. The research proposes a design parameter wish list for the development of measurement systems in micro-grid environments. The desired features include synchronization capabilities, mobility, time-domain measurements, easy data processing, ruggedness, scalability, real-time monitoring, and compatibility with various communication protocols and standards.

The developed measurement system utilizes a Picoscope USB oscilloscope, Raspberry Pi 4 as a node-local control device, and other components such as a battery bank, network jack, and USB connector. The system is designed to be scalable, allowing for the addition of multiple measurement points and devices as needed. Software plays a crucial role in the measurement system, enabling control, data processing, and visualization of the collected data. Python is used as the scripting language, with the main script communicating with the Raspberry Pi via a LAN and interfacing with the Picoscope API libraries coded in C.

The applications of the developed measurement system are demonstrated through two case studies. The first case study focuses on long-duration monitoring of power quality issues in a modern building, highlighting the improvements achieved after corrective measures were implemented. The second case study explores short-duration high-bandwidth measurements, specifically examining the effects of inrush currents caused by switching on LED lights. The results obtained from Simulink simulations of inrush currents provide valuable insights into the behavior of power systems and help in understanding the impact of switching events on the system's performance. By comparing the simulation results with the measured data, researchers can validate the accuracy of the Simulink models and gain confidence in their predictive capabilities.

The simulation results often align well with the measured data, demonstrating the effectiveness of the Simulink models in replicating real-world scenarios. The currents generated by different combinations of activated loads,

such as LED clusters, show similar characteristics in both the simulation and measurement data. The maximum peak current values and the rising edge current slopes exhibit close agreement between the simulation and measurements, usually within a deviation of 20%. Furthermore, the simulations enable researchers to explore the effects of increasing the number of devices or loads in the system. It is observed that as more devices are switched simultaneously, the inrush currents intensify, reaching levels that may cause voltage sags and compromise the performance and reliability of the system. These findings highlight the importance of considering the cumulative effect of stacking low-power devices, as the combined inrush currents can exceed the limits specified by standards. Additionally, simulations help in assessing the response of the system's control system and evaluating the effectiveness of mitigation strategies. By varying parameters and incorporating different components, researchers can analyze the impact of various factors on inrush currents and explore potential solutions, such as the use of low-pass filters or staged switching.

8.1.2 *Analysis and Core Problems Identified*

The analysis of real-world systems and the effects of rapid load changes in the context of renewable power sources and distributed generation have revealed several core problems and implications. The research findings highlight the complex nature of power sources in distributed energy systems, categorizing them into inertial sources (e.g., rotating generators) and static sources (e.g., inverters). Inertial sources exhibit slower response times to load changes due to their higher inertia, while inverters can respond quickly but have lower inertia. The implementation of both supply configurations, particularly in micro-grids, has resulted in increased EMC issues. Rapid load fluctuations during load activation, connection/disconnection, and operational mode changes have raised concerns about system stability and reliability. Failures and damage to generators and inverter power supplies have been observed due to impulse-type currents, such as inrush or pulsed continuous currents from specific loads.

Grid instability and compromised energy security pose significant risks, leading to financial losses, disruption in communications, and even safety hazards. Examples from South Africa (mentioned in introduction) and Indonesia (case study) demonstrate the need for reliable and independent micro-grid systems to address these challenges. However, the existing literature on the effects of rapid load changes on isolated micro-grids is limited, mainly relying on simulation-based studies that do not fully capture real-world complexities. Case studies conducted in Indonesia and the Netherlands shed light on the response of micro-grid systems to rapid load changes. Inverter-based micro-grids showed voltage dips, frequency deviations, and unstable operation during load fluctuations. Generator-based systems faced challenges in meeting sudden load increases or decreases, leading to voltage and frequency deviations, protective system activations, and potential mechanical damage.

Implications for micro-grid design and operation include the need for improved system inertia, control mechanisms, protective systems, and updated standards for testing and evaluating power supplies. The research emphasizes the importance of empirical studies integrating laboratory experiments with real-world measurements to gain a comprehensive understanding of the effects of rapid load changes on micro-grids. The analysis of real-world systems and the effects of rapid load changes highlights the critical challenges and core problems associated with the transition to renewable power sources and distributed generation. Further research and development efforts should focus on addressing these challenges to ensure the stability, reliability, and efficiency of micro-grid systems in diverse operating environments.

8.1.3 *Proposed Solutions*

The use of an AEF coupled with an FPGA-based control system as a solution to address the challenges of rapid load changes in low-inertia AC micro-grids. The AEF provides pulse-like currents instead of relying solely on the power supply, thereby protecting the power supply from high-frequency current components and reducing transient effects. The research demonstrates through sim-

ulations and experimental validation that the AEF effectively mitigates the adverse effects of high rising-edge load changes and minimizes the impact on the micro-grid's electromagnetic environment. The response time of the AEF is measured to be within 14 μs.

The research highlights the significance of high derivatives of power as indicators of stability challenges in low-inertia micro-grids. Rapid load changes can cause voltage and frequency deviations, jeopardizing system reliability. The AEF is shown to be promising in compensating for impulse-like load currents with its fast reaction time, effectively counteracting the first slope of the load change. The FPGA-based control system achieves a reaction time of 14 μs, providing precise current measurements and rapid response.

8.1.4 *Wrap Up*

Through the development of a sophisticated measurement system, extensive measurements on real-world systems, and the proposal of a solution involving an AEF coupled with an FPGA-based control system, significant contributions have been made to enhance the reliability of power supplies in isolated micro-grids.

The measurement system developed in this study has demonstrated its effectiveness in capturing and analyzing data with high bandwidth, enabling a comprehensive understanding of EMC issues arising from rapid load changes. The measurements on real-world diesel generators and inverters have revealed the weaknesses and lack of inertia in isolated micro-grids, which are crucial factors contributing to the EMC-related reliability challenges faced by these systems.

While this research has made significant progress in addressing the challenges of rapid load changes in isolated micro-grids, there is still room for further investigation and improvement. Future research should focus on validating the proposed solution under diverse grid environments and exploring its scalability for larger micro-grid systems. Additionally, more comprehensive studies incorporating laboratory experiments and real-world measure-

ments are needed to gain a deeper understanding of the effects of rapid load changes on micro-grid stability and reliability.

And remember, while we strive for stable micro-grids, let us not forget to embrace the excitement of rapid load changes—the roller coaster ride of power fluctuations!

8.2 *Recommendations for Future Work*

Based on the findings and insights gained from this research, several avenues for future work can be explored to further enhance the understanding and mitigation of rapid load changes in isolated micro-grids. The following recommendations are provided to guide future research efforts:

8.2.1 *1. Experimental Validation and Field Testing*

To validate the effectiveness of the proposed solution in real-world scenarios, it is essential to conduct experimental validation and field testing. This would involve implementing the AEF and FPGA-based control system in practical micro-grid installations and evaluating their performance under various load conditions. Long-term monitoring and data collection in different geographic locations and climatic conditions can provide valuable insights into the system's robustness and scalability.

8.2.2 *2. System-Level Studies*

Future research should focus on conducting comprehensive system-level studies to evaluate the impact of rapid load changes on the overall stability and reliability of micro-grids. This would involve considering multiple interconnected micro-grids, exploring the interactions between different components, and analyzing the system's response to dynamic load variations. By integrating advanced modeling and simulation techniques with real-world measurements, a deeper understanding of the complex dynamics and control strategies of micro-grids can be achieved.

8.2.3 3. *Optimal Control Strategies*

Investigating and developing optimal control strategies for micro-grids can significantly improve their ability to handle rapid load changes. Future work should explore advanced control techniques, such as predictive control algorithms, adaptive control, and machine learning-based approaches, to enhance the dynamic response and stability of micro-grid systems. These strategies can optimize power flow management, enable seamless transitions between different operational modes, and improve the resilience of micro-grids against rapid load fluctuations.

8.2.4 4. *Integration of Energy Storage Systems*

The integration of energy storage systems, such as batteries or supercapacitors, in micro-grids can provide additional flexibility and enhance their ability to handle rapid load changes. Future research should focus on investigating the optimal sizing, placement, and control strategies for energy storage systems to mitigate the impact of load fluctuations. This includes exploring the coordination between storage systems and the proposed AEF-FPGA control system to maximize their combined effectiveness in stabilizing micro-grid operation.

8.2.5 5. *Standards and Guidelines*

Developing updated standards and guidelines specific to the EMC challenges of isolated micro-grids can facilitate the design, testing, and evaluation of power supplies in these systems. Future work should involve collaboration with regulatory bodies, industry stakeholders, and standardization organizations to establish comprehensive guidelines addressing the impact of rapid load changes on micro-grid performance (such as the proposal for updating IEC 61000-4-30). This includes defining EMC requirements, testing methodologies, and certification procedures tailored to the unique characteristics of isolated micro-grids. By pursuing these recommended future research directions, the understanding and mitigation of rapid load changes in isolated micro-grids can be further advanced, leading to more resilient and efficient energy systems that can effectively accommodate the growing

penetration of renewable energy sources and distributed generation. Remember, the quest for knowledge and innovation never ends, and there are always exciting frontiers to explore in the realm of micro-grid research!

9 List Of Publications



9.1 Peer Reviewed Journals

1. **Synchronous multi-point low-frequency EMI measurement and applications**, **Alexander Matthee**, Niek Moonen, Frank Leferink, IEEE Letters on Electromagnetic Compatibility Practice and Applications, vol. 4, pp. 120–124, 2022
2. **Transient Response of Generator- and Inverter-Based Micro-grids to Rapid Load Changes**, **Alexander Matthee**, Niek Moonen, Ilman Sulaeman, Frank Leferink, IEEE Transactions on Electromagnetic Compatibility (Accepted with Major revisions 03-04-2024)
3. **Active EMI Filter for Enhanced EMC in Low-Inertia Micro-Grids with Rapid Load Transitions and Non-linear Loads**, **Alexander Matthee**, Niek Moonen, Frank Leferink, IEEE Transactions on Electromagnetic Compatibility (Submitted April 2024)

9.2 Peer Reviewed Conference Papers

1. **Versatile High-Sample Frequency Power Quality Measurement Device**, **Alexander Matthee**, Niek Moonen, Frank Leferink, 2020 IEEE International Symposium on Electromagnetic Compatibility and Signal Power Integrity, EMCSI 2020, pp. 213–215, 2020
2. **Sub-Millisecond Transient Analysis with Multi-Point Measurement in Weak Grids**, **A. Matthee**, N. Moonen, F. Leferink, 2020 International Symposium on Electromagnetic Compatibility, EMC EUROPE 2020
3. **Lightning Strike EMP Effect On Local Grids**, **Alexan-**

der Matthee, Peter William Futter, Robert Vogt-Ardatjew, Frank Leferink, 2021 Asia-Pacific International Symposium on Electromagnetic Compatibility (APEMC)

4. **Versatile LabVIEW-FPGA-based Testbench for Electromagnetic Interference Evaluation in VSDs**, Douglas Nascimento, Robert Smoleński, Piotr Leżyński, **Alexander Matthee**, Niek Moonen, Frank Leferink, 2022 International Symposium on Electromagnetic Compatibility – EMC Europe
5. **Active Transient EMI Stabilization**, B. Ihsan, **A. Matthee**, F. Leferink, T. D.Rachmilda, and D. Hamdani, Proceedings of the 2021 Asia-Pacific International Symposium on Electromagnetic Compatibility, APEMC 2021, pp. 2021–2024, 2021
6. **Micro-Grid Inrush Current Stability Analysis**, **A. Matthee**, N. Moonen, and F. Leferink, 2021 Joint IEEE International Symposium on Electromagnetic Compatibility Signal and Power Integrity, and EMC Europe, EMC/SI/PI/EMC Europe, 2021, pp. 440–444, 2021
7. **Standardized Impedance: Microgrid Perspective for Inrush Current Compliance**, **A. Matthee**, N. Moonen, and F. Leferink, IEEE International Symposium on Electromagnetic Compatibility, vol. 2022-Septe, pp. 783–787, 2022
8. **Superposition of Transient Switching Currents for Non-Linear Devices**, **A. Matthee**, N. Moonen, and F. Leferink, 2022 IEEE International Symposium on Electromagnetic Compatibility and Signal/Power Integrity, EMCSI 2022, pp.216–220, 2022
9. **Concentration of Multi-Point Measurement Data for DC-150 kHz EMI Analysis**, **Alexander Matthee**, Niek Moonen, Frank Leferink, 2023 IEEE 7th Global Electromagnetic Compatibility Conference (GEMC-CON)
10. **Risk of EMI due to Necessary Modification in a Remote Microgrid in Indonesia**, Ilman Sulaeman, **Alexander Matthee**, Hafsah Halidah, Kholid Akhmad, Niek Moonen, Jelena Popovic, Frank Leferink, 2023 International Symposium on Electromagnetic Compatibility – EMC Europe

A Appendix A

A.1 HDL Coder Sub-blocks

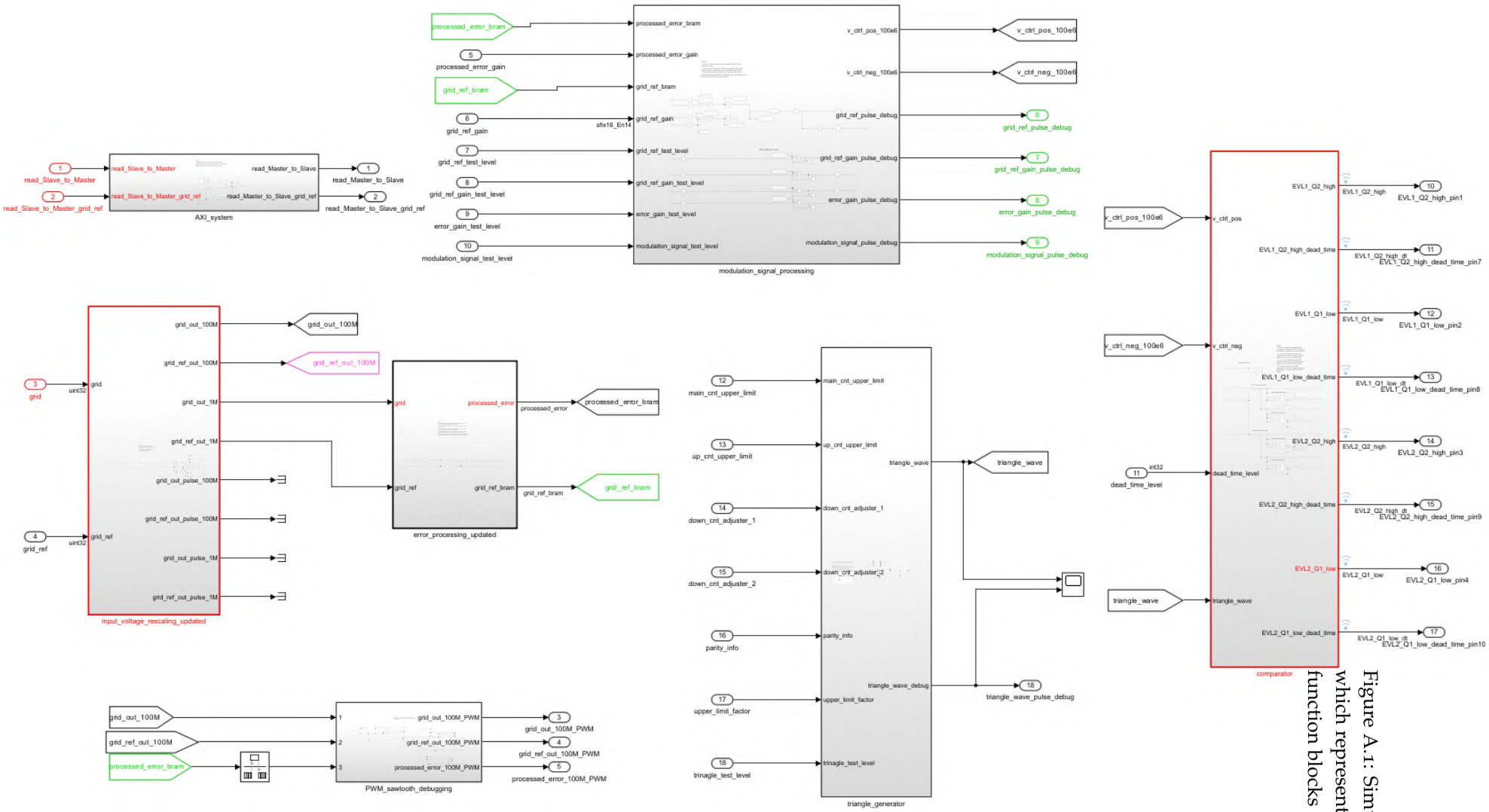


Figure A.1: Simulink sub-block which represents all internal function blocks

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Biography



Alexander Matthee was born in 1989 in Cape Town, South Africa. He attended Cape Peninsula University of technology (CPUT) and graduated with a diploma in electrical engineering in 2010. After, he attended the University of Stellenbosch graduating with a bachelor's degree in electrical engineering in 2014.

Working full time as an application design engineer at Azoteq, he continued with his master's degree part time at Stellenbosch graduating in 2017 (Cum Laude). His master's thesis topic included design, manufacturing and testing of a magnetic gear for dry cooling (ACC) power plant applications. In April 2017 he started working at Altair FEKO group as a software integrity engineer until his appointment as a PhD candidate and researcher at the University of Twente in October 2019.

His PhD research topic, with the EEMCS group at the University of Twente, forms part of the ETOPIA project in the field of electromagnetic compatibility focused on conducted emissions.